Does Artificial Intelligence Democratize Science? Evidence from AlphaFold*

Francois Bares† Anantha Divakaruni‡ Ludovic Phalippou§

October 2026

Abstract

Scientific output in resource-intensive fields is concentrated in a small number of elite universities. We study whether an open-access artificial intelligence tool reduces this concentration, using the public releases of AlphaFold, the deep-learning system that predicts protein structures. In the two years after AlphaFold was released openly, the share of the average low-ranked university in top-journal protein research rose by 50 percent relative to that of the average top-ranked university. The gains are absent outside protein research and in lower-tier journals. Papers from lower-ranked universities also become more novel and depart further from their authors' prior work. Lower-ranked universities with in-house structural biologists rely less on collaboration with top-ranked universities. The evidence suggests that open-access artificial intelligence lowers the human and financial resources required to produce frontier science.

JEL classifications: O33, O31, I23, L17, O34

Keywords: artificial intelligence; science; inequality; structural biology; AlphaFold

*We are grateful to Pierre Azoulay, Kyle Myers, Jacqueline Lane, Fabrizio Dell'Acqua, Alex Chan, Hans Hvide, Savvas Savvides, Rob van der Kant, Emiel Michiels, and Lucrezia Vittoria Viti for their invaluable feedback and support. We also thank seminar participants at Harvard University, University of Bergen, and University of Illinois for useful comments. An earlier version of the paper was circulated under the title "AI-qualizing Science".

†Saïd Business School, University of Oxford, UK. Email: francois.bares.dphil@said.oxford.edu

‡Department of Economics, University of Bergen, Norway. Email: anantha.divakaruni@uib.no

§Saïd Business School, University of Oxford, UK. Email: ludovic.phalippou@sbs.ox.ac.uk

1 Introduction

Research at the scientific frontier is produced by a small number of scientists and universities. The 1% most cited scientists receive about 20% of all citations, and their share has been rising (Nielsen and Andersen, 2021). This concentration tends to persist because funding, recognition, and influence accrue to those who already have them (Azoulay et al., 2014; Bol et al., 2018; Brogaard et al., 2024; Merton, 1968), and because leading scientists also serve as journal editors and grant reviewers (Azoulay et al., 2019; Nicholson and Ioannidis, 2012; Packalen and Bhattacharya, 2018).

This paper asks whether artificial intelligence (AI) weakens or reinforces this concentration. Elite universities may adopt a powerful new tool first and use it best, because they have more skilled researchers, more computing power, and more influence over what gets published, and this would widen existing gaps. Alternatively, a tool that substitutes for scarce inputs may allow universities that lack them to compete at the frontier. The answer matters for the rate and direction of scientific progress. In standard growth models, the flow of new ideas rises with the number of researchers working at the frontier (Bloom et al., 2020; Jones, 1995), and most researchers work outside elite universities. A tool that lets them reach the frontier could therefore raise the rate of discovery, and may even shift the focus of their research. In addition, because such tools are increasingly built by a few firms, the terms on which these firms release them pose a key question for science policy.

We study this question in protein research. To understand how a protein drives cellular function, how it causes disease, and how a drug can target it, researchers need to know its structure in three-dimensional (3D) space. Until recently, they obtained these structures mainly through very costly experiments which also require specialized facilities, trained structural biologists, and heavy computation (Cheng et al., 2015; Garman, 2014; Nogales, 2016; Stevens, 2003). These inputs are concentrated in a few universities, and so is protein research: before AlphaFold, the 12 highest-ranked universities produced 25% of the protein research published in top journals, and the 50 highest-ranked produced nearly half.[1] AlphaFold, an AI model developed by Google DeepMind, predicts a protein's structure without an experiment, and its predictions have been freely available since 2021. AlphaFold can therefore substitute for the inputs that elite universities hold, which makes protein research a natural test of whether AI weakens the concentration of science.

Using academic publication data from OpenAlex, we measure each university's share of the protein articles published in top journals. We rank universities by their citation score

[1] Top journals are the 17 biochemistry, molecular biology, and multidisciplinary journals with the highest SCImago Journal Rank, a prestige-weighted citation measure. Market shares are computed over the articles that the 500 highest-ranked universities published in these journals.

before AlphaFold and group them into quartiles. The top quartile contains 12 universities and the bottom quartile 364, and each produced 25% of top-journal protein articles before AlphaFold. A difference-in-differences design compares the change in each quartile's share after AlphaFold with the change in the top quartile's share. Quartile shares were flat in the six years before AlphaFold. In the two years after DeepMind released AlphaFold 2 and a free database of its predictions in July 2021, the share of the average bottom-quartile university rose by 50% relative to the top quartile, and the gains rise as university rank falls.

Further evidence that AlphaFold drives these gains comes from its release in two versions. AlphaFold 1 won the main protein-structure prediction competition in December 2018, but DeepMind only published its methodology without releasing a usable model for researchers (Senior et al., 2020). AlphaFold 2 came with a free database of predictions. The two releases yield a sharper prediction. Structural protein research, which determines protein structures, could benefit from AlphaFold 1, because its specialists could build on the published method. Non-structural protein research, which uses structures to study protein function, disease, and drug targets, could benefit mainly from AlphaFold 2, because its researchers need ready-made predictions. The data show this pattern. After AlphaFold 1, the bottom quartile gained share mainly in structural protein research. After AlphaFold 2, its gains extended to non-structural protein research, where the share of the average bottom-quartile university rose by 58% relative to the top quartile.

We also find more direct evidence that lower-ranked universities gained these shares by using AlphaFold. Mentions and citations of AlphaFold and similar tools in top-journal protein papers were rare until the release of AlphaFold 2 and then rose sharply, reaching about 30% of structural protein papers in 2023. Lower-ranked universities used the tool disproportionately: in 2023, the bottom quartile wrote 34% of the structural protein papers that mention or cite these tools.

Next, we run placebo tests. General-purpose AI tools, such as large language models, reached every field over the same period, and specialized AI tools appeared in some fields, such as in materials science. If these tools, and not AlphaFold, had raised the output of lower-ranked universities, the same pattern would appear in other fields. We therefore apply the same empirical tests to four fields: microbiology outside protein research, materials science, economics and finance, and the social sciences as a whole. In none of them do lower-ranked university market shares move after 2018, not even in materials science.

We also rule out three alternate explanations. First, AlphaFold might have raised the volume of research at lower-ranked universities more than its quality. However, their

share in lower-tier protein journals remains unchanged, implying that AlphaFold helped lower-ranked universities produce research meeting the standards of top journals. Second, lower-ranked universities might have gained share only by joining projects led by elite laboratories. The gains persist, however, even when restricting credit to papers where lower-ranked researchers hold first or last authorship, the roles that conduct the research and lead the project in the life sciences. Third, the pandemic closed laboratories in 2020 and 2021 and shifted research toward COVID-19. Excluding COVID-19 research leaves the estimates virtually unchanged, and dropping 2020 and 2021 makes them larger.

The results are also robust to our main specification choices. They hold when we measure output by the number of top-journal protein articles instead of shares, and when we estimate OLS or Poisson regressions instead of fractional logit regressions. They also hold when we exclude the six top-ranked review journals in our sample, which mostly publish reviews commissioned by their editors, and when we rank universities by their pre-AlphaFold share of top-journal protein articles instead of their citation score.

We then ask how AlphaFold helped lower-ranked universities, and find that it replaced inputs they lacked. The first is access to computing infrastructure, as determining protein structures experimentally requires intensive computation (Cheng et al., 2015; McKeon and Boudou, 2024a; Nogales, 2016). Before AlphaFold, half of top-quartile universities had access to high-performance computing (HPC), compared with fewer than one in ten bottom-quartile universities.[2] Although running AlphaFold also requires such computational capacity, downloading its predictions does not (Google DeepMind, 2021; Jumper et al., 2021). Bottom-quartile universities without HPC accordingly raised their output more significantly than those with such resources. Following the open release of AlphaFold 2, their number of top-journal protein articles rose by 129% relative to top-quartile universities without HPC. By comparison, bottom-quartile universities with HPC increased their output by 72% relative to top-quartile institutions with HPC, with the difference between these two relative gains being significant at the 10% level.

The second input is the experiment itself, together with the structural biologists who perform it, of whom the average top-quartile university had about five times as many as the average bottom-quartile university before AlphaFold. A prediction can most readily replace this input when a protein already has a documented structure, established either directly or through a related protein determined prior to AlphaFold. Researchers can immediately evaluate such a prediction against existing evidence, whereas a prediction lacking prior data remains a hypothesis needing experimental confirmation (Akdel et al., 2022; Terwilliger

[2]A university has HPC if it hosted a system on the Top500 list of the world's most powerful supercomputers before AlphaFold.

et al., 2024). For example, predicting an unmapped segment of a well-studied protein allows researchers to cross-check the output against its known structural features, whereas predicting an entirely novel protein requires direct experimental validation. If AlphaFold substituted for experiments, the gains of lower-ranked universities should therefore concentrate on proteins with a documented structure.

Following the release of AlphaFold 2, bottom-quartile universities increased their top-journal publications in non-structural protein research by about 19% more for proteins with a previously documented structure than for other proteins, relative to the top quartile. Correspondingly, their share of top-journal publications rose by about 90% more for these documented proteins. These relative gains expand under a stricter definition of a documented structure that requires a greater share of the paper's studied proteins to have prior structural data. Furthermore, the publication gains are largest among universities that lacked in-house structural biologists prior to AlphaFold, which faced the greatest constraints in conducting the experiments that AlphaFold's predictions now substitute for. By contrast, this gap is absent in structural protein research, where novel experimental work remains the primary contribution, and no such effect is observed in either field after AlphaFold 1.

AlphaFold also changed what lower-ranked universities publish in protein science and with whom they collaborate. In both structural and non-structural protein research, papers by researchers at bottom-quartile universities moved further from their authors' prior work and drew on rarer concepts, relative to the top quartile. Their disruptiveness, which measures how often later papers cite them without citing the work they build on, rose only modestly. No such changes occur in the four placebo fields.

Following the release of AlphaFold 2, protein papers from bottom-quartile universities were ten percentage points less likely to include a structural biologist, relative to those from top-quartile institutions. AlphaFold's predictions thus substituted for a portion of the structural expertise these teams previously required, whether sourced from within their own faculties or through external co-authors. Collaboration with top-quartile institutions, however, diverged based on a university's pre-existing structural expertise. Following the release of AlphaFold 1, bottom-quartile universities with a higher share of structural biologists reduced their co-authorship with top-quartile peers. Conversely, those with few or no structural biologists increased their top-quartile co-authorships, plausibly because AlphaFold made elite structural biologists more productive and valuable partners for teams seeking complementary expertise.

Our paper contributes to the economics of AI. A growing literature measures how AI changes productivity in tasks such as writing, programming, image recognition, and

customer support, mostly in experiments or within a single firm (Brynjolfsson et al., 2025; Caplin et al., 2026; Cui et al., 2024; Fügener et al., 2026; Krakowski et al., 2026; Noy and Zhang, 2023). Several of these studies find the largest gains for less experienced or less skilled workers (Brynjolfsson et al., 2025; Noy and Zhang, 2023). In science, Hao et al. (2026) find that scientists who adopt AI tools publish about three times as many papers, while AI narrows the range of topics studied and concentrates citations on fewer papers. Qian (2026), Hill and Stein (2026), and Gans and Mohnen (2026) study AlphaFold at the level of individual proteins and find that predictions complemented rather than replaced experimental structure determination, though evidence remains mixed on whether research shifted toward proteins without a known structure.

We show that the pattern observed among individual workers extends across entire institutions: lower-ranked universities experienced the largest relative gains in top-journal output following AlphaFold's open release. Furthermore, while Hao et al. (2026) observe that AI concentrates citations on a smaller set of papers, AlphaFold reduced the institutional concentration of top-journal protein research away from elite universities. Our findings align with the protein-level evidence on complementarity, while also delineating where predictions actively substitute for physical experiments. In structural protein research, where physical experimentation remains essential to determine structures, the output gains of lower-ranked universities do not differ significantly between proteins with and without a documented structure. However, in non-structural protein research, which uses known structures as inputs, these gains concentrate heavily on proteins with a documented structure, where a prediction can effectively replace an experiment.

We also contribute to the economics of scientific discovery, in particular to the work on how changes in the cost of research inputs affect who produces science. Agrawal and Goldfarb (2008) show that Bitnet, an early university computer network, raised collaboration and publication by researchers at middle-tier universities by lowwring communication costs. Murray et al. (2016) find that cheaper access to genetically engineered mice brought new researchers into the field, and Furman and Teodoridis (2020) find that the free availability of Microsoft's Kinect, a motion-sensing device, made outsider researchers more competitive. CRISPR, by contrast, initially favored institutions already at the frontier of genetic research (Thompson and Zyontz, 2017). These earlier shocks lowered communication or material costs, whereas AlphaFold substitutes for physical capital and specialized expertise. Our results also show that the Matthew effect (Azoulay et al., 2014; Merton, 1968) can weaken when a technology lowers the resources needed to produce top-quality research.

Our findings suggest that the terms on which AI tools are released affect who benefits from them. The gains came after DeepMind made AlphaFold 2 and its predictions freely

available, and they were larger at universities without HPC, which could download predictions without running the model. The successor, AlphaFold 3, was released with restricted access in May 2024. After calls from the research community, DeepMind released the AlphaFold 3 code for academic use in November 2024. As AI models for science become more expensive to train and run, access decisions by a few firms may determine whether these models narrow or widen the gap between elite and other institutions.

2 Data and Sample Construction

We combine publication records, journal and university rankings, university characteristics, and protein databases into a multi-year panel of universities. Below, we outline our procedure for identifying top protein research journals, assigning articles to institutions, measuring university characteristics, and linking papers to individual proteins.

2.1 Journals and Publications

To identify top journals in protein research, we compile a candidate list of 2,170 "biochemistry, genetics, and molecular biology" journals and 175 interdisciplinary journals that publish protein science. We rank these journals by their average SCImago Journal Rank (SJR), a prestige-weighted metric that reflects both citation volume and the status of citing journals. SJR provides a field-normalized, prestige-weighted, and transparently documented measure, making it preferable to the traditional Journal Impact Factor. Regardless of the specific metric used, our selection remains virtually identical, consistently capturing top-tier journals such as *Cell*, *Nature*, and *Science* (Section B.2). To account for annual fluctuations, we calculate each journal's average SJR over the six-year period (2013–2018) preceding AlphaFold's release. In total, there are 17 journals with an average SJR of 10 or higher, forming our benchmark set of top journals in protein research (Table A2).

We obtain data on all articles published in these 17 journals from OpenAlex (the successor to Microsoft Academic Graph). OpenAlex provides rich metadata for each publication, including titles, abstracts, citation networks, publishing journals, and author affiliations down to the departmental level. We favor OpenAlex over alternative databases such as Scopus or Web of Science due to its reproducible data snapshots, preprint coverage, disambiguated entity records, and the granular topic classifications supporting our field definitions (Section B.1). Each article is assigned to one of three mutually exclusive categories:

- *Structural biology*: if at least one author is a structural biologist or the paper covers any of 238 identified structural biology topics. This category represents the most specialized subset of protein research, focusing on determining three-dimensional protein structures.
- *Non-structural protein*: if none of the authors is a structural biologist and the paper's topics fall under one or more of 1,066 protein research topics outside structural biology. This category covers research on protein function, disease mechanisms, and drug targets, in which structures serve as an input rather than as the main contribution.
- *Non-protein*: for all other publications in these journals that do not involve research on proteins.

Our primary sample comprises 10,581 protein research articles published across the 17 top journals between 2013 and 2023 (see Table A10). Within this sample, 5,409 articles focus on structural biology, while 5,172 represent non-structural protein research.

2.2 Universities and Market Shares

Next, we rank universities by their average total normalized citation score (TNCS) in the category "biomedical and health sciences" over 2013–2018, as computed by the CWTS Leiden Ranking. The Leiden Ranking covers about 1,000 universities worldwide, of which we retain the 500 with the highest average scores. It is the standard subfield-normalized scientometric ranking. Computing the ranking over the pre-AlphaFold window yields a fixed institutional stratification that cannot be mechanically affected by AlphaFold itself (see Section B.3 for details and robustness to alternative rankings). We rank universities based on their average scores, with the score for university u defined as follows:

$$Score_u = \frac{1}{6} \sum_{t=2013}^{2018} TNCS_{u,t} \tag{1}$$

where $TNCS_{u,t}$ denotes the total normalized citation score in year t. To examine publication trends across major research institutions, we focus on the top 500 universities derived from this metric. Restricting the panel to the top 500 captures the overwhelming majority of high-impact protein research while preventing unstable annual counts from institutions that publish only occasionally in the top journals. The top 12 universities are Harvard, Johns Hopkins, UCSF, Toronto, UPenn, Washington, UCL, Stanford, Duke, Michigan, UCLA and Oxford (Table A4).

Each author is assigned to a single university (the highest-ranked university for authors with multiple affiliations). Authors with no university affiliation, such as those based only at hospitals, companies, or independent research centers, are dropped. The research output of each university u in year t is measured as the annual count of articles published in our list of top journals, where at least one author is affiliated with that university. We denote this output $P_{u,t}^{all}$ when we count all authors listed on a publication and $P_{u,t}^{main}$ when we count only the two main authors, listed first and last[3]. The share of top-journal publications attributed to university u in year t, which we term its *market share*, is calculated as follows.

$$MktS_{u,t}^{all} = \frac{P_{u,t}^{all}}{\sum_{i=1}^{500} P_{i,t}^{all}} \tag{2}$$

$$MktS_{u,t}^{main} = \frac{P_{u,t}^{main}}{\sum_{i=1}^{500} P_{i,t}^{main}} \tag{3}$$

Each measure uses full counting: a multi-university article contributes one count to every represented university. The denominator applies the same rule, so market shares sum to 100% in each year.

Top-journal publication output is heavily concentrated among leading institutions. Figure 1 displays the cumulative distribution of $MktS_{u,t}^{all}$ across universities ranked by TNCS score. The top 10 universities generate 18% of all the top-journal publications. The top 50 generate nearly half of all publications. The top 150 universities produce almost three times as many top-journal publications as the next 350 universities (Figure 1(A)). Figure 1(B) shows that the distribution is similar if we only take into account the main authors (first and last authors) of each publication. The curve rises steeply among the highest-ranked universities and flattens further down the distribution. Universities with higher citation rankings therefore tend to hold significantly larger shares of top-journal output.

We then order universities by their average TNCS over 2013–2018 and divide them into four quartiles of roughly equal output in top-journal protein research before AlphaFold. Quartile boundaries are set so that each quartile accounts for about a quarter of the distinct author–journal pairs in top-journal protein research during 2013–2018. The top quartile consists of the top 12 universities by average TNCS score, the following 36 universities make up the second quartile, the next 88 make up the third, and 364 make up the fourth (lower) quartile. Geographically, top-quartile institutions are concentrated in the United States. The second quartile includes universities from the US and Europe, along with

[3]See section B.4 for a formal definition of $P_{u,t}^{all}$ and $P_{u,t}^{main}$

three in Australia. Half of third quartile institutions are in Europe, and a quarter in Asia (notably China, Japan, and South Korea). The fourth quartile is more globally dispersed, with concentrations in Europe and East Asia and representation from South America, the Middle East, and Africa. See Figure A1 for a visual summary.

2.3 Author and University Characteristics

We construct several university- and author-level variables to use as controls in our regression analyses. We classify an author as a structural biologist if OpenAlex explicitly identifies them as such, or if their departmental affiliation contains keywords such as "structural biology", "computational biology", "biophysics", or "protein folding". Author gender and ethnicity are inferred from names using the R packages `genderizeR` and `rethnicity`, respectively. We aggregate these author attributes to the university-year level as proportions of total biology faculty, while controlling separately for overall biology faculty size. Additional controls record institutional composition, including the shares of authors affiliated with government agencies, research laboratories, industry, or medical centers, as well as geographic shares across the U.S., Europe, and Asia.

Information on high-performance computing (HPC) comes from Top500.org, which ranks the world's 500 most powerful supercomputers twice a year. We classify a university as having HPC access if the university hosted at least one Top500-listed system in any year between 2013 and 2018. The classification uses only years before AlphaFold 1 and stays fixed through 2023, so AlphaFold cannot affect which universities count as having HPC access. For universities with HPC access, we also record computing power, measured in petaflops per second (10^{15} floating-point operations per second).

Descriptive statistics presented in Table A3 show how unevenly research resources were distributed across universities before AlphaFold . The average top-quartile university had 614 biology faculty members and a 6% share of structural biologists, against 178 faculty members and 3.9% in the bottom quartile. The gap in computing was larger still: 6 of the 12 top-quartile universities had HPC access, against 9% of the 364 bottom-quartile universities. Lower-ranked universities therefore had far less computing capacity and structural expertise than top-ranked universities.

2.4 Linking Articles to Proteins

Lastly, we link the articles in our sample to the specific proteins they study. While university publication counts capture aggregate shifts in market share, protein-level data allow us to examine the scientific content driving those gains. Specifically, this

linkage enables us to distinguish between research on proteins with previously known structures and research on proteins with unknown structures, where predictions still require experimental validation (Akdel et al., 2022; Terwilliger et al., 2024). Identifying these proteins allows us to test whether or not publication gains were concentrated among papers leveraging existing structural knowledge.

Identifying these proteins is challenging, as articles do not report molecular targets in standardized metadata fields. A protein can appear under a gene symbol, full name, abbreviation, or species-specific synonym. To resolve these uncertainties, we map articles to the reviewed section of UniProtKB (Swiss-Prot), which contains over 570,000 expert-curated protein entries, through three independent channels. First, we invert the literature citations that UniProt curators attach to each protein entry. Second, we use PubTator3, a National Library of Medicine tool that tags gene mentions in PubMed articles with standard identifiers. Third, we apply biomedical named-entity recognition models to article abstracts, matching protein mentions to their corresponding reviewed entries. An article is linked to a protein when any of the three channels establishes a match. This pipeline connects 2,859 of the 10,581 protein research articles to 48,314 distinct proteins, achieving a 27% coverage rate (see Table A10). Linkage coverage is balanced across institutional tiers, spanning 25.2% to 27.5% across university quartiles (see Table A12). While database curation lags result in a modest reduction in coverage, particularly for post-AlphaFold publications, these delays occur at a similar rate in every quartile. This uniform distribution ensures that differential linkage success or attrition across university tiers does not drive the quartile comparisons presented in Section 4.3.

For each linked protein, we record whether its structure was known before the AlphaFold database release on 22 July 2021.[4] Roughly a third of the linked proteins (34.6%) meet this criterion (Table A11). Appendix C describes the complete procedure used to link papers to proteins.

3 Publication Rates around AlphaFold

3.1 Changes in Aggregate Market Shares

We plot the aggregate market shares of each university quartile over time in Figure 2. Panels 2(A) and 2(B) report structural-biology and non-structural protein publications,

[4]A protein's structure counts as known if, before that date, the Protein Data Bank held an experimental structure of the protein itself, or SWISS-MODEL or UniProt recorded a solved template sharing at least 30% of its amino-acid sequence.

respectively, counting all authors. Panels 2(C) and 2(D) report the same publications counting only main authors. In all four panels, each quartile's market share is flat before the first release of AlphaFold. The figures also highlight three milestones in AlphaFold's development: (i) AlphaFold 1 (*AF1*) won the 13th Critical Assessment of Protein Structure Prediction (CASP13) in December 2018; (ii) AlphaFold 2 (*AF2*) won CASP14 in November 2020; and (iii) the open release of AlphaFold 2's code and database in July 2021 included predictions for about 350,000 proteins, with the database subsequently expanding to over 214 million predicted structures. Section D.3 and D.5 provide further details.

We observe a notable shift within structural biology publications after the December 2018 CASP13 result (Figure 2(A)). The bottom quartile gained share mainly at the expense of the top quartile. The bottom quartile's share rose from 24.5% in 2018 to about 28% in 2019–2021. The rise continued after the July 2021 open release of AlphaFold 2's code and database. By 2023, the bottom quartile's share reached 29%, while the shares of the top and second quartiles fell to about 22% and 23%, respectively. A similar picture emerges when only main (first and last) authors are considered (Equation (3)). The bottom quartile's share rose from about 25% before AlphaFold to 29.5% in structural biology, and 30.5% in non-structural protein research by 2023. The top quartile's share fell from about 24% to about 20% in both (see Figures 2(C) and 2(D)).

Non-structural protein publications follow a different timing (Figure 2(B)). After AlphaFold 1, the bottom quartile's share rises by only about 1 percentage point. After the open release of AlphaFold 2, the share rises sharply, from 26% in 2021 to 30% in 2022 and 2023. The third quartile also gains share, rising from 24% in 2021 to about 26.5% in 2023. The top quartile's share falls from 25% before AlphaFold to about 21% in 2023, the largest decline of any quartile. The second quartile's share falls from 25.5 to 22.5%.

A potential concern is that the increase in lower-quartile publications reflects the broader wave of AI tools released during this period, including large language models and machine learning applications that may have benefited lower-quartile institutions across multiple fields. Specialized AI tools have also emerged in materials science (Cheng et al., 2026; Choudhary et al., 2022; Merchant et al., 2023) and other domains, raising the possibility that the observed effects are not unique to AlphaFold. We test this alternative explanation by examining market share changes in top-journal publications across microbiology (excluding protein research), economics and finance, materials science, and social sciences. Figure 3 reports the results. Quartile market shares in these fields do not change after 2018. For microbiology, we use the top journals in the field after removing the journals in our protein sample. The change in market shares after AlphaFold is therefore specific to protein research.

Alternatively, the gains by lower-quartile universities in top journals could reflect increased overall research output following AlphaFold, rather than improved research quality. We test this hypothesis by examining protein-related publications in lower-ranked journals (SJR between 1 and 10) for all four university quartiles. The corresponding yearly market shares are shown in Figure A2. We do not observe any pre-trend and no changes in market share after AlphaFold.

Overall, these results show that the increase in top-journal market share among lower-quartile universities after AlphaFold is confined to protein research and to top journals within that field. The gains do not simply reflect increased publication output: the share of publications in lower-ranked journals among lower-quartile universities remains unchanged. Moreover, the absence of comparable patterns in other fields makes broader trends in scientific publishing or concurrent AI developments an unlikely explanation and points instead to a field-specific effect of AlphaFoldon protein research.

3.2 Regression Analysis

To further test whether the changes in market shares observed in Figure 2 can be attributed to the introduction of AlphaFold, we employ a two-way fixed effects regression framework. Our analysis is conducted at the university level using a yearly panel of universities from 2013 to 2023. Each university is assigned to its pre-AlphaFold quartile as in Figure 2. This quartile-based approach allows us to estimate the average post-AlphaFold effect within each group, capturing shifts in publication dynamics while accounting for both time-invariant university characteristics and common yearly shocks. Specifically, for each university u in year t, we estimate the following equation:

$$y_{ut} = \beta_0 + \sum_{q=2}^{4} Quartile_{q,u} \cdot PostAF1_t \cdot \beta_q + \mathbf{X}'_{ut}\delta + \alpha_u + \gamma_t + \epsilon_{ut} \tag{4}$$

where $Quartile_{q,u}$ is a binary variable equal to 1 if university u belongs to quartile q (zero otherwise). $PostAF1$ is a binary variable set to 1 for all years from 2019 onward, capturing the period after the December 2018 CASP13 result. The vector $\mathbf{X}_{u,t}$ includes university-level control variables, with δ representing the corresponding coefficients. Fixed effects α_u control for time-invariant university characteristics, while γ_t captures common shocks affecting all universities in a given year.

We estimate the impact of AlphaFold on university research output using three variations

of the outcome variable y_{ut}. Each variation captures university publication activity in a different form. First, we define y_{ut} as the number of top-journal publications (in log) by all authors at university u in year t, to assess absolute changes in publication output. Second, we assess relative changes by defining y_{ut} as the university's annual market share based on all authors, using equation (2). Third, we define y_{ut} as the university's annual market share based on main authors (first and last) only, using equation (3).

We first estimate these models on the full sample of protein research publications. Panel 1(A) of Table 1 shows estimates for the three outcome variables[5]. The bottom-quartile interaction terms are positive and significant at the 1% level in all three columns. The second- and third-quartile estimates are smaller and less precise. In model (1), the average bottom-quartile university increases its yearly top-journal publications after AlphaFold by 46% relative to the top quartile (95% confidence interval 30 to 64%). The effect is monotonic. The second- and third-quartile universities exhibit the same upward trend, but with smaller magnitudes.

Given that market shares are bounded between zero and one, we estimate models (2) and (3) using a fractional logit specification (Papke and Wooldridge, 1996). This approach fits a logistic conditional mean via quasi-maximum likelihood, ensuring fitted values remain strictly within the unit interval. Consistency requires only that the conditional expectation function is correctly specified, placing no restrictive assumptions on the underlying distribution of the dependent variable. The estimated coefficients capture differences on the logit index rather than linear percentage-point changes in market share. For small individual university shares, however, the logit index closely approximates the natural logarithm of the share.[6] A point estimate β therefore yields a proportional change of $e^{\beta} - 1$ in a university's expected market share.

The 0.20 estimate in model (2) implies a 23% increase in the expected annual market share of a bottom-quartile university after AlphaFold, relative to top-quartile peers (95% confidence interval 8% to 39%).[7] The corresponding gains for the second and third quartiles are 6% and 14%. Neither is significant at the 5% level. Summed over all its universities, the bottom quartile's share of top-journal protein publications rose from 30.9% in 2013–2018 to 34.4% in 2019–2023. Over the same years, the top quartile's share fell from 20.1% to 17.9%.[8]

[5]Tables A5 and Table A6 report robustness to different combinations of controls and fixed effects.

[6]The average top-quartile university holds about 1.7% of top-journal protein publications before AlphaFold. At that share, $\ln[p/(1-p)]$ and $\ln p$ differ by less than 0.02.

[7]The estimates in Table 1 are robust to the exclusion of review journals. Table A8 in the appendix drops six review journals from the 17 top journals: *Nature Reviews Molecular Cell Biology*, *Trends in Cell Biology* and four *Annual Review* journals. Every coefficient in Table A8 lies within 0.01 of the corresponding coefficient in Table 1. The bottom-quartile market-share coefficient equals 0.20 in both tables.

[8]These shares count each article once for every university on its author list, as in equation (2). Shares

Restricting the share definition to first and last authors in model (3) yields relative gains of 10%, 14%, and 24% for the second, third, and bottom quartiles.

We also estimate the models separately for structural biology and for non-structural protein research (Panels 1(B) and 1(C)). The estimated effect is similar in both domains. Relative to the top quartile, the market share of the average bottom-quartile university rises by 27% in each domain (95% confidence interval 10% to 47% in both). The impact of AlphaFold therefore extends beyond structural biology. Note that the results reported here reflect average outcomes before and after AlphaFold's release: annual trends are discussed in Section E.2.

Table A7 compares the baseline estimates with results obtained using alternative estimators for each outcome. Column (1) estimates publication counts directly using Poisson pseudo-maximum likelihood (PPML), rather than applying the estimator to their logarithm (Santos and Tenreyro, 2006). Column (2) applies PPML to the number of unique author–publication pairs. Columns (3) and (4) reproduce the fractional-logit estimates from models (2) and (3), respectively, and report average marginal effects beneath each coefficient (Papke and Wooldridge, 1996). Column (5) estimates the outcome in model (2) using OLS rather than fractional logit. Across these specifications, the estimates show a clear gradient: the lower the university's baseline quartile, the larger its increase in top protein research output following AlphaFold . The only exception is the number of unique author–publication pairs in column (2), for which this monotonic pattern breaks down. This result is informative because it indicates that the larger increase in publication output among lower-quartile universities is not driven by a mechanical increase in the number of co-authors per paper.[9]

4 Mechanisms

4.1 The Role of Accessibility

Having established that AlphaFold increased the share of top-journal protein publications from lower-quartile universities, we now examine mechanisms that may explain this shift. AI-based protein-structure prediction has grown substantially since 2014, with the number of such tools submitted to CASP competitions rising steadily over the past decade

summed within a quartile therefore differ from the shares in Figure 2, which count each article once per quartile and assign authors to quartiles by their pre-AlphaFold affiliations.

[9]The results are also robust to COVID-era publishing dynamics. Excluding publications linked to COVID-19 topics yields virtually identical quartile estimates, while dropping the 2020–2021 period entirely produces larger point estimates, indicating that our baseline results are conservative (see Figure A10).

(Figure A3). AlphaFold's breakthrough performance at CASP13 in 2018 accelerated this trend. Despite the subsequent proliferation of competing AI approaches, AlphaFold has maintained a dominant position, with more than 60% of prediction models submitted to recent CASP competitions either directly using AlphaFold or building on its methodology. This pattern raises a key question: does the observed democratization in top-journal protein publications stem primarily from AlphaFold 's technical superiority, or does its distinctive open accessibility play an additional role?

Although AlphaFold 1 won CASP13 in December 2018 and its methodology was published in 2019 (Senior et al., 2019), it was not until January 2020 that DeepMind released the code and trained weights (Senior et al., 2020). Even then, the release was limited because input features were provided solely for the CASP13 targets. Researchers seeking to apply AlphaFold 1 to new proteins had to therefore construct these features themselves, requiring structural biology expertise and substantial computational resources. Its successor, AlphaFold 2, won CASP14 in November 2020, and in July 2021 DeepMind released its full code together with a public database of predicted structures. The database initially contained structures for about 350,000 proteins and has since expanded to more than 214 million. The two releases thus differed not only in predictive accuracy but also in ease of use and accessibility.[10] This staggered sequence allows us to examine whether the broader release of AlphaFold 2 was associated with a stronger democratizing effect than AlphaFold 1's initial breakthrough, while recognizing that any larger effect after the second release may reflect both lower adoption barriers and improved predictive accuracy.

We therefore compare publication changes before and after the open release of AlphaFold, and assess whether adoption barriers under the closed-source model limited the democratizing effect of AlphaFold. We re-estimate our baseline specification with two period indicators in place of the single *PostAF1* indicator. *PostAF1* equals one in 2019–2021, the years between the December 2018 CASP13 result and the open release. *PostAF2* equals one in 2022–2023, the first two full years after DeepMind released the code of AlphaFold 2 and its structure database in July 2021.[11] The pre-AlphaFold years, 2013–2018, are the reference period.

The results reported in Table 2 show that the gains of lower-quartile universities come almost entirely after the open release of AlphaFold 2. For all top-journal protein

[10]On the hardest CASP targets, the lead of AlphaFold over the best non-AlphaFold group in median GDT_score (an accuracy metric from 0 to 100) grew from 6.5 points in CASP13 to 26.3 points in CASP14. Appendix D.5 documents the key improvements in structural prediction accuracy and accessibility of each AlphaFold release (see Figures D1 and D2).

[11]We assign 2021 to PostAF1 because, given publication lags, research published that year largely predates the July 2021 database release.

publications, the market share of the average bottom-quartile university rises by 8% relative to the top quartile in 2019–2021. The 2019–2021 estimate is not statistically significant. In 2022–2023, the increase is 50%.[12] Summed over all its universities, the bottom quartile's share of top-journal protein publications is 30.9% in 2013–2018, 31.7% in 2019–2021, and 38.6% in 2022–2023. Over the same periods, the top quartile's share falls from 20.1% to 19.6% and then to 15.2%. Third-quartile universities gain 32% after the open release. Second-quartile universities show no significant change in either period. Model (3), which credits only first and last authors, shows the same timing, with gains of 11% in 2019–2021 and 49% in 2022–2023. The timing differs between structural biology and other protein research. In structural biology, the market share of the average bottom-quartile university already rises by 20% relative to the top quartile in 2019–2021. In 2022–2023, the increase in structural biology is 40%. In other protein research, the corresponding increase in 2019–2021 is 11% and is not statistically significant. In 2022–2023, the increase in other protein research is 58%.

The concentration of the gains following the open release is consistent with lower adoption barriers, although the higher predictive accuracy of AlphaFold 2 may also have contributed. The early gains in structural biology are consistent with AlphaFold 1 being used primarily by specialists who could build on its published methods and construct the required inputs. The open release broadened access to researchers who use protein structures as inputs rather than produce them. Lower-quartile universities could use the predicted structures directly from the public database, without requiring in-house structural-biology expertise, potentially extending the benefits of AlphaFold beyond institutions with the capacity to implement the technology independently.[13]

4.2 The Role of Resource Constraints

We next examine whether AlphaFold reduces the dependence of protein research on expensive computing infrastructure. Historically, the field has required substantial computing power for complex tasks like molecular dynamics simulations, structural model fitting, drug discovery, and functional genomics (Cheng et al., 2015; Garman, 2014; McKeon and Boudou, 2024b; Nogales, 2016). This capacity is heavily unequally

[12]The concentration of gains after the open release of AlphaFold 2 is also robust to the exclusion of the six review journals (see Table A9 in the appendix). Excluding the six review journals, the bottom-quartile market-share coefficient is 0.07 in 2019–2021 and 0.41 in 2022–2023. None of the coefficients in Table A9 differs by more than 0.02 from the corresponding coefficient in Table 2.

[13]These timing dynamics are not an artifact of pandemic-era publishing shocks. The concentration of gains in the post-AlphaFold 2 period remains robust when excluding COVID-19-related publications, and persists when omitting the 2020–2021 pandemic years from the sample entirely (Figure A11).

distributed: before AlphaFold , half of top-quartile universities (6 of 12) had access to high-performance computing (HPC), compared to just 9% of bottom-quartile universities (Table A3). Researchers can utilize AlphaFold 2 in two ways, which could potentially yield distinct empirical predictions for our study. Running the model locally still requires a modern graphics processor and over two terabytes of genetic databases (Google DeepMind, 2021; Jumper et al., 2021). By contrast, the open AlphaFold database delivers predicted structures without requiring any local computation. If computing power remained a barrier after the open release, research gains among lower-quartile universities should be concentrated at institutions with HPC. Conversely, if the open database effectively removed this barrier, universities without HPC should benefit at least as much as those with it.

To test this, Table 3 estimates the specification from Table 2 separately for universities with and without HPC access, comparing lower-quartile universities to top-quartile institutions within their respective hardware tier. The results show that after the open release of AlphaFold 2, bottom-quartile universities without HPC gained at least as much ground as those with it. Their overall market share rose by 62% relative to the top quartile, compared to a 48% gain for HPC-equipped institutions (Panel A, columns 3 and 4). This parity persists when isolating main-author credits, with gains of 58 and 46%, respectively (columns 5 and 6). Neither difference is statistically significant. Publication counts, however, rose significantly faster at universities without HPC: an increase of 28% (versus 6%) in 2019–2021, and 129% (versus 72%) in 2022–2023 (columns 1 and 2). This equalizing pattern remains consistent outside structural biology (Panel C), where market shares rose by 63% without HPC and 57% with it, and publication counts by 136% and 82%, respectively. Within structural biology (Panel B), the differences between the two groups are not statistically significant, though non-HPC institutions saw slightly smaller market-share gains (39% versus 52%) but larger increases in publication counts (144% versus 95%). Finally, these post-release benefits extend beyond the bottom tier, with third-quartile universities also gaining regardless of HPC access (Panel A, columns 1–4).

Access to computing power was therefore not a limiting factor in determining which universities benefited from the open database release under AlphaFold 2. Institutions without HPC gained at least as much market share as those with it and achieved even larger growth in publication counts. By delivering predicted structures directly, the open database eliminated the need to run AlphaFold locally, placing non-HPC universities on an equal footing with their HPC-equipped peers. Yet while equalized access explains why non-HPC institutions kept pace, computing power alone cannot explain why lower-quartile universities gained relative to top-quartile ones overall. The next section therefore pivots from HPC as an input to analyzing the specific proteins studied in each paper.

4.3 The Role of Prior Protein Information

Sections 4.1 and 4.2 show that the gains of lower-quartile universities emerged primarily after the open release of AlphaFold 2 and were at least as large at universities without high-performance computing. We now ask whether AlphaFold also substituted for a second costly input: experimental structure determination, which requires specialized facilities and trained structural biologists (Cheng et al., 2015; Garman, 2014; Nogales, 2016). Protein-level studies suggest that AlphaFold predictions can both replace experiments and serve as starting points for new experimental work (Gans and Mohnen, 2026; Hill and Stein, 2026; Qian, 2026). The scope for substitution depends on what is already known about the protein being studied. When the structure of the same protein, or of a closely related protein, was determined experimentally before AlphaFold , researchers can compare the prediction with existing evidence and use it without new laboratory work. For proteins without such prior structural information, by contrast, the prediction remains a hypothesis until experimentally validated (Akdel et al., 2022; Terwilliger et al., 2024). Research on these proteins should therefore remain more dependent on the university's own capacity to conduct experimental work. Consistent with this, top-quartile universities employ a larger share of structural biologists than bottom-quartile universities (Table A3). If AlphaFold enabled lower-quartile universities to substitute predictions for experimental work, the gains of each lower quartile relative to the top quartile should be concentrated in papers studying proteins with prior structural information. We test this prediction by turning from research inputs to research content: we identify the proteins studied in each paper and compare each lower quartile with the top quartile separately for papers studying proteins with and without prior structural information.

Implementing this test requires a publication–protein linkage that standard bibliometric data do not provide. Establishing this link is challenging because protein names are non-unique, species-dependent, and reported inconsistently. The most reliable linkage available, used in Hill and Stein (2026) among others, is curated in Swiss-Prot, where each link is individually validated by biocurators. These links are precise but cover only a small part of the protein space. We retain this channel and complement it with linkages based on PubTator3's normalized gene identifiers and biomedical named-entity recognition over abstracts. Crucially, we link publications only to experimental structures and homology templates available before AlphaFold 2's public database release on 22 July 2021. This ensures that our research classification is not mechanically affected by AlphaFold itself.

To identify the proteins studied in each paper, we match every publication against the reviewed section of UniProtKB, the reference catalog of over 570,000 curated protein entries. The difficulty is that papers can state the proteins they study in numerous ways in

the text. The proteins must therefore be inferred from how the text refers to them. The inference is further complicated by how proteins are named. A single protein goes by several names: its gene symbol, its full recommended name, an abbreviation, or any of its recorded synonyms. A single name, in turn, can refer to several proteins, because most proteins have counterparts in other species and each counterpart is a separate entry in the catalog. We therefore build the link between papers and proteins ourselves, using three distinct and novel methods.

The first method exploits the literature citations that UniProt curators attach to each protein's entry. Each entry cites the publications that document the protein's properties. Read in reverse, this record links a paper to every protein whose entry cites it. Links of this kind are precise: each exists because a human curator judged that the paper reports experimental evidence on the given protein. Curation, however, is a slow process. Because entries are updated years after the underlying papers appear, the publications least likely to be linked are the most recent ones: Those are precisely the post-AlphaFold papers that identify our effect.

Our second method replaces the curator's reading with PubTator3, an automated annotation system run by the National Library of Medicine. PubTator3 scans the PubMed literature weekly and tags each gene a paper mentions with its standard gene identifier. Every entry in the UniProt catalog records the identifier of its own gene, which turns the match into an exact lookup and puts a paper published last month on the same footing as one published a decade ago. The third method requires only the paper's abstract. Models trained to recognize protein names mark each mention, each mention is searched in the catalog, and only the best match is retained. Because a link needs nothing beyond a name in the abstract, this method reaches papers that neither curators nor PubTator3 have processed.

A paper is linked to a protein when at least one of the three methods establishes a match. In our data, each successive channel links papers that the preceding channels leave unlinked, which is why we use all three rather than relying on curation alone.[14] The union links at least one protein to 2,859 of the 10,581 protein research papers in the sample, yielding a coverage rate of 27.0% (Table A10). Coverage is broadly similar across university quartiles, reducing concern that the results are driven by differences in linkage coverage by university rank.

We then ask what was already known about each of these proteins. A protein has a known structure if, before the AlphaFold protein structure database was released on 22 July 2021, its own three-dimensional structure had been solved experimentally, or a usable

[14] Appendix C sets out the three channels, the matching rules, and the limitations of the procedure.

homology template existed. The second case rests on a basic regularity of biology: proteins with similar amino-acid sequences fold into similar shapes. The solved structure of a close relative can therefore stand in for the target protein, with the share of sequence the two have in common measuring how far the substitute can be trusted. We treat a template as usable at 30% sequence identity or above, the conventional threshold below which a related structure no longer supports reliable modeling. Of the 48,314 linked proteins, 18.5% had their own experimental structure, 16.1% had only a homology template, and 65.4% had neither (Table A11). Because a paper can study one or several proteins, we classify each linked paper based on the share of its linked proteins that had a prior structure. A paper is classified as structure-available when at least half of its linked proteins had a prior structure before AlphaFold . Otherwise, it is classified as a no-prior-structure paper.

Results and interpretation We add to the two-period specification of Table 2 an interaction with an indicator for papers with a prior structure (Table 6). The triple interaction terms measure how much more a quartile gains on papers with a prior structure than on papers without one, relative to the top quartile. Post-AlphaFold 1, none of these differences is statistically significant. The point estimates for the third and fourth quartiles favor papers without a prior structure. After AlphaFold 2, the pattern reverses in non-structural protein research. In this field, the publications of the average bottom-quartile university rise by about 19% more on papers with a prior structure than on papers without one, relative to the top quartile. The corresponding difference in market share is about 90%. Second-quartile universities show larger differences on both measures. For third-quartile universities, the difference is smaller and significant only for publications. Within structural biology, no difference is statistically significant after AlphaFold 2, with negative point estimates for the third- and fourth-quartile market shares. In the pooled sample of all protein research, which combines the two fields, only the second-quartile difference is significant.

Figure 4 and A14 illustrate this reversal between the two periods. The reversal is robust to alternative thresholds for classifying a paper as drawing on prior structures, as shown in Figure A15. These thresholds further highlight the dynamics: the stricter the threshold for classifying a paper as having a prior structure, the larger the post-AlphaFold 2 gains for lower-quartile universities.

The post-AlphaFold 2 gains on papers with prior structures in non-structural protein research point to the role of the AlphaFold database as a low-cost structural input. Researchers in these fields may use a structure as an input into functional or applied studies rather than as the principal contribution. When an existing experimental structure or homology template supports it, an AlphaFold prediction can be credible without

experimental validation. Lower-quartile universities may benefit most because they have less experimental capacity. For proteins with no prior structural information, experimental confirmation retains more of its value, and top universities retain more of their advantage. This interpretation explains why lower-quartile gains are smaller for papers without prior structures and in structural biology, where the value of experimentation is greater.

Two additional results further support this interpretation. Table A13 splits universities by whether a structural biologist appeared on any of their pre-AlphaFold top-journal papers. After AlphaFold 2, the lower-quartile gains at universities without such an author are larger for papers with a prior structure. The difference in lower-quartile gains between papers with and without prior structures is also greater for universities without a structural biologist pre-AlphaFold . Team composition changed as well. Table A14 shows that, after AlphaFold 2, papers from bottom-quartile universities are 10.2 percentage points less likely to include a structural biologist as a co-author.

Mechanism summary Overall, the evidence presented in this section yields a consistent conclusion. Between the releases of AlphaFold 1 and 2, market-share gains among lower-quartile universities were modest and confined to structural biology, where specialists could adapt the published methods of AlphaFold 1. These gains expanded dramatically only after AlphaFold 2 democratized access through a publicly accessible database. By lowering the capital and expertise barriers to obtaining protein structures, AlphaFold 2 leveled the playing field for lower-quartile universities, which historically lacked extensive experimental facilities, compute resources, and human expertise. Computing access, however, was not the binding constraint: universities without HPC gained at least as much as those with it. Linking paper-level data to target proteins reveals the precise channel of substitution: post-database gains outside structural biology were heavily concentrated among proteins with pre-existing experimental structures (or close homologs). For these proteins, researchers could readily validate AlphaFold predictions against established structures, bypassing the need to conduct new experiments or await external determinations.

5 Research Reorientation and Team Formation

The preceding section shows that AlphaFold made lower-quartile universities more competitive by reducing their dependence on structural-biology expertise and on new experimental structure determination. Their gains were at least as large where high-performance computing was absent. We now ask whether this relaxation of constraints was

accompanied by changes in the research portfolios and team formation of lower-quartile universities. We examine two dimensions of this adjustment. First, we assess research orientation using three metrics: *pivot behavior*, which captures shifts toward areas farther from an author's prior work; *novelty*, which captures engagement with less common concepts; and *disruptiveness*, which captures changes in subsequent citation patterns. Second, we examine whether reliance on collaborations with top-quartile universities changed after AlphaFold.

5.1 Qualitative Measures of Research

We examine three qualitative metrics that reflect distinct dimensions of research activity at the author paper level. These indicators offer insight into how AlphaFold may be reshaping scientific inquiry, enabling researchers to pursue new directions, introduce more original ideas, and contribute more meaningfully to ongoing debates in the field.[15] Analyzing these metrics across university quartiles helps assess whether the rising presence of lower-quartile institutions in top journals is accompanied by changes in the character and potential impact of their research output.

The three metrics are defined as follows:

- **Pivot score**: This measure quantifies the extent to which a given paper by a researcher differs from their prior work based on the journals cited. It is calculated as 1 minus the cosine similarity between the distribution of journals referenced in the focal paper and those in the author's previous publications over the past three years (Hill et al., 2025) (see detailed description of this measure in section B.7). The pivot score ranges from 0 (no departure from prior work) to 1 (implying complete departure). A higher pivot score indicates a greater shift in research direction, which may be facilitated by AlphaFold's provision of novel structural methodologies or predictions.

[15]To motivate the analysis, we present some examples of how AlphaFold impacted various fields in protein research: AlphaFold has enabled the discovery of new 3D shapes (folds) in parts of viruses that bind to sugar molecules (glycan-binding domains), shifting research from studying rigid, crystal-like arrangements (paracrystalline) to more flexible, varied forms (polymorphic assembly in rotavirus studies (Hu et al., 2022)). In drug discovery, AlphaFold sped up the identification of mind-affecting drugs (psychotropic agonists) that target a brain protein involved in mood (TAAR1), achieving a higher success rate (hit rate) than predictions based on similar known structures (traditional homology models) (Díaz-Holguín et al., 2024). Additionally, AlphaFold supported predictions across all proteins in an organism (proteome-wide) for changes in DNA that alter protein building blocks (missense variant effects), improving understanding of their disease-causing potential (pathogenicity) and aiding gene-based medicine (clinical genomics) (Cheng et al., 2023). AlphaFold has also been instrumental to uncovering the structure of an immune signaling complex (IL-27), revealing new ways cells communicate during inflammation (Jin et al., 2022). Access to AlphaFold also accelerated the discovery of a small molecule inhibitor for CDK20, a protein linked to cancer, by generating accurate structures for rapid screening (Ren et al., 2023).

- **Novelty**: This metric evaluates originality in a paper by quantifying the rarity of concepts associated with its cited references, using OpenAlex topic classifications. As such, it follows an entity-based approach to estimating novelty (Azoulay et al., 2011; Hill and Stein, 2025; Liu et al., 2022; Luo et al., 2022), leveraging OpenAlex's concept classification. For each reference, concepts with relevance scores above 0.5 are included, where the relevance score reflects centrality to the reference content. Novelty is then calculated as the average inverse of the works count for these concepts, with works count indicating prevalence in the broader literature (refer to Section B.7 for a detailed description of this metric.). This approach denotes novelty because reliance on less common concepts signals that the paper builds on underrepresented ideas or unusual knowledge combinations, setting it apart from incremental advances in mainstream subject areas. Novelty matters in the context of AlphaFold because predicted 3D structures reduce the need for experiments requiring substantial resources, allowing researchers to study proteins that were previously inaccessible.

- **Disruptiveness**: This variable measures the degree to which a paper affects subsequent research, based on citation patterns in which future works cite the focal paper but not its predecessors (Funk and Owen-Smith, 2017; Park et al., 2023); see Section B.7 for details. A positive disruptiveness score indicates that the paper draws citations primarily to itself, implying that it reduces the prominence of earlier work. Accurate protein structures supplied by AlphaFold may facilitate this form of research, particularly for scientists at lower-quartile institutions.

  Importantly, these measures describe intellectual orientation and citation impact, not the structural novelty of the focal protein. A paper can cite unfamiliar journals or rare concepts while studying a protein with prior structural information. Research reorientation can therefore coexist with the preceding result that lower-quartile gains concentrate where AlphaFold predictions can be used without new experimental validation.

Changes in pivot behavior

Analysis of pivot scores reveals significant shifts in research direction across university quartiles following the introduction of AlphaFold. Prior to AlphaFold, researchers at top-quartile universities consistently exhibited higher pivot scores, indicating a greater propensity to make substantial shifts in their research direction. This pattern likely stems from their access to superior resources, funding, and collaborative networks, which

facilitate exploration across diverse subfields of protein research. However, post-AlphaFold, this disparity narrows, with lower-quartile universities experiencing larger increases in pivot scores.

In structural biology, pivot scores increased across all university quartiles following AlphaFold, with larger gains observed in lower quartiles (Figure 5(A)). We quantify this effect using the regression specification in Equation 4, which estimates the differential impact of AlphaFold on pivot scores across quartiles while controlling for university fixed effects and time trends. Results from Model (1) in Table 4 show that, relative to the top quartile, bottom-quartile universities experienced a 2.6% increase in pivot scores, third-quartile universities a 1.7% increase, and second-quartile universities a 1% increase, all highly statistically significant.

We observe similar trends in non-structural protein research. Pivot scores increased across all university quartiles following AlphaFold, but with a more pronounced pattern favoring lower-quartile institutions relative to those observed in structural biology (Figure 5(B)). We once again estimate this effect using Equation 4. Results from Model (4) in Table 4 show that, relative to the top quartile, second-quartile universities experienced an increase of 3.4% in pivot scores and fourth-quartile universities an increase of 3.3%, both highly statistically significant.

Taken together, these patterns suggest that AlphaFold, by offering a widely accessible tool for protein structure prediction, has helped lower barriers to entry in both structural and non-structural protein research. Once structural predictions became a low-cost input, lower-quartile researchers could pursue questions farther from their prior citation portfolios. These findings are consistent with prior work showing that technological shocks can broaden participation in science by reducing costs and enabling the exploration of ideas that are more novel or distant from researchers' existing trajectories (Agrawal and Goldfarb, 2008; Furman and Teodoridis, 2020).

Changes in novelty

Analysis of novelty scores reveals distinct patterns across university quartiles following the introduction of AlphaFold. These patterns likely stem from AlphaFold's capacity to provide access to uncommon protein structures and predictions, allowing researchers to pursue hypotheses that were previously difficult to explore due to various constraints.

In structural biology, novelty scores increased across all university quartiles post-AlphaFold, with significant gains in lower quartiles (Figure 5(C)). Regression estimates using Equation 4 shown in Model (2) of Table 4 show that, relative to the top quartile, second-quartile universities experienced a 2% increase, while third- and fourth-quartile

universities both saw a 1.5% rise in novelty. These shifts underscore AlphaFold's pivotal role in enabling researchers at resource-constrained institutions to address novel structural challenges, including the prediction of previously unresolved protein folds.

In the case of non-structural protein research, changes in novelty patterns are more pronounced, with novelty scores rising sharply for lower-quartile institutions (Figure 5(D)). The corresponding regression results presented in Model (5) of Table 4 indicate no significant effect for the second quartile but a substantial 1% (2.7%) increase for the third (bottom) quartile, relative to the top quartile. These results suggest that by leveraging AlphaFold-derived structural insights, researchers at lower-quartile institutions are able to integrate accurate and new protein structures into non-structural domains such as drug discovery and functional genomics despite lacking experimental resources.[16]

Changes in disruptiveness

Disruptiveness, a citation-based metric, evaluates whether a paper consolidates (negative values) or disrupts (positive values) the existing citation network by attracting citations independently of its references. Our findings show that protein papers published in top journals post-AlphaFold have modestly redirected future studies away from prior literature. Due to citation lags, the full impact of AlphaFold on citation patterns may take years to emerge.

In structural biology, disruptiveness scores exhibit small but statistically significant increases across all university quartiles, with no clear trend favoring lower quartiles (Figure 5(E)). Regression estimates from Model (3) in Table 4 indicate average annual increases of 1.3% for lower-quartile (fourth-quartile) universities, 1.2% for second-quartile, and 0.5% for third-quartile universities, relative to the top quartile. These modest gains suggest that

[16]For example, researchers at McMaster University (ranked 69 in our sample) and others used ai predictions to study how proteins interact based on their genetic sequences, showing that the sequence can be more important than the 3D structure for predicting these interactions in genomics (Hosseini et al., 2024). This approach allowed them to incorporate uncommon ideas about protein behavior that were hard to test previously, boosting the originality of their findings, and helped better understand how enzymes repair DNA in bacteria. Similarly, researchers at the University of Alberta (ranked 84 in our sample) incorporated AlphaFold predictions to identify molecular markers for better medical diagnosis and treatment, enabling exploration of rare protein functions in bacterial genomes by adding notes to predicted structures in their system, which enhanced the novelty of their research by uncovering new biological insights (Jackson et al., 2025). Additionally, researchers at Universidade de Lisboa (ranked 315 in our sample) used AlphaFold to model a protein structure (CDS2), revealing potential spots for drugs to target in cancer-related pathways, and to design small binders for a receptor involved in cell processes for targeted breakdown (Huang et al., 2025). Their approach introduced original concepts in combining structure with function for new therapeutic ideas. Lastly, researchers at University of Coimbra (ranked 372 in our sample) participated in a study that applied AlphaFold to predict a protein structure (TEX264), identifying parts involved in clearing damaged DNA, which brought fresh perspectives to DNA repair mechanisms and increased the work's originality by linking structure to rare cellular cleanup processes (Lascaux et al., 2024).

AlphaFold supports incremental disruptions by providing accessible structural predictions that address challenges in traditional experimental methods, such as predicting complex protein folds previously difficult to resolve.

In non-structural protein research, disruptiveness effects are much weaker, with scores showing some overall increase (Figure 5(F)). Regression results in Model (6) of Table 4 reveal a small positive effect of 0.6% for lower-quartile (fourth-quartile) universities and a small negative effect of -0.6% for third-quartile universities, relative to the top quartile, both marginally significant at the 10% level. These subtle shifts indicate that AlphaFold's structural insights have not yet significantly altered citation dynamics in non-structural protein domains. This may be due to the initial integration of structural predictions into studies aligning with existing research trajectories rather than immediately disrupting them. The limited impact could also reflect the time-lagged nature of citation-based metrics, where significant changes in research patterns often take years to fully emerge in the literature.

5.2 Collaboration Patterns

Collaboration represents another pathway through which AlphaFold, as a technological shock, may have reshaped the competitive dynamics of protein research publishing in top journals. Prior work shows that such shocks can influence collaboration patterns (Agrawal and Goldfarb, 2008; Murray et al., 2016; Teodoridis, 2018), which in turn have important implications for publication outcomes. High-impact research is increasingly produced through multi-university collaborations, which tend to involve institutions of comparable prestige (Jones et al., 2008; Milojević, 2014; Wu et al., 2019; Wuchty et al., 2007). Overcoming this tendency toward institutional homophily can be particularly beneficial for researchers at lower-quartile universities, who often see improved publication outcomes when collaborating with peers from top-quartile institutions (Agrawal and Goldfarb, 2008; Khor and Yu, 2016; Kwiek, 2015; Li et al., 2019). Such collaborations offer access to advanced infrastructure, technical expertise, and greater visibility. Empirical evidence also indicates that the citation impact of multi-university teams tends to converge toward that of the more prestigious partner, effectively enhancing the scholarly reach of lower-quartile institutions (Jones et al., 2008).

Accordingly, we examine whether AlphaFold increases the likelihood that researchers from lower-quartile institutions collaborate with peers from top-quartile universities—a dynamic that could disproportionately benefit the former and help account for their observed gains in top-journal market share. We estimate equation 4, using as the outcome

variable the fraction of a university's top journal publications in protein science that include at least one co-author from a top-quartile institution in a given year. In addition to the standard two-way interaction term $Quartile_{q,u} \times PostAF1_t$, we include a triple interaction with the share of structural biologists in the university's biology faculty for that year to distinguish changes in collaboration patterns among structural biologists and non-structural biologists.

Results are shown in Table 5. In the baseline specifications (models 1, 3, and 5), collaborations with top-quartile authors generally decline among lower-quartile universities following AlphaFold. The decline is most pronounced for fourth-quartile institutions and is also evident for second-quartile universities in all three samples, while the third-quartile estimate in the full protein sample is close to zero. Taken together, these estimates suggest a broad move toward greater self-reliance among lower-quartile universities, especially those with larger in-house structural biology capacity.

Models 2, 4, and 6 include the triple interaction with the share of structural biologists. In these specifications, the coefficient on $Quartile_{q,u} \times PostAF1_t$ is positive for the second, third, and fourth quartiles. Because the model includes the triple interaction, this coefficient measures the post-AlphaFold change for a university with no structural biologists. The coefficients on $Quartile_{q,u} \times PostAF1_t \times Structural\ Biologists(\%)$ are negative and statistically significant. This pattern points to substantial heterogeneity in collaboration responses. Collaborations with top-quartile institutions rise primarily among lower-quartile universities with limited in-house structural biology expertise, and this increase weakens as structural biology capacity rises. One possible interpretation is that AlphaFold improves the productivity of structural biologists, who are disproportionately concentrated at top-quartile institutions, allowing them to participate in a larger number of projects.[17] Lower-quartile researchers with fewer in-house structural biologists may therefore become more likely to collaborate with top-quartile structural biologists in the post-AlphaFold period.

6 Conclusions

The persistent concentration of scientific output within elite institutions has long been attributed to unequal access to resources, infrastructure, skill, and reputational capital. Our findings provide empirical support for the idea that AI can disrupt these structural

[17] Improvements in resolving the 3D structrure of the human nuclear pore complex, one of the largest protein assemblies in human cells, demonstrates how AlphaFold predictions accelerate model building and experimental structure solution (see Figure D2, Panel (B) in Appendix D.5).

advantages. The introduction of AlphaFold represents an exogenous technological shock that reduced the skill and resource intensity historically required for cutting-edge protein research.

The introduction of AlphaFold represents an exogenous technological shock that reduced the skill and resource intensity historically required for cutting-edge protein research, particularly in structural biology. This shock enabled researchers at lower-ranked institutions to increase their presence in top journals, suggesting that AI can function as a scientific equalizer. Notably, the effects are both quantitative and qualitative: lower-ranked universities not only increased their publication volumes and market shares in top journals but also exhibited significant improvements in research quality relative to elite institutions, thereby narrowing pre-existing institutional gaps in innovation and impact within protein science.

These results align with prior work on the redistributive effects of technological shocks in science. Prior studies have shown that when experimental costs fall or computational tools become more accessible, the institutional distribution of research can shift (Furman and Stern, 2011; Furman and Teodoridis, 2020; Murray et al., 2016). Whereas such shocks manifest gradually and are limited in scope, AlphaFold's effects have been rapid and large, possibly pointing to a different mechanism: one linked to the general-purpose and open-access nature of modern AI tools. Beyond AlphaFold , Hao et al. (2026) document how quickly and broadly other AI tools -from simple random forest to large language models- have been adopted across the natural sciences. They find that researchers using AI increased their publication output by a factor of three. We complement their study by showing that this increase in productivity benefited lower-ranked scientists.

Our findings also contribute to long-standing debates about the *Matthew* effect in science. Elite universities have historically leveraged cumulative advantages in funding, talent, editorial control, and collaboration networks to reinforce their dominance (Azoulay et al., 2014; Merton, 1968). The changes we observe after AlphaFold suggest that such feedback loops can be interrupted when new tools lower the resources needed to produce top-quality research. For instance, post-AlphaFold , lower-ranked universities with stronger structural-biology capacity exhibited reduced reliance on collaborations with top-ranked partners to publish protein research in top journals, as evidenced by an overall decrease in their propensity for such partnerships. This reduced dependence is consistent with AlphaFold broadening access to structural insights and allowing lower-ranked institutions to pursue projects more autonomously. Importantly, AlphaFold's open accessibility played a central role: unlike many scientific breakthroughs constrained by proprietary datasets or infrastructure, AlphaFold 2 provided a freely available computational model and a

comprehensive, open-access protein structure database, effectively diluting traditional barriers to entry in structural biology research.

Whether future AI innovations will follow AlphaFold's trajectory remains to be seen. As AI tools become more complex and expensive to train, the risk increases that control over their development and deployment will be centralized within a small number of firms. AlphaFold itself was developed by Google DeepMind and made freely available, but future iterations of protein prediction models may be gated by proprietary data, computational scale, and commercial licensing. This raises the concern that AlphaFold's democratizing effects may be temporary, potentially giving way to new hierarchies rooted in unequal access to advanced AI infrastructure and expertise. Furthermore, while AlphaFold lowers the cost of protein structure determination, downstream activities such as experimental validation and drug discovery remain capital-intensive and skewed toward elite institutions, which disproportionately hold advanced facilities, proprietary datasets, and interdisciplinary expertise. As such, the increased publication volume and improved research quality we document for lower-ranked universities may not immediately lead to comparable advances in practical applications, such as faster development of new therapies or more equitable access to research funding and infrastructure.

Future research should examine whether the rebalancing of top publication shares observed post-AlphaFold represents a lasting shift, translating into more durable changes in scientific influence, collaboration networks, and funding allocation. It is also important to assess whether these democratizing effects extend beyond protein science to other domains such as genomics, materials science, and climate modelling, where AI tools may similarly empower under-represented institutions or, conversely, introduce new forms of exclusion. Beyond top-journal publications, other aspects of inequality should be studied. For example, Hao et al. (2026) finds that the adoption of AI leads to greater concentration of citations among the most influential papers.

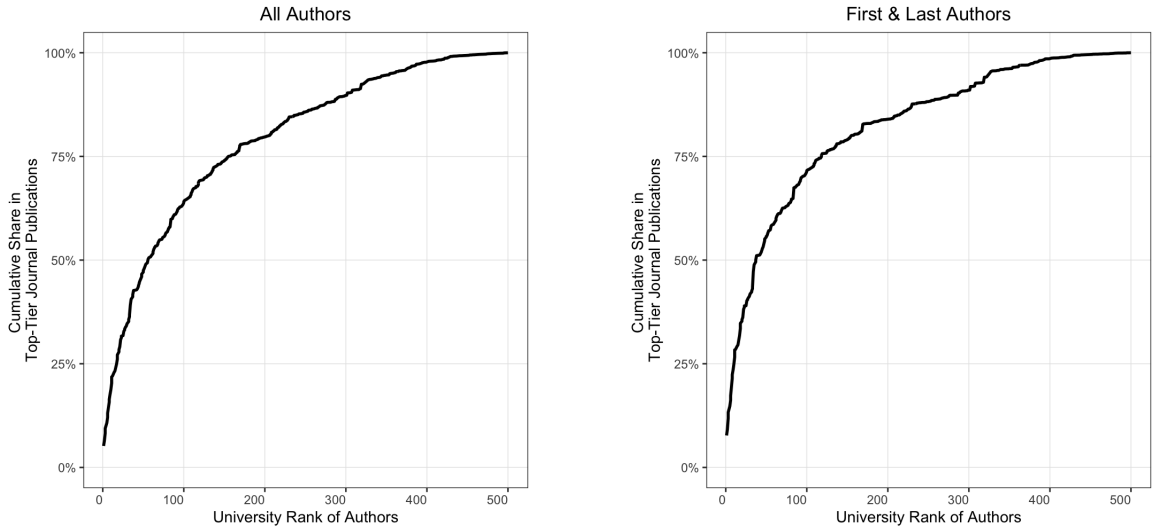
Recent developments surrounding AlphaFold 3 highlight the fragility of such gains. It was initially released with restricted access in May 2024, which prompted widespread calls from the scientific community for greater access. It was ultimately made open access in November 2024. This episode underscores the importance of longitudinal studies on the evolving role of AI in research ecosystems, particularly regarding how access and governance decisions shape equitable innovation.

As AI continues to transform scientific discovery, global policy responses are accelerating, and many initiatives, such as the U.S. AI Action Plan, have emphasized equitable access to AI technologies. In this context, a deeper understanding of AI's potential to either reinforce or reduce institutional hierarchies will be essential for designing inclusive science and

innovation policies that broaden participation and enhance societal benefit.

Figure 1: **Cumulative market share of top 500 universities before release of AlphaFold**

Figure illustrates the cumulative distribution of top-journal publication for the top 500 universities. Panel (A) shows the distribution when publications are allocated based on the affiliation of any of the paper authors; Panel (B) shows the distribution when publications are allocated based on the affiliation of the first or last author. The university rankings are determined by the average *total normalized citation score* (TNCS) of each university in the "biomedical and health sciences" category between 2013 and 2018.



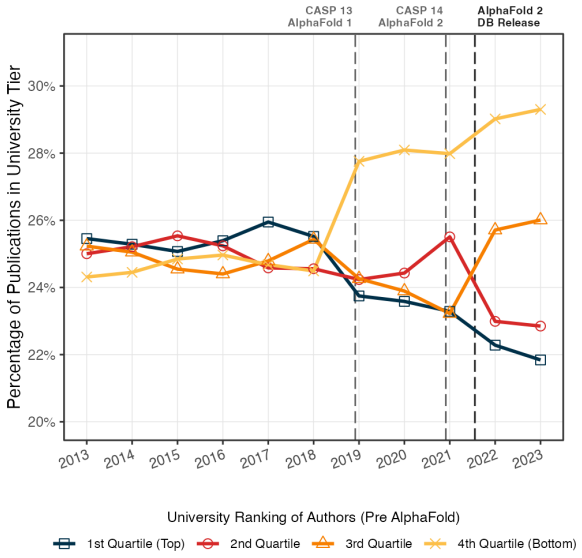
(A) Counting all authors

(B) Counting only first and last author

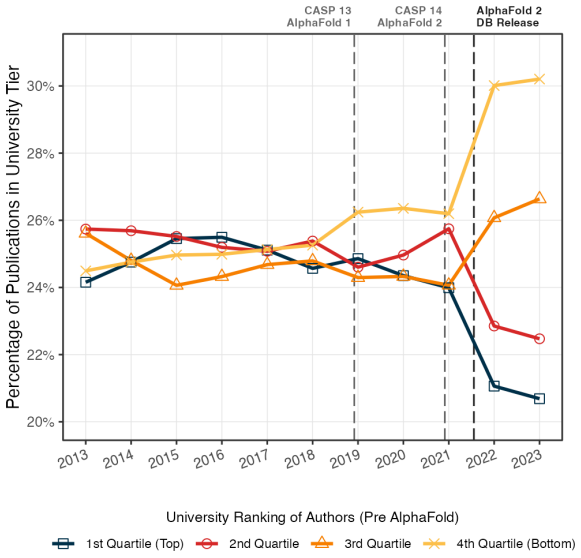
Figure 2: **Protein publications in top journals by university quartiles**

The figure reports, for each year, the share of top-journal protein publications attributable to each university quartile. Top journals are those with an average SJR of at least 10 during 2013–2018. The sample is limited to authors affiliated with the top 500 universities. Universities are ordered by pre-AlphaFold TNCS and grouped into consecutive quartiles with approximately equal pre-AlphaFold top-journal publication shares. Each author is assigned to the highest-ranked quartile among the universities listed on their top-journal papers before AlphaFold. Authors with no top-journal paper before AlphaFold are assigned to the quartile of the university they list on the paper. A paper counts once for each quartile represented among its authors. A quartile's share is its paper count divided by the sum of the four counts. Panels A and B count all authors. Panels C and D count only main (first and last) authors. Panels A and C cover structural biology. Panels B and D cover non-structural protein research. Dashed lines mark CASP13 (December 2018), CASP14 (November 2020) and the release of the AlphaFold 2 code and database (July 2021).

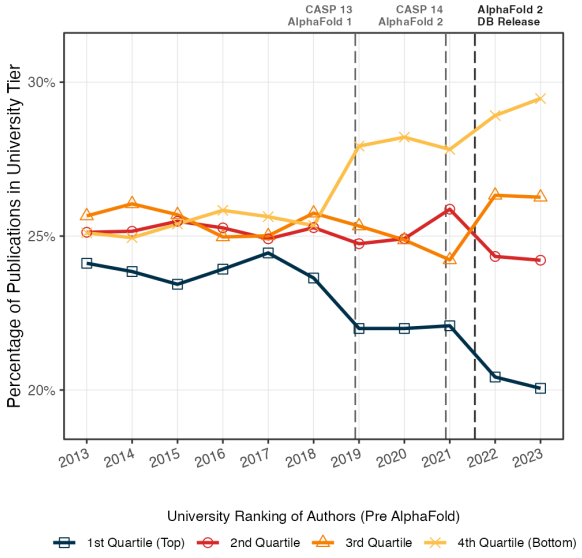
(A) Structural biology, all authors



(B) Non-structural protein, all authors



(C) Structural biology, main authors



(D) Non-structural protein, main authors

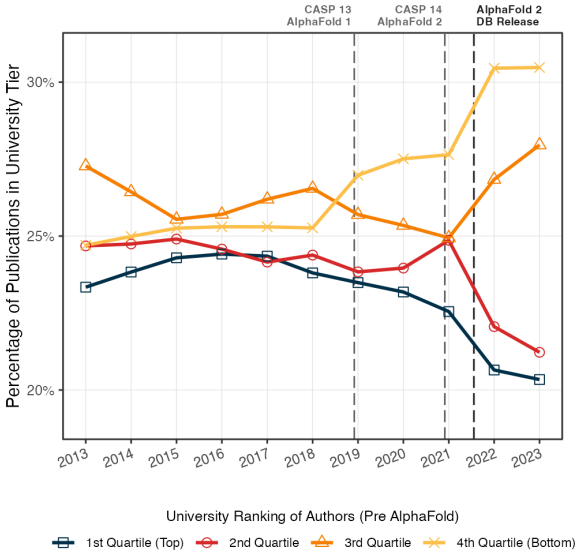
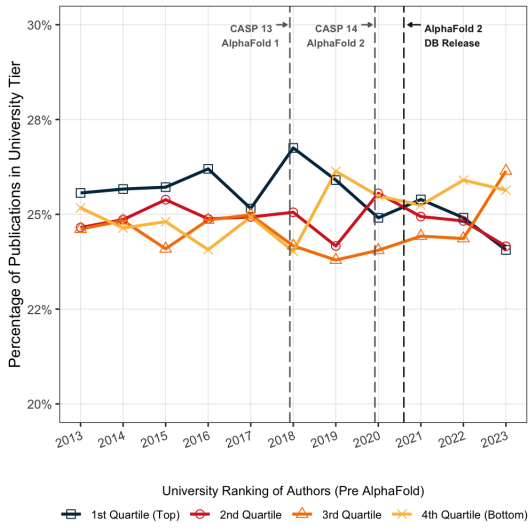


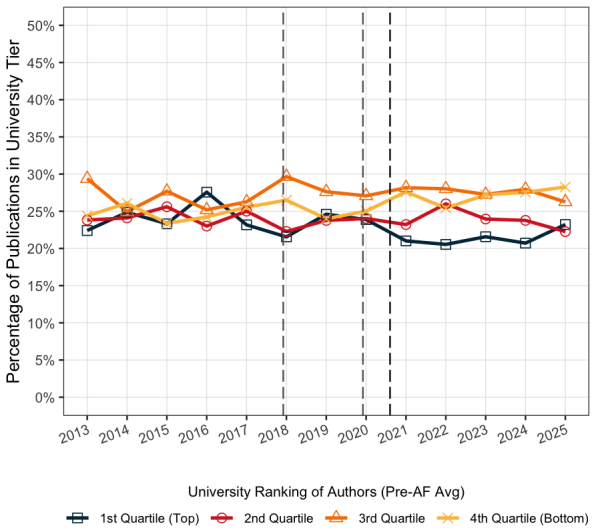
Figure 3: **Publications in other fields in top journals by university quartiles**

Figure illustrates the percentage of publications in top journals (SJR ≥ 10) held by each university quartile over time. Universities are ranked based on their field-specific TNCS. As in protein research, quartiles are fixed based on each university's average rank in the pre-AlphaFold period (before 2019) rather than being redefined annually. Panel (A) shows trends for Microbiology (excluding protein research) publications. Panel (B) shows trends for Economics journals. Panel (C) for Materials Science journals and Panel (D) for the Social Sciences. University quartile composition is fixed within each panel but defined separately across panels. A top quartile university for Microbiology might not be in the top quartile for Materials Science. Unlike in protein research, there are no visible changes in top journal share after 2018.

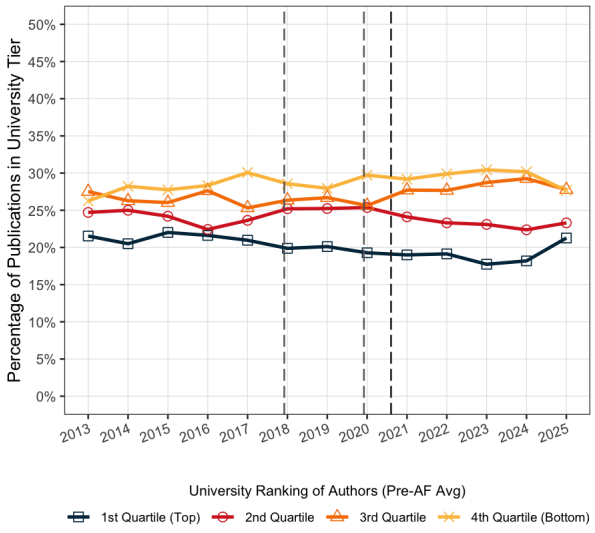
(A) Microbiology (no protein)



(B) Economics & Finance



(C) Materials Science



(D) Social Sciences

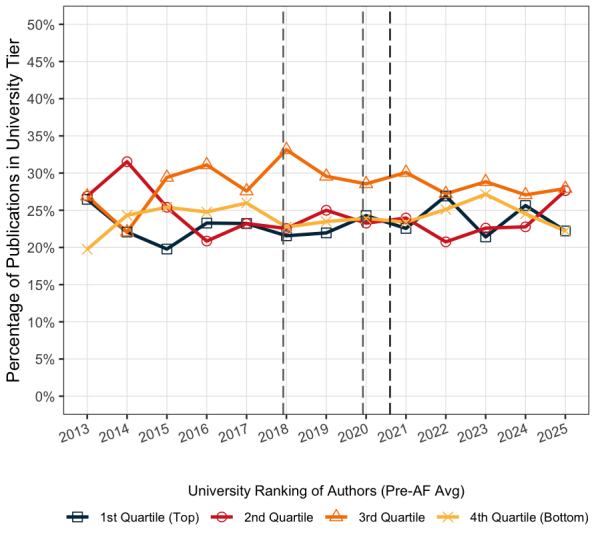


Figure 4: **Bottom-quartile coefficients by prior protein information and period**
Each point compares the bottom quartile with the top-quartile reference group. Bars are 95% confidence intervals.

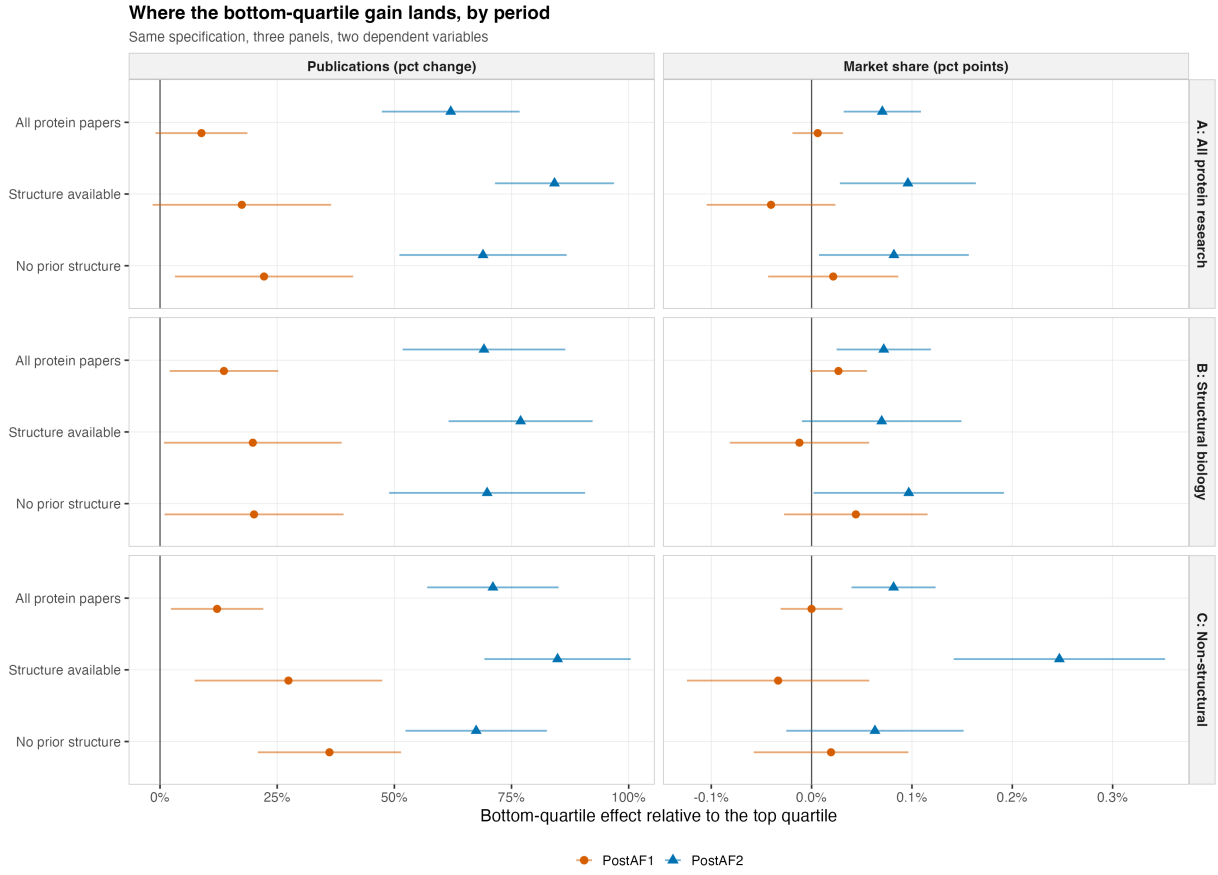
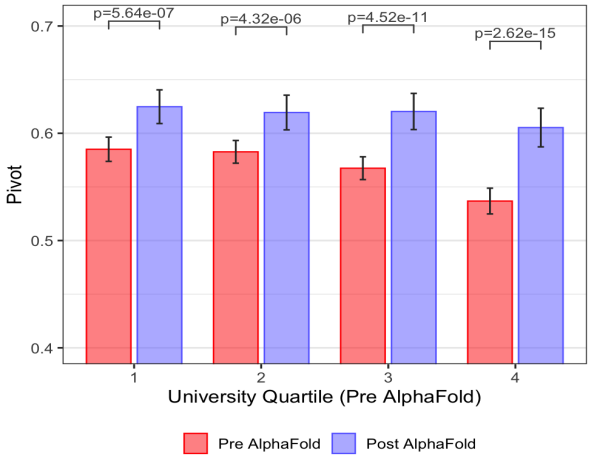
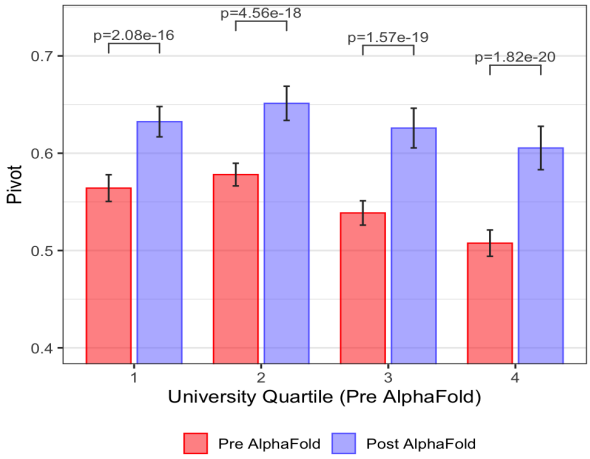


Figure 5: **Research direction measures in top journal publications by university quartile**
Figure shows pivot scores, novelty, and disruptiveness measures based on three-year publication histories of protein researchers that published in top journals (SJR ≥ 10). *p*-values shown indicate the statistical significance of pre- vs. post-AlphaFold differences within each quartile. Panels (A) and (B) show pivot scores for structural biology and non-structural protein publications. Panel (C) and (D) show novelty measures. Panel (E) and (F) show disruptiveness measures. Universities are divided into four quartiles based on their aggregate share of publications in top journals during 2013-18.

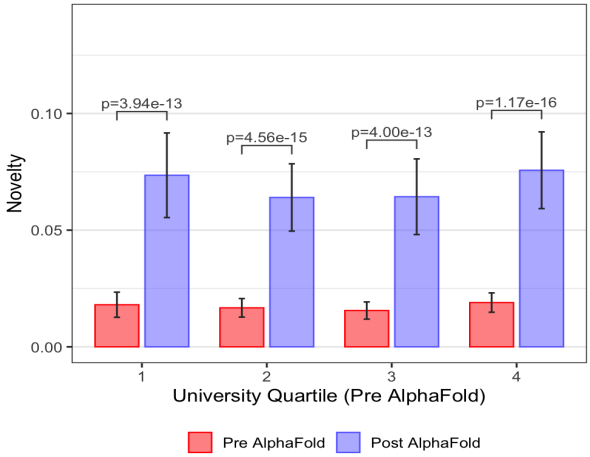
(A) Pivot: Structural biology pubs.



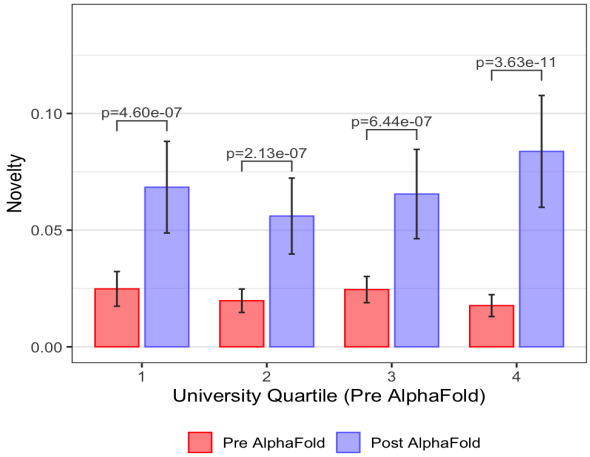
(B) Pivot: Non-structural protein pubs.



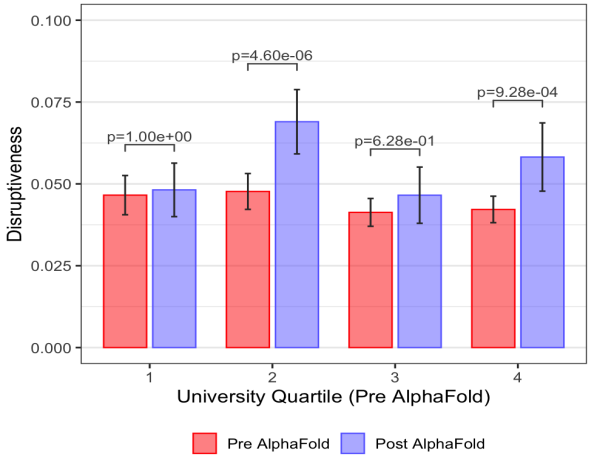
(C) Novelty: Structural biology pubs.



(D) Novelty: Non-structural protein pubs.



(E) Disruptiveness: Structural biology pubs.



(F) Disruptiveness: Non-structural protein pubs.

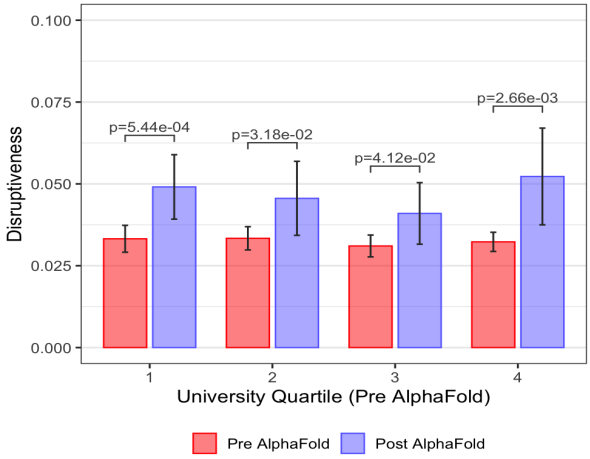


Table 1: **Protein publications in top journals around AlphaFold**

PostAF1 is a binary variable set to one for papers published after the December 2018 CASP13 result. The sample is limited to all protein research papers (both structural biology and non-structural protein research) published in top journals with an SJR score of 10 or higher and to authors affiliated with the top 500 universities prior to the CASP13 results. Universities are ranked based on their average TNCS in the period preceding the CASP13. The highest-ranked universities placed in the top quartile (*University Tier = 1st Quartile*) are used as the reference group. Standard errors are clustered by university and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

(A) All protein publications

Dependent Variables:	Log(1 + No. of Publications)	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)
Model:	(1)	(2)	(3)
University Tier = 2nd Quartile × PostAF1	0.12*	0.06	0.09
	(0.07)	(0.07)	(0.07)
University Tier = 3rd Quartile × PostAF1	0.23***	0.13*	0.13*
	(0.06)	(0.07)	(0.07)
University Tier = 4th Quartile × PostAF1	0.38***	0.20***	0.22***
	(0.06)	(0.07)	(0.07)
Controls	✓	✓	✓
Year FE	✓	✓	✓
University FE	✓	✓	✓
Observations	5,500	5,500	5,500
R^2/Pseudo R^2	0.83	0.93	0.93

(B) Structural biology publications

Dependent Variables:	Log(1 + No. of Publications)	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)
Model:	(1)	(2)	(3)
University Tier = 2nd Quartile × PostAF1	0.19**	0.13*	0.14*
	(0.09)	(0.07)	(0.08)
University Tier = 3rd Quartile × PostAF1	0.29***	0.16**	0.15**
	(0.08)	(0.07)	(0.08)
University Tier = 4th Quartile × PostAF1	0.46***	0.24***	0.23***
	(0.07)	(0.07)	(0.08)
Controls	✓	✓	✓
Year FE	✓	✓	✓
University FE	✓	✓	✓
Observations	5,500	5,500	5,500
R^2/Pseudo R^2	0.82	0.92	0.92

(C) Non-structural protein publications

Dependent Variables:	Log(1 + No. of Publications)	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)
Model:	(1)	(2)	(3)
University Tier = 2nd Quartile × PostAF1	0.19**	0.08	0.11
	(0.08)	(0.08)	(0.08)
University Tier = 3rd Quartile × PostAF1	0.30***	0.16**	0.16**
	(0.07)	(0.08)	(0.07)
University Tier = 4th Quartile × PostAF1	0.44***	0.24***	0.29***
	(0.06)	(0.07)	(0.07)
Controls	✓	✓	✓
Year FE	✓	✓	✓
University FE	✓	✓	✓
Observations	5,500	5,500	5,500
R^2/Pseudo R^2	0.79	0.90	0.91

Table 2: **Protein publications in top journals around AlphaFold: AF1 versus AF2**

PostAF1/AF2 is a categorical variable with the pre-period as the reference group, where *PostAF1* is set to one for papers published after the December 2018 CASP13 result but before the July 2021 open release of AlphaFold 2's code and database, and *PostAF2* is set to one for papers published after that open release. The sample includes all protein research papers (both structural and non-structural biology) published in top journals with an SJR score of 10 or higher and to authors affiliated with the top 500 universities prior to the CASP13 result. Universities are ranked based on their average TNCS in the period preceding CASP13. The highest ranked universities placed in the top quartile (*University Tier = 1st Quartile*) are used as the reference group. Standard errors are clustered by university, and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

(A) All protein publications

Dependent Variables:	Log(1 + No. of Publications)	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)
Model:	(1)	(2)	(3)
University Tier = 2nd Quartile × PostAF1	0.08	0.05	0.07
	(0.07)	(0.07)	(0.07)
University Tier = 2nd Quartile × PostAF2	0.19**	0.09	0.13
	(0.09)	(0.09)	(0.09)
University Tier = 3rd Quartile × PostAF1	0.11*	0.04	0.04
	(0.06)	(0.06)	(0.06)
University Tier = 3rd Quartile × PostAF2	0.40***	0.28***	0.27***
	(0.09)	(0.09)	(0.09)
University Tier = 4th Quartile × PostAF1	0.16***	0.07	0.10*
	(0.05)	(0.06)	(0.06)
University Tier = 4th Quartile × PostAF2	0.70***	0.41***	0.40***
	(0.08)	(0.09)	(0.09)
Controls	✓	✓	✓
Year FE	✓	✓	✓
University FE	✓	✓	✓
Observations	5,500	5,500	5,500
R^2/Pseudo R^2	0.83	0.93	0.93

(B) Structural biology publications

Dependent Variables:	Log(1 + No. of Publications)	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)
Model:	(1)	(2)	(3)
University Tier = 2nd Quartile × PostAF1	0.16**	0.13*	0.10
	(0.08)	(0.07)	(0.08)
University Tier = 2nd Quartile × PostAF2	0.22*	0.13	0.20*
	(0.12)	(0.11)	(0.11)
University Tier = 3rd Quartile × PostAF1	0.19***	0.11*	0.11
	(0.07)	(0.07)	(0.07)
University Tier = 3rd Quartile × PostAF2	0.45***	0.24**	0.23**
	(0.12)	(0.11)	(0.11)
University Tier = 4th Quartile × PostAF1	0.24***	0.18***	0.15**
	(0.06)	(0.06)	(0.07)
University Tier = 4th Quartile × PostAF2	0.79***	0.34***	0.36***
	(0.11)	(0.11)	(0.11)
Controls	✓	✓	✓
Year FE	✓	✓	✓
University FE	✓	✓	✓
Observations	5,500	5,500	5,500
R^2/Pseudo R^2	0.82	0.92	0.92

(C) Non-structural protein publications

Dependent Variables:	Log(1 + No. of Publications)	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)
Model:	(1)	(2)	(3)
University Tier = 2nd Quartile × PostAF1	0.16*	0.10	0.13
	(0.08)	(0.08)	(0.08)
University Tier = 2nd Quartile × PostAF2	0.24**	0.07	0.09
	(0.10)	(0.10)	(0.10)
University Tier = 3rd Quartile × PostAF1	0.19***	0.09	0.09
	(0.07)	(0.08)	(0.07)
University Tier = 3rd Quartile × PostAF2	0.46***	0.29***	0.29***
	(0.09)	(0.10)	(0.10)
University Tier = 4th Quartile × PostAF1	0.24***	0.10	0.17**
	(0.06)	(0.07)	(0.07)
University Tier = 4th Quartile × PostAF2	0.74***	0.46***	0.48***
	(0.08)	(0.10)	(0.10)
Controls	✓	✓	✓
Year FE	✓	✓	✓
University FE	✓	✓	✓
Observations	5,500	5,500	5,500
R^2/Pseudo R^2	0.80	0.91	0.90

Table 3: **Protein publications in top journals around AlphaFold: differences by institutional access to high-performance computing**

The table estimates the specification of Table 2 separately for universities with and without access to high-performance computing (HPC). A university has HPC access if it hosted at least one system on the Top500 list in any year between 2013 and 2018. The status is fixed for the whole sample period. Each column compares quartiles within the same HPC group: the reference group is top-quartile universities with HPC in columns (1), (3) and (5), and without HPC in columns (2), (4) and (6). *PostAF1/AF2* is a categorical variable with the pre-period as the reference group, where *PostAF1* is set to one for papers published after the December 2018 CASP13 result but before the July 2021 open release of AlphaFold 2's code and database, and *PostAF2* is set to one for papers published after that open release. Panel A covers all protein research papers, Panel B structural biology papers and Panel C non-structural protein papers published in top journals with an average SJR of at least 10 during 2013–2018. The sample is limited to authors affiliated with the top 500 universities. Universities are ranked by their average TNCS in 2013–2018. Columns (1)–(2) are estimated by OLS and columns (3)–(6) by fractional logit. Standard errors are clustered by university and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

(A) All protein publications

Dependent Variables:	Log(1 + No. of Publications)		Market Share (% Publications, All Authors)		Market Share (% Publications, Main Authors)	
HPC Access at Institution	Yes	No	Yes	No	Yes	No
Model:	(1)	(2)	(3)	(4)	(5)	(6)
University Tier = 2nd Quartile × PostAF1	0.05	0.09	0.01	0.14*	0.02	0.15**
	(0.09)	(0.07)	(0.10)	(0.07)	(0.11)	(0.07)
University Tier = 2nd Quartile × PostAF2	0.13	0.24**	0.06	0.16	0.12	0.18
	(0.12)	(0.12)	(0.13)	(0.14)	(0.13)	(0.17)
University Tier = 3rd Quartile × PostAF1	0.08	0.09	0.05	0.02	0.06	0.03
	(0.08)	(0.06)	(0.09)	(0.06)	(0.09)	(0.06)
University Tier = 3rd Quartile × PostAF2	0.33***	0.42***	0.28**	0.29*	0.30**	0.26
	(0.11)	(0.12)	(0.12)	(0.15)	(0.12)	(0.18)
University Tier = 4th Quartile × PostAF1	0.06	0.25***	0.08	0.13**	0.11	0.14**
	(0.08)	(0.04)	(0.10)	(0.05)	(0.10)	(0.05)
University Tier = 4th Quartile × PostAF2	0.54***	0.83***	0.39***	0.48***	0.38***	0.46***
	(0.11)	(0.10)	(0.13)	(0.15)	(0.13)	(0.17)
Controls	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓	✓
Observations	1,191	4,309	1,191	4,309	1,191	4,309
R^2/Pseudo R^2	0.85	0.81	0.95	0.91	0.95	0.90

(B) Structural biology publications

Dependent Variables:	Log(1 + No. of Publications)		Market Share (% Publications, All Authors)		Market Share (% Publications, Main Authors)	
HPC Access at Institution	Yes	No	Yes	No	Yes	No
Model:	(1)	(2)	(3)	(4)	(5)	(6)
University Tier = 2nd Quartile × PostAF1	0.15	0.17**	0.09	0.20***	0.08	0.17*
	(0.12)	(0.08)	(0.12)	(0.07)	(0.12)	(0.09)
University Tier = 2nd Quartile × PostAF2	0.17	0.28*	0.13	0.18	0.15	0.39**
	(0.20)	(0.15)	(0.15)	(0.15)	(0.14)	(0.16)
University Tier = 3rd Quartile × PostAF1	0.19*	0.12	0.14	0.08	0.15	0.10
	(0.10)	(0.07)	(0.09)	(0.07)	(0.10)	(0.08)
University Tier = 3rd Quartile × PostAF2	0.38**	0.46***	0.25	0.25	0.22	0.38**
	(0.19)	(0.14)	(0.16)	(0.16)	(0.15)	(0.16)
University Tier = 4th Quartile × PostAF1	0.16*	0.29***	0.20**	0.19***	0.18*	0.15*
	(0.09)	(0.06)	(0.10)	(0.06)	(0.10)	(0.08)
University Tier = 4th Quartile × PostAF2	0.67***	0.89***	0.42***	0.33**	0.40***	0.47***
	(0.18)	(0.12)	(0.15)	(0.15)	(0.15)	(0.16)
Controls	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓	✓
Observations	1,844	3,656	1,844	3,656	1,844	3,656
R^2/Pseudo R^2	0.84	0.79	0.94	0.90	0.93	0.89

(C) Non-structural protein publications

Dependent Variables:	Log(1 + No. of Publications)		Market Share (% Publications, All Authors)		Market Share (% Publications, Main Authors)	
HPC Access at Institution	Yes	No	Yes	No	Yes	No
Model:	(1)	(2)	(3)	(4)	(5)	(6)
University Tier = 2nd Quartile × PostAF1	0.15	0.16*	0.09	0.20**	0.12	0.20**
	(0.11)	(0.08)	(0.11)	(0.08)	(0.12)	(0.09)
University Tier = 2nd Quartile × PostAF2	0.14	0.31**	0.01	0.13	0.10	0.10
	(0.13)	(0.15)	(0.14)	(0.17)	(0.14)	(0.17)
University Tier = 3rd Quartile × PostAF1	0.14	0.19***	0.10	0.09	0.10	0.06
	(0.09)	(0.07)	(0.11)	(0.08)	(0.11)	(0.07)
University Tier = 3rd Quartile × PostAF2	0.40***	0.47***	0.30**	0.25	0.29**	0.24
	(0.10)	(0.13)	(0.12)	(0.18)	(0.12)	(0.18)
University Tier = 4th Quartile × PostAF1	0.12	0.35***	0.10	0.20***	0.17	0.22***
	(0.08)	(0.05)	(0.11)	(0.06)	(0.11)	(0.06)
University Tier = 4th Quartile × PostAF2	0.60***	0.86***	0.45***	0.49***	0.50***	0.47***
	(0.10)	(0.12)	(0.14)	(0.17)	(0.13)	(0.17)
Controls	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓	✓
Observations	1,844	3,656	1,844	3,656	1,844	3,656
R^2/Pseudo R^2	0.82	0.77	0.93	0.86	0.93	0.85

Table 4: **Research patterns in top-journal publications around AlphaFold**

Table presents regression estimates examining qualitative changes in the research produced following the December 2018 CASP13 result. Dependent variables are pivot scores, novelty measures, and disruptiveness scores for publications in top journals (SJR ≥ 10), averaged at the university year level. *PostAF1* is a binary indicator for papers published after the CASP13 result in December 2018. Sample includes structural biology publications (columns 1-3) and non-structural protein publications (columns 4-6) by authors affiliated with top 500 universities. University quartiles based on pre-AlphaFold publication shares in top journals (2013-2018), with top quartile as reference group. Novelty and disruptiveness are min–max normalized within each calendar quarter across all universities to the [0,1] range; the pivot score is naturally bounded on the same range and is not additionally scaled. Coefficients are therefore differences in scale units, not percentage changes in raw measures. Standard errors clustered by university in parentheses. ***, **, * denote significance at 1%, 5%, 10% levels.

	Structural Biology Pubs.			Non-Structural Protein Pubs.		
Dependent Variables:	Pivot Score	Novelty	Disruptiveness	Pivot Score	Novelty	Disruptiveness
Model:	(1)	(2)	(3)	(4)	(5)	(6)
University Tier = 2nd Quartile × PostAF1	0.0104**	0.0196***	0.0117***	0.0345***	0.0026	-0.0010
	(0.0025)	(0.0020)	(0.0003)	(0.0013)	(0.0015)	(0.0012)
University Tier = 3rd Quartile × PostAF1	0.0166***	0.0152***	0.0046**	0.0028	0.0112***	-0.0056*
	(0.0026)	(0.0020)	(0.0010)	(0.0025)	(0.0011)	(0.0022)
University Tier = 4th Quartile × PostAF1	0.0263***	0.0154**	0.0126***	0.0326***	0.0268***	0.0065*
	(0.0021)	(0.0032)	(0.0015)	(0.0034)	(0.0018)	(0.0026)
Year FE	✓	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓	✓
Observations	5,500	5,500	5,500	5,500	5,500	5,500
R^2	0.03	0.18	0.07	0.02	0.16	0.06

Table 5: **Collaborations with top-quartile universities in top journal publications around AlphaFold**

Table presents regression estimates examining changes in collaboration patterns following the December 2018 CASP13 result. The dependent variable is the share of top-journal publications produced by university u in year t that were co-authored with at least one author from a top-quartile university. For each sample, we run two specifications: One where we only include the usual double interaction term between *PostAF1* and *University Tier* (column 1,3,5) and one where we include the triple interaction term between *PostAF1*, *University Tier*, and the share of structural biologists at the university (column 2,4,6) in year t. University quartiles based on pre-AlphaFold publication shares in high-impact journals (2013-2018), with 1st quartile as reference group. *PostAF1* is a binary indicator for papers published after the CASP13 result in December 2018. Quartile 1 is the reference group and the controls are the covariates shown in Table A3. Standard errors clustered by university in parentheses. ***, **, * denote significance at 1%, 5%, 10% levels.

Dependent Variable:	Collaboration 1st Quartile (%)					
	All Protein Pubs.		Structural Biology Pubs.		Non-Structural Protein Pubs.	
Model:	(1)	(2)	(3)	(4)	(5)	(6)
University Tier = 2nd Quartile × PostAF1	-0.0124***	0.0738**	-0.0491***	0.1535***	-0.0337**	0.1704***
	(0.0004)	(0.0199)	(0.0021)	(0.0210)	(0.0081)	(0.0163)
University Tier = 3rd Quartile × PostAF1	0.0006	0.1093***	-0.0657**	0.1678***	-0.0506**	0.1544***
	(0.0091)	(0.0175)	(0.0118)	(0.0146)	(0.0151)	(0.0166)
University Tier = 4th Quartile × PostAF1	-0.0566**	0.0377*	-0.1295***	0.0937***	-0.1162***	0.0665**
	(0.0124)	(0.0131)	(0.0146)	(0.0102)	(0.0174)	(0.0158)
University Tier = 2nd Quartile × PostAF1 × Structural Biologists (%)		-1.240**		-2.750***		-2.788***
		(0.2747)		(0.3011)		(0.2796)
University Tier = 3rd Quartile × PostAF1 × Structural Biologists (%)		-1.432**		-3.030***		-2.815***
		(0.3628)		(0.3633)		(0.3428)
University Tier = 4th Quartile × PostAF1 × Structural Biologists (%)		-1.079**		-2.745***		-2.377***
		(0.3363)		(0.3140)		(0.3096)
Controls	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓	✓
Observations	5,500	5,500	5,500	5,500	5,500	5,500
R^2	0.19	0.20	0.29	0.29	0.30	0.30

Table 6: **The gap between the two protein strata across periods**

A protein counts as having prior known information if an experimental structure of its own, or a homology template of at least 30 pct sequence identity, is dated before the AlphaFold Protein Structure Database release on 22 July 2021. Papers are assigned to mutually exclusive strata at a 50 pct threshold: a paper is Structure available when at least half of its linked proteins meet that test and No prior structure otherwise. The stratum enters every interaction as an indicator for structure-available papers, so a positive coefficient means the lower-quartile differential is larger for papers with a pre-existing structure available. Publication-count coefficients are log-point differences for Log(1 + publications); market-share coefficients are fractional-logit index differences, not percentage-point effects.

Dependent Variables:	Log(1 + No. of publications)			Market share (% of publications)		
	All protein research	Structural biology	Non-structural	All protein research	Structural biology	Non-structural
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
2nd Quartile × PostAF1	0.04	0.03	0.11	-0.04	-0.04	-0.01
	(0.12)	(0.13)	(0.11)	(0.15)	(0.17)	(0.16)
3rd Quartile × PostAF1	0.19*	0.17*	0.29***	0.15	0.16	0.14
	(0.10)	(0.10)	(0.08)	(0.12)	(0.13)	(0.13)
4th Quartile × PostAF1	0.22**	0.20**	0.36***	0.09	0.18	0.07
	(0.01)	(0.01)	(0.08)	(0.14)	(0.15)	(0.15)
2nd Quartile × PostAF2	0.11	0.19	0.24**	-0.21	-0.03	-0.21
	(0.11)	(0.13)	(0.1)	(0.19)	(0.21)	(0.2)
3rd Quartile × PostAF2	0.44***	0.44***	0.5***	0.09	0.03	0.04
	(0.1)	(0.11)	(0.08)	(0.17)	(0.21)	(0.19)
4th Quartile × PostAF2	0.69***	0.7***	0.67***	0.35**	0.39**	0.24
	(0.09)	(0.11)	(0.08)	(0.16)	(0.2)	(0.17)
2nd Quartile × PostAF1 × Structure available	0.0948	0.14	0.00	0.00	0.09	-0.17
	(0.13)	(0.14)	(0.13)	(0.18)	(0.2)	(0.19)
3rd Quartile × PostAF1 × Structure available	-0.09	-0.03	-0.09	-0.24	-0.17	-0.18
	(0.12)	(0.12)	(0.13)	(0.18)	(0.2)	(0.22)
4th Quartile × PostAF1 × Structure available	-0.05	-0.00	-0.09	-0.26	-0.23	-0.2
	(0.11)	(0.11)	(0.12)	(0.19)	(0.19)	(0.21)
2nd Quartile × PostAF2 × Structure available	0.35**	0.16	0.23***	0.45**	0.15	0.78***
	(0.14)	(0.14)	(0.1)	(0.21)	(0.24)	(0.22)
3rd Quartile × PostAF2 × Structure available	0.13	0.06	0.16*	0.01	-0.07	0.35
	(0.11)	(0.11)	(0.08)	(0.20)	(0.24)	(0.24)
4th Quartile × PostAF2 × Structure available	0.15	0.07	0.17**	0.04	-0.12	0.65***
	(0.10)	(0.11)	(0.08)	(0.18)	(0.22)	(0.2)
Fixed-effects						
University	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	11,154	10,890	10,802	10,340	9,900	9,504
R^2/Pseudo R^2	0.58	0.58	0.5	0.65	0.61	0.53

References

Agrawal, Ajay and **Goldfarb**, Avi, "Restructuring Research: Communication Costs and the Democratization of University Innovation", *American Economic Review*, 2008, *98* (4), 1578–1590.

Akdel, Mehmet, **Pires**, Douglas, **Pardo**, Eduard, **Jänes**, Jürgen, **Zalevsky**, Arthur, **Mészáros**, Bálint, **Bryant**, Patrick, **Good**, Lydia, **Laskowski**, Roman, **Pozzati**, Gabriele, **Shenoy**, Aditi, **Zhu**, Wensi, **Kundrotas**, Petras, **Serra**, Victoria, **Rodrigues**, Carlos, **Dunham**, Alistair, **Burke**, David, **Borkakoti**, Neera, **Velankar**, Sameer, **Frost**, Adam, **Basquin**, Jérôme, **Lindorff-Larsen**, Kresten, **Bateman**, Alex, **Kajava**, Andrey, **Valencia**, Alfonso, **Ovchinnikov**, Sergey, **Durairaj**, Janani, **Ascher**, David, **Thornton**, Janet, **Davey**, Norman, **Stein**, Amelie, **Elofsson**, Arne, **Croll**, Tristan, and **Beltrao**, Pedro, "A structural biology community assessment of AlphaFold2 applications", *Nature Structural & Molecular Biology*, 2022, *29* (11), 1056–1067.

Azoulay, Pierre, **Fons-Rosen**, Christian, and **Graff Zivin**, Joshua, "Does Science Advance One Funeral at a Time?", *American Economic Review*, 2019, *109* (8), 2889–2920.

Azoulay, Pierre, **Graff Zivin**, Joshua, and **Manso**, Gustavo, "Incentives and creativity: evidence from the academic life sciences", *RAND Journal of Economics*, 2011, *42* (3), 527–554.

Azoulay, Pierre, **Stuart**, Toby, and **Wang**, Yanbo, "Matthew: Effect or Fable?", *Management Science*, 2014, *60* (1), 92–109.

Bloom, Nicholas, **Jones**, Charles, **Van Reenen**, John, and **Webb**, Michael, "Are Ideas Getting Harder to Find?", *American Economic Review*, 2020, *110* (4), 1104–1144.

Bol, Thijs, **Vaan**, Mathijs de, and **Rijt**, Arnout van de, "The Matthew effect in science funding", *Proceedings of the National Academy of Sciences*, 2018, *115* (19), 4887–4890.

Brogaard, Jonathan, **Engelberg**, Joseph E., **Eswar**, Sapnoti K., and **Van Wesep**, Edward D., "On the Causal Effect of Fame on Citations", *Management Science*, 2024, *70* (10), 7187–7214.

Brynjolfsson, Erik, **Li**, Danielle, and **Raymond**, Lindsey, "Generative AI at Work", *Quarterly Journal of Economics*, 2025, *140* (2), 889–942.

Caplin, Andrew, **Deming**, David, **Li**, Shangwen, **Martin**, Daniel, **Marx**, Philip, **Weidmann**, Ben, and **Ye**, Kadachi, "The ABC's of Who Benefits from Working with AI: Ability, Beliefs, and Calibration", *Management Science*, 2026, *72* (7), 5843–5852.

Cheng, Jun, **Novati**, Guido, **Pan**, Joshua, **Bycroft**, Clare, **Žemgulytė**, Akvilė, **Applebaum**, Taylor, **Pritzel**, Alexander, **Wong**, Lai, **Zielinski**, Michal, **Sargeant**, Tobias, et al., "Accurate proteome-

wide missense variant effect prediction with AlphaMissense", *Science*, 2023, *381* (6664), eadg7492.

Cheng, Mouyang, **Fu**, Chu-Liang, **Okabe**, Ryotaro, **Chotrattanapituk**, Abhijatmedhi, **Boonkird**, Artittaya, **Hung**, Nguyen, and **Li**, Mingda, "Artificial intelligence-driven approaches for materials design and discovery", *Nature Materials*, 2026, 1–17.

Cheng, Yifan, **Grigorieff**, Nikolaus, **Penczek**, Pawel, and **Walz**, Thomas, "A primer to single-particle cryo-electron microscopy", *Cell*, 2015, *161* (3), 438–449.

Choudhary, Kamal, **DeCost**, Brian, **Chen**, Chi, **Jain**, Anubhav, **Tavazza**, Francesca, **Cohn**, Ryan, **Park**, Cheol, **Choudhary**, Alok, **Agrawal**, Ankit, **Billinge**, Simon, **Holm**, Elizabeth, **Ong**, Shyue, and **Wolverton**, Chris, "Recent advances and applications of deep learning methods in materials science", *npj Computational Materials*, 2022, *8* (1), 59.

Cui, Zheyuan, **Demirer**, Mert, **Jaffe**, Sonia, **Musolff**, Leon, **Peng**, Sida, and **Salz**, Tobias, *The Effects of Generative AI on High Skilled Work: Evidence from Three Field Experiments with Software Developers*, SSRN Working Paper, 2024.

Díaz-Holguín, Alejandro, **Saarinen**, Marcus, **Vo**, Duc, **Sturchio**, Andrea, **Branzell**, Niclas, **Cabeza de Vaca**, Israel, **Hu**, Huabin, **Mitjavila-Domènech**, Núria, **Lindqvist**, Annika, **Baranczewski**, Pawel, et al., "AlphaFold accelerated discovery of psychotropic agonists targeting the trace amine–associated receptor 1", *Science Advances*, 2024, *10* (32), eadn1524.

Fügener, Andreas, **Walzner**, Dominik, and **Gupta**, Alok, "Roles of Artificial Intelligence in Collaboration with Humans: Automation, Augmentation, and the Future of Work", *Management Science*, 2026, *72* (1), 538–557.

Funk, Russell and **Owen-Smith**, Jason, "A Dynamic Network Measure of Technological Change", *Management Science*, 2017, *63* (3), 791–817.

Furman, Jeffrey and **Stern**, Scott, "Climbing atop the Shoulders of Giants: The Impact of Institutions on Cumulative Research", *American Economic Review*, 2011, *101* (5), 1933–1963.

Furman, Jeffrey and **Teodoridis**, Florenta, "Automation, Research Technology, and Researchers' Trajectories: Evidence from Computer Science and Electrical Engineering", *Organization Science*, 2020, *31* (2), 330–354.

Gans, Joshua and **Mohnen**, Myra, "How AI Impacts the Quest for Knowledge: Evidence from AlphaFold2", 2026.

Garman, Elspeth, "Developments in X-ray Crystallographic Structure Determination of Biological Macromolecules", *Science*, 2014, *343* (6175), 1102–1108.

Google DeepMind, *AlphaFold 2: Source Code and Documentation*, https://github.com/google-deepmind/alphafold, 2021.

Hao, Qianyue, **Xu**, Fengli, **Li**, Yong, and **Evans**, James, "Artificial Intelligence Tools Expand Scientists' Impact but Contract Science's Focus", *Nature*, 2026, *649* (8099), 1237–1243.

Hill, Ryan and **Stein**, Carolyn, "Race to the Bottom: Competition and Quality in Science", *Quarterly Journal of Economics*, 2025, *140* (2), 1111–1185.

— *How Artificial Intelligence Shapes Science: Evidence from AlphaFold*, Working Paper w35143, National Bureau of Economic Research, 2026.

Hill, Ryan, **Yin**, Yian, **Stein**, Carolyn, **Wang**, Xizhao, **Wang**, Dashun, and **Jones**, Benjamin, "The pivot penalty in research", *Nature*, 2025, *642* (8069), 999–1006.

Hosseini, SeyedMohsen, **Golding**, Brian, and **Ilie**, Lucian, "Seq-InSite: sequence supersedes structure for protein interaction site prediction", *Bioinformatics*, 2024, *40* (1), btad738.

Hu, Liya, **Salmen**, Wilhelm, **Sankaran**, Banumathi, **Lasanajak**, Yi, **Smith**, David, **Crawford**, Sue, **Estes**, Mary, and **Prasad**, Venkataram, "Novel fold of rotavirus glycan-binding domain predicted by AlphaFold2 and determined by X-ray crystallography", *Communications Biology*, 2022, *5* (1), 419.

Huang, Buwei, **Abedi**, Mohamad, **Ahn**, Green, **Coventry**, Brian, **Sappington**, Isaac, **Tang**, Cong, **Wang**, Rong, **Schlichthaerle**, Thomas, **Zhang**, Jason, **Wang**, Yujia, et al., "Designed endocytosis-inducing proteins degrade targets and amplify signals", *Nature*, 2025, *638* (8051), 796–804.

Jackson, Hayley, **Oler**, Eponine, **Torres-Calzada**, Claudia, **Kruger**, Ray, **Hira**, Amandeep, **López-Hernández**, Yamilé, **Pandit**, Devanshi, **Wang**, Jiaxuan, **Yang**, Kellie, **Fatokun**, Omolola, et al., "MarkerDB 2.0: a comprehensive molecular biomarker database for 2025", *Nucleic Acids Research*, 2025, *53* (D1), D1415–D1426.

Jin, Yibo, **Fyfe**, Paul, **Gardner**, Scott, **Wilmes**, Stephan, **Bubeck**, Doryen, and **Moraga**, Ignacio, "Structural insights into the assembly and activation of the IL-27 signaling complex", *EMBO Reports*, 2022, *23* (10), e55450.

Jones, Benjamin, **Wuchty**, Stefan, and **Uzzi**, Brian, "Multi-University Research Teams: Shifting Impact, Geography, and Stratification in Science", *Science*, 2008, *322* (5905), 1259–1262.

Jones, Charles I., "R & D-Based Models of Economic Growth", *Journal of Political Economy*, 1995, *103* (4), 759–784.

Jumper, John, **Evans**, Richard, **Pritzel**, Alexander, **Green**, Tim, **Figurnov**, Michael, **Ronneberger**, Olaf, **Tunyasuvunakool**, Kathryn, **Bates**, Russ, **Žídek**, Augustin, **Potapenko**, Anna, **Bridgland**, Alex, **Meyer**, Clemens, **Kohl**, Simon, **Ballard**, Andrew, **Cowie**, Andrew, **Romera-Paredes**, Bernardino, **Nikolov**, Stanislav, **Jain**, Rishub, **Adler**, Jonas, **Back**, Trevor, **Petersen**, Stig, **Reiman**, David, **Clancy**, Ellen, **Zielinski**, Michal, **Steinegger**, Martin, **Pacholska**, Michalina, **Berghammer**, Tamas, **Bodenstein**, Sebastian, **Silver**, David, **Vinyals**, Oriol, **Senior**, Andrew, **Kavukcuoglu**, Koray, **Kohli**, Pushmeet, and **Hassabis**, Demis, "Highly accurate protein structure prediction with AlphaFold", *Nature*, 2021, *596* (7873), 583–589.

Khor, A. and **Yu**, G., "Influence of international co-authorship on the research citation impact of young universities", *Scientometrics*, 2016, *107* (3), 1095–1110.

Krakowski, Sebastian, **Haftor**, Darek, **Luger**, Johannes, **Pashkevich**, Natallia, and **Raisch**, Sebastian, "Human-Centered Artificial Intelligence: A Field Experiment", *Management Science*, 2026, *72* (1), 57–72.

Kwiek, Marek, "The Internationalization of Research in Europe: A Quantitative Study of 11 National Systems From a Micro-Level Perspective", *Journal of Studies in International Education*, 2015, *19* (4), 341–359.

Lascaux, Pauline, **Hoslett**, Gwendoline, **Tribble**, Sara, **Trugenberger**, Camilla, **Antičević**, Ivan, **Otten**, Cecile, **Torrecilla**, Ignacio, **Koukouravas**, Stelios, **Zhao**, Yichen, **Yang**, Hongbin, et al., "TEX264 drives selective autophagy of DNA lesions to promote DNA repair and cell survival", *Cell*, 2024, *187* (20), 5698–5718.

Li, Weihua, **Aste**, Tomaso, **Caccioli**, Fabio, and **Livan**, Giacomo, "Early coauthorship with top scientists predicts success in academic careers", *Nature Communications*, 2019, *10* (1), 5170.

Liu, Meijun, **Bu**, Yi, **Chen**, Chongyan, **Xu**, Jian, **Li**, Daifeng, **Leng**, Yan, **Freeman**, Richard, **Meyer**, Eric, **Yoon**, Wonjin, **Sung**, Mujeen, **Jeong**, Minbyul, **Lee**, Jinhyuk, **Kang**, Jaewoo, **Min**, Chao, **Song**, Min, **Zhai**, Yujia, and **Ding**, Ying, "Pandemics are catalysts of scientific novelty: Evidence from COVID-19", *Journal of the Association for Information Science and Technology*, 2022, *73* (8), 1065–1078.

Luo, Zhuoran, **Lu**, Wei, **He**, Jiangen, and **Wang**, Yuqi, "Combination of research questions and methods: A new measurement of scientific novelty", *Journal of Informetrics*, 2022, *16* (2), 101282.

McKeon, John and **Boudou**, Justine, *Innovation Under Resource Constraints: Supercomputing in Scientific Research*, en, SSRN Scholarly Paper, Rochester, NY, 2024a.

— *Innovation Under Resource Constraints: Supercomputing in Scientific Research*, SSRN Working Paper, 2024b.

Merchant, Amil, **Batzner**, Simon, **Schoenholz**, Samuel, **Aykol**, Muratahan, **Cheon**, Gowoon, and **Cubuk**, Ekin, "Scaling deep learning for materials discovery", *Nature*, 2023, *624* (7990), 80–85.

Merton, Robert, "The Matthew Effect in Science", *Science*, 1968, *159* (3810), 56–63.

Milojević, Staša, "Principles of scientific research team formation and evolution", *Proceedings of the National Academy of Sciences*, 2014, *111* (11), 3984–3989.

Murray, Fiona, **Aghion**, Philippe, **Dewatripont**, Mathias, **Kolev**, Julian, and **Stern**, Scott, "Of Mice and Academics: Examining the Effect of Openness on Innovation", *American Economic Journal: Economic Policy*, 2016, *8* (1), 212–252.

Nicholson, Joshua and **Ioannidis**, John, "Conform and be funded", *Nature*, 2012, *492* (7427), 34–36.

Nielsen, Mathias and **Andersen**, Jens, "Global citation inequality is on the rise", *Proceedings of the National Academy of Sciences*, 2021, *118* (7), e2012208118.

Nogales, Eva, "The Development of Cryo-EM into a Mainstream Structural Biology Technique", *Nature Methods*, 2016, *13* (1), 24–27.

Noy, Shakked and **Zhang**, Whitney, "Experimental evidence on the productivity effects of generative artificial intelligence", *Science*, 2023, *381* (6654), 187–192.

Packalen, Mikko and **Bhattacharya**, Jay, *Does the NIH Fund Edge Science?*, Working Paper, National Bureau of Economic Research, 2018.

Papke, Leslie and **Wooldridge**, Jeffrey, "Econometric Methods for Fractional Response Variables with an Application to 401(k) Plan Participation Rates", *Journal of Applied Econometrics*, 1996, *11* (6), 619–632.

Park, Michael, **Leahey**, Erin, and **Funk**, Russell, "Papers and patents are becoming less disruptive over time", *Nature*, 2023, *613* (7942), 138–144.

Qian, Jiarui, *AI in the Lab: AlphaFold2's Impacts on Human-Produced Knowledge*, SSRN Working Paper 6133666, 2026.

Ren, Feng, **Ding**, Xiao, **Zheng**, Min, **Korzinkin**, Mikhail, **Cai**, Xin, **Zhu**, Wei, **Mantsyzov**, Alexey, **Aliper**, Alex, **Aladinskiy**, Vladimir, **Cao**, Zhongying, et al., "AlphaFold accelerates artificial intelligence powered drug discovery: efficient discovery of a novel CDK20 small molecule inhibitor", *Chemical Science*, 2023, *14* (6), 1443–1452.

Santos, Silva and **Tenreyro**, Silvana, "The Log of Gravity", *The Review of Economics and Statistics*, 2006, *88* (4), 641–658.

Senior, Andrew, **Evans**, Richard, **Jumper**, John, **Kirkpatrick**, James, **Sifre**, Laurent, **Green**, Tim, **Qin**, Chongli, **Žídek**, Augustin, **Nelson**, Alexander, **Bridgland**, Alex, et al., "Improved protein structure prediction using potentials from deep learning", *Nature*, 2020, *577* (7792), 706–710.

Senior, Andrew, **Evans**, Richard, **Jumper**, John, **Kirkpatrick**, James, **Sifre**, Laurent, **Green**, Tim, **Qin**, Chongli, **Žídek**, Augustin, **Nelson**, Alexander, **Bridgland**, Alex, **Penedones**, Hugo, **Petersen**, Stig, **Simonyan**, Karen, **Crossan**, Steve, **Kohli**, Pushmeet, **Jones**, David, **Silver**, David, **Kavukcuoglu**, Koray, and **Hassabis**, Demis, "Protein structure prediction using multiple deep neural networks in the 13th Critical Assessment of Protein Structure Prediction (CASP13)", *Proteins: Structure, Function, and Bioinformatics*, 2019, *87* (12), 1141–1148.

Stevens, Raymond, "The Cost and Value of Three-Dimensional Protein Structure", *Drug Discovery World*, 2003, *4* (3), 35–48.

Teodoridis, Florenta, "Understanding Team Knowledge Production: The Interrelated Roles of Technology and Expertise", *Management Science*, 2018, *64* (8), 3625–3648.

Terwilliger, Thomas, **Liebschner**, Dorothee, **Croll**, Tristan, **Williams**, Christopher, **McCoy**, Airlie, **Poon**, Billy, **Afonine**, Pavel, **Oeffner**, Robert, **Richardson**, Jane, **Read**, Randy, and **Adams**, Paul, "AlphaFold predictions are valuable hypotheses and accelerate but do not replace experimental structure determination", *Nature Methods*, 2024, *21* (1), 110–116.

Thompson, Neil and **Zyontz**, Samantha, *Decomposing the "Tacit Knowledge Problem:" Codification of Knowledge and Access in CRISPR Gene-Editing*, SSRN Working Paper, 2017.

Wu, Lingfei, **Wang**, Dashun, and **Evans**, James, "Large teams develop and small teams disrupt science and technology", *Nature*, 2019, *566* (7744), 378–382.

Wuchty, Stefan, **Jones**, Benjamin, and **Uzzi**, Brian, "The Increasing Dominance of Teams in Production of Knowledge", *Science*, 2007, *316* (5827), 1036–1039.

Appendix

Contents

Appendix A Supplementary Figures and Tables **A1**

A.1 Supplementary Figures A1

A.2 Supplementary Tables A15

Appendix B Sample Selection and Methods **B1**

B.1 Publications B1

B.2 Journals B1

B.3 University Rankings B2

B.4 Measuring Research Output B3

B.5 Additional Data B4

B.6 Empirical Design B4

B.7 Measures of Research Direction and Impact B5

B.8 Sample Construction for Placebo Analysis B7

Appendix C Identifying the Proteins Studied by Each Publication **C1**

Appendix D Literature & Background **D1**

D.1 Concentration in Scientific Research D1

D.2 Structural Biology D2

D.3 The Advent of AlphaFold D2

D.4 Role of AlphaFold in Structural Biology D4

D.5 Protein Structure Determination Before and After AlphaFold D4

D.6 Descriptive Statistics D8

Appendix E Additional Results **E1**

E.1 AlphaFold Adoption E1

E.2 Dynamic Estimates E2

E.3 Publication Outcomes and Research Patterns Outside Protein Science E2

A Supplementary Figures and Tables

A.1 Supplementary Figures

Figure A1: **Geographical distribution of universities by quartile**
The figure shows the location of the top 500 universities in our sample. Colors denote quartiles. Universities are ordered by pre-AlphaFold TNCS and grouped into consecutive quartiles with approximately equal pre-AlphaFold top-journal publication shares. Circle diameters

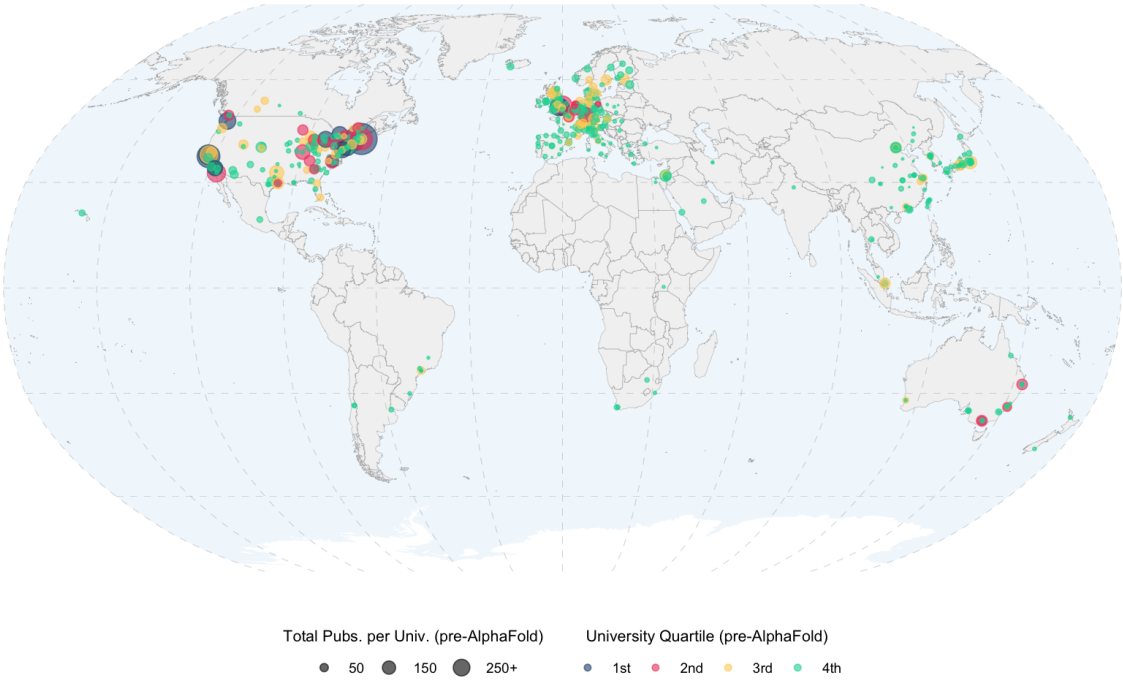
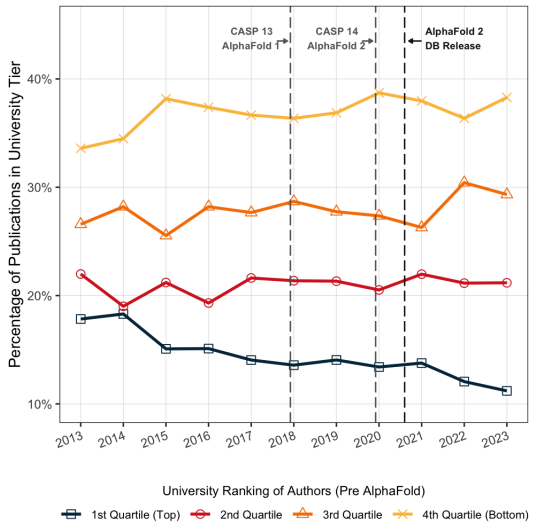


Figure A2: **Publications in lower-ranked journals by university quartile**

The figure reports the percentage of protein research publications in lower-ranked journals, defined as those with an average SJR between 1 and 10 during 2013–2018. The analysis is limited to authors affiliated with the top 500 universities, ordered by pre-AlphaFold TNCS and grouped into consecutive quartiles with approximately equal pre-AlphaFold top-journal publication shares.

(A) Structural biology publications



(B) Non-structural protein publications

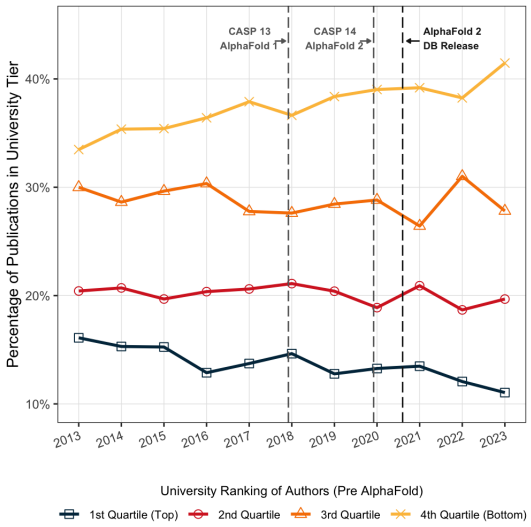


Figure A3: **AI tools introduced during CASP competitions**

The figure reports the number and share of prediction tools using AI at each CASP competition, together with the share of participating models that

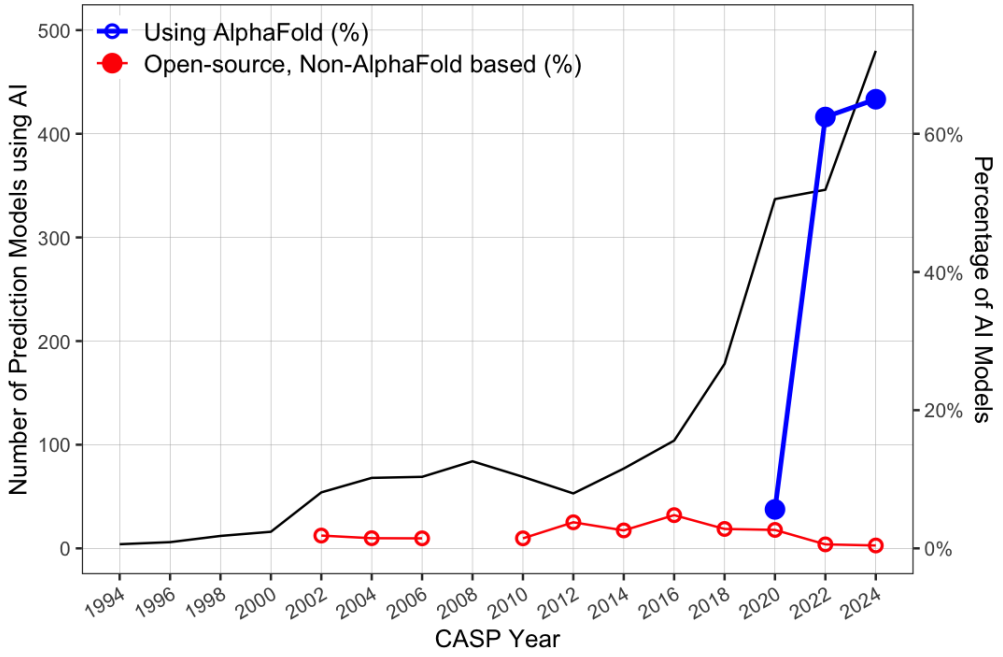


Figure A4: **Distribution of top-journal publications over time**

The figure reports publication volume over time for all top-journal publications, defined as those in journals with SJR ≥ 10, in the sample.

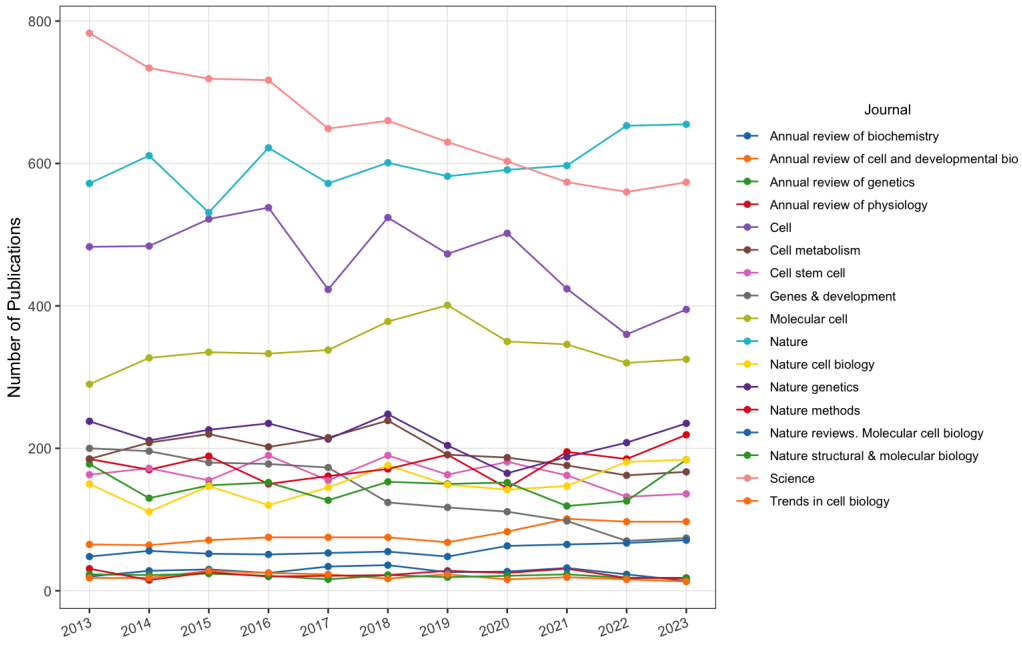


Figure A5: **Timeline of AI development in structural biology**

The development of AI systems in structural biology occurred in two distinct phases. In the early phase, following the December 2018 CASP13 announcement, adoption likely required substantial domain knowledge and ICT skills. In the later phase, the July 2021 open release of AlphaFold 2 (AF2) code and database made adoption easier and likely reached a wider group. A third version, AlphaFold 3 (AF3), was subsequently released.

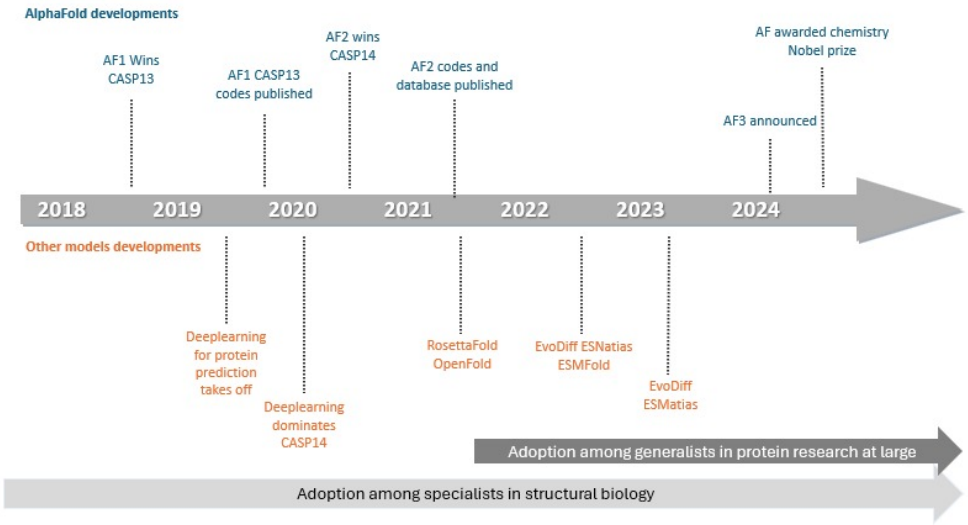
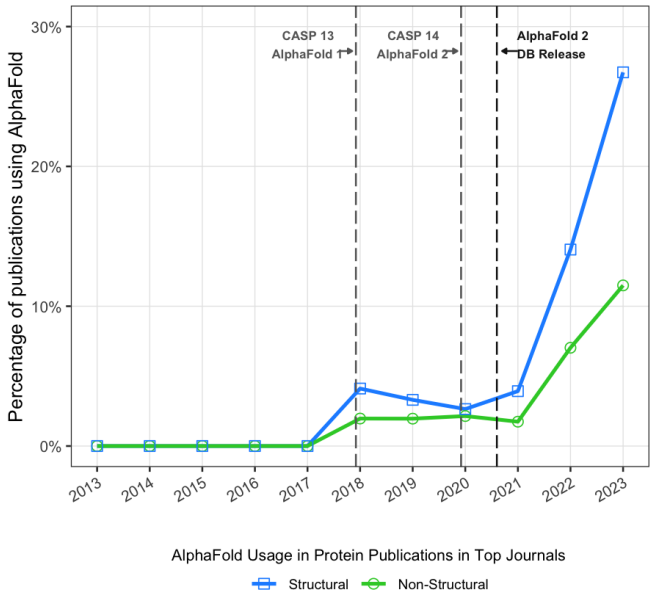


Figure A6: **AlphaFold mentions in top protein publications**

Figure shows details on AlphaFold adoption in top journals (SJR ≥ 10), specifically the share of protein research papers published that mention explicit use of AlphaFold in their abstract or related AI tools, such as RoseTTAFold. Panel (A) shows this share for structural and non-structural protein research. Panel (B) breaks down the adoption rate by subfield of structural biology.

(A) All protein publications



(B) AlphaFold mentions in top subfields within structural biology

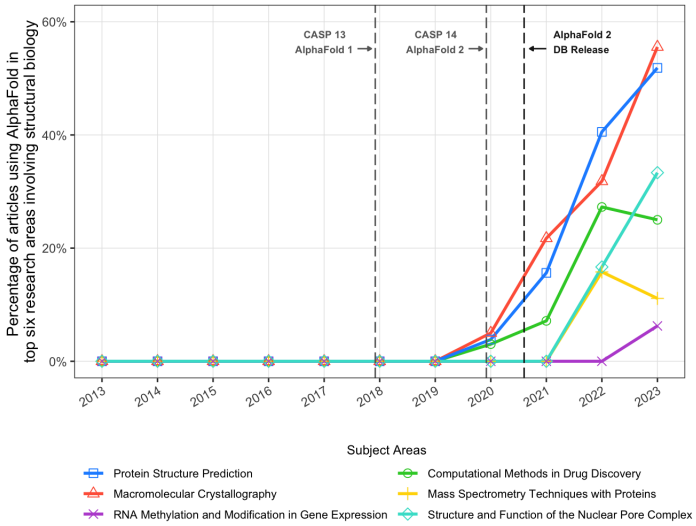
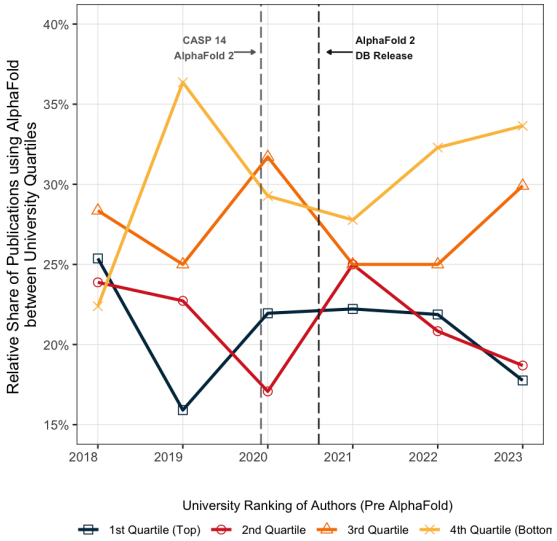


Figure A7: **AlphaFold mentions in top protein publications by university quartile**

The figure reports the share of publications using AlphaFold attributable to each of the four university quartiles. These shares are computed for structural-biology publications in Panel (A) and non-structural protein publications in Panel (B).

(A) Structural biology publications



(B) Non-structural protein publications

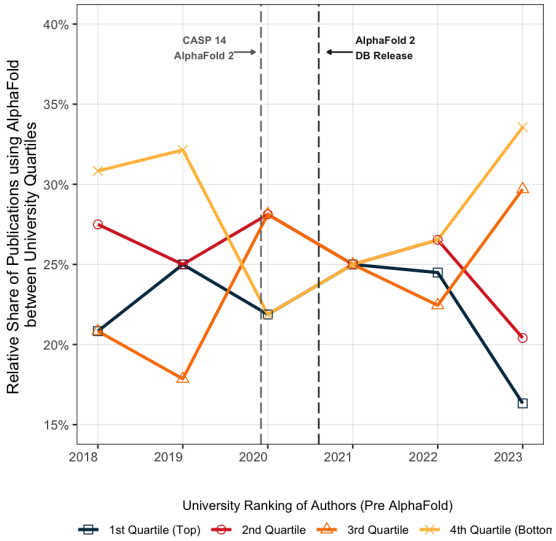
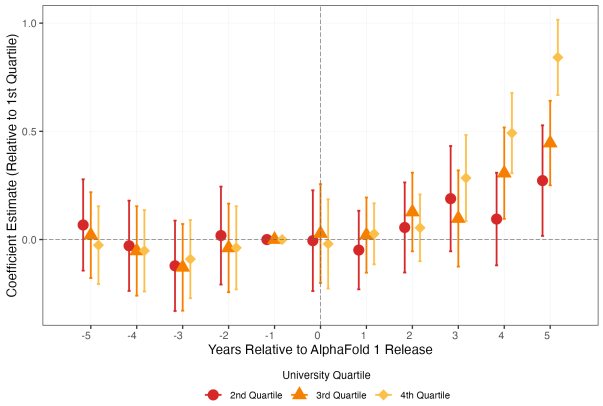


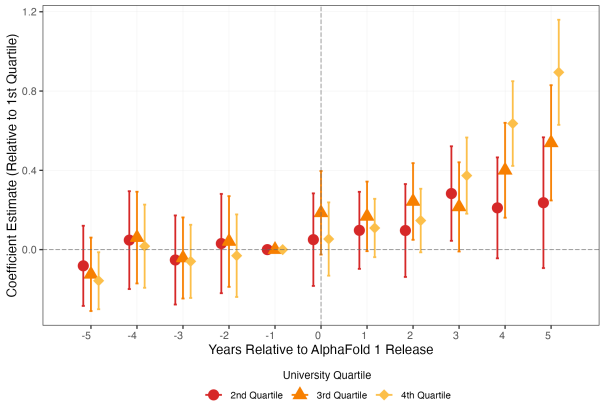
Figure A8: **Protein publications by universities in top journals around AlphaFold : dynamic estimates**

The figure reports annual estimates from regressions of the number of top-journal protein publications on university-quartile indicators and the controls in Equation 4. The plot shows point estimates with 95% confidence intervals. The dotted line marks the release of AlphaFold. The analysis includes all protein publications in Panel (A), structural-biology publications in Panel (B), and non-structural protein publications in Panel (C). Universities are ordered by pre-AlphaFold TNCS and grouped into consecutive quartiles with approximately equal pre-AlphaFold top-journal publication shares.

(A) All protein publications



(B) Structural biology publications



(C) Non-structural protein publications

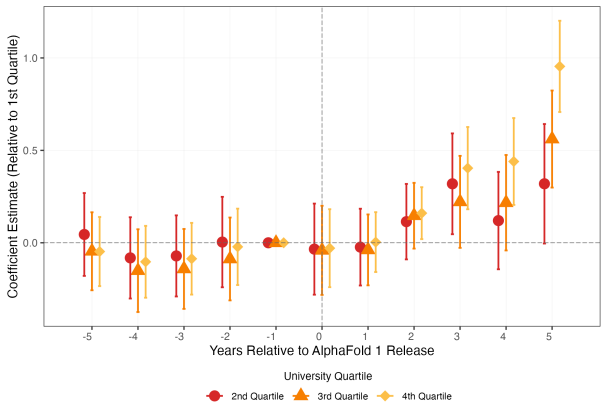
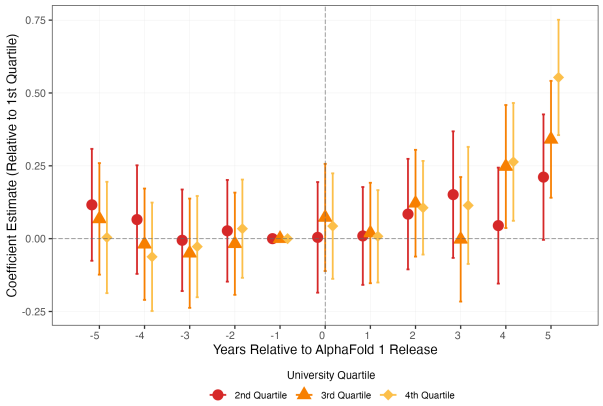


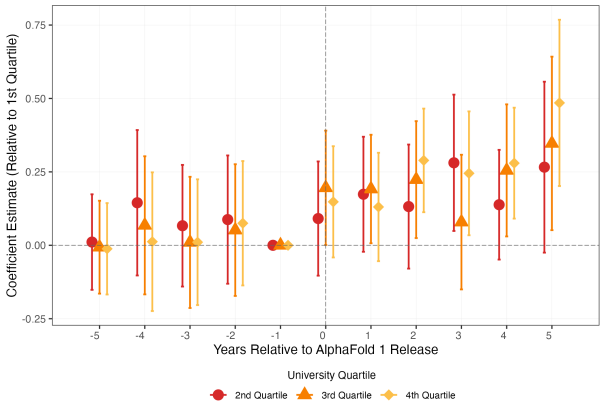
Figure A9: **Market shares of universities in top-journal protein publications around AlphaFold : dynamic estimates**

The figure reports annual estimates from regressions of top-journal market share on university-quartile indicators and the controls in the main specification. The plot shows point estimates with 95% confidence intervals. The dotted line marks the release of AlphaFold. The analysis includes all protein publications in Panel (A), structural-biology publications in Panel (B), and non-structural protein publications in Panel (C). Universities are ordered by pre-AlphaFold TNCS and grouped into consecutive quartiles with approximately equal pre-AlphaFold top-journal publication shares.

(A) All protein publications



(B) Structural biology publications



(C) Non-structural protein publications

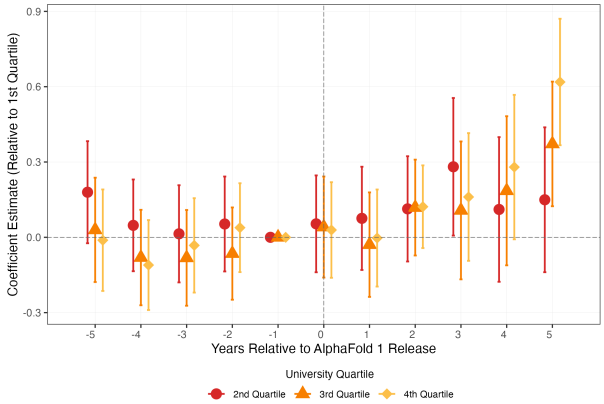
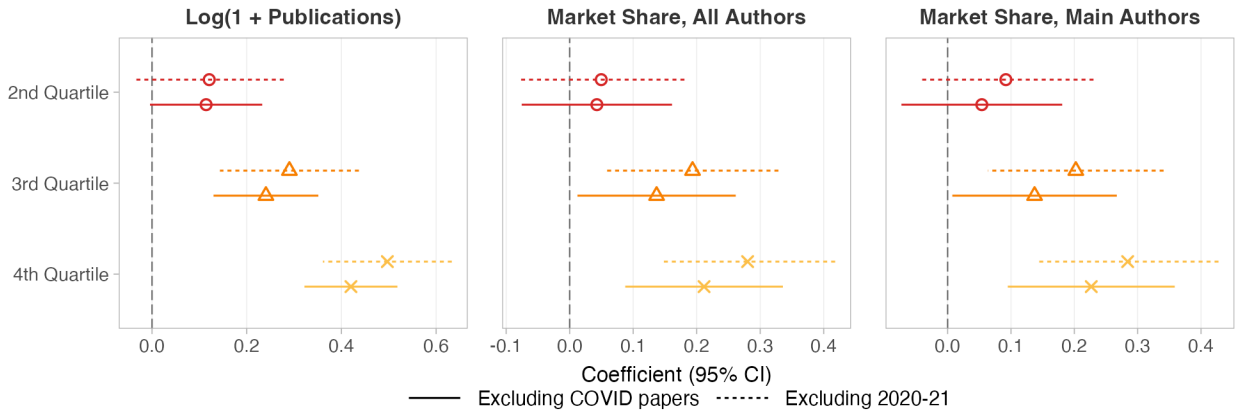


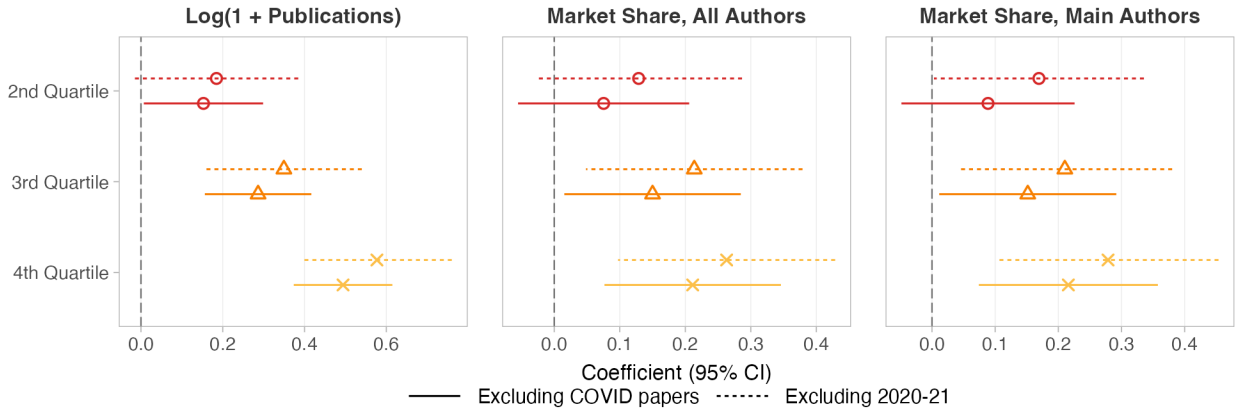
Figure A10: **University-quartile gains in top-journal protein publications, excluding COVID-19 research and the pandemic years**

The figure reports the quartile × PostAF1 estimates from Equation 4 for two restricted samples: one excluding publications assigned to COVID-19 topics, and one excluding the pandemic years 2020–2021. Point estimates are shown with 95% confidence intervals, with standard errors clustered by university. The first quartile is the reference group. Panel (A) covers all protein publications, Panel (B) structural-biology publications, and Panel (C) non-structural protein publications.

(A) All protein publications



(B) Structural biology publications



(C) Non-structural protein publications

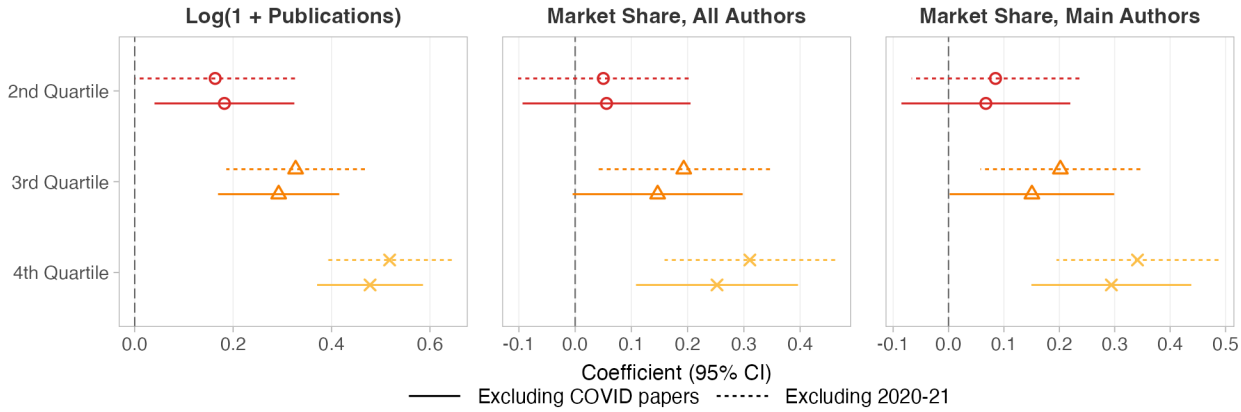
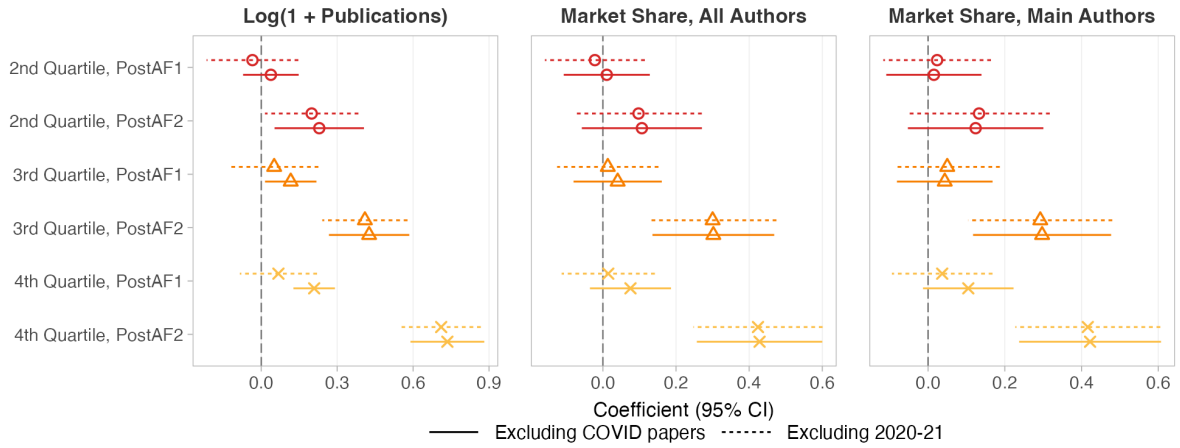


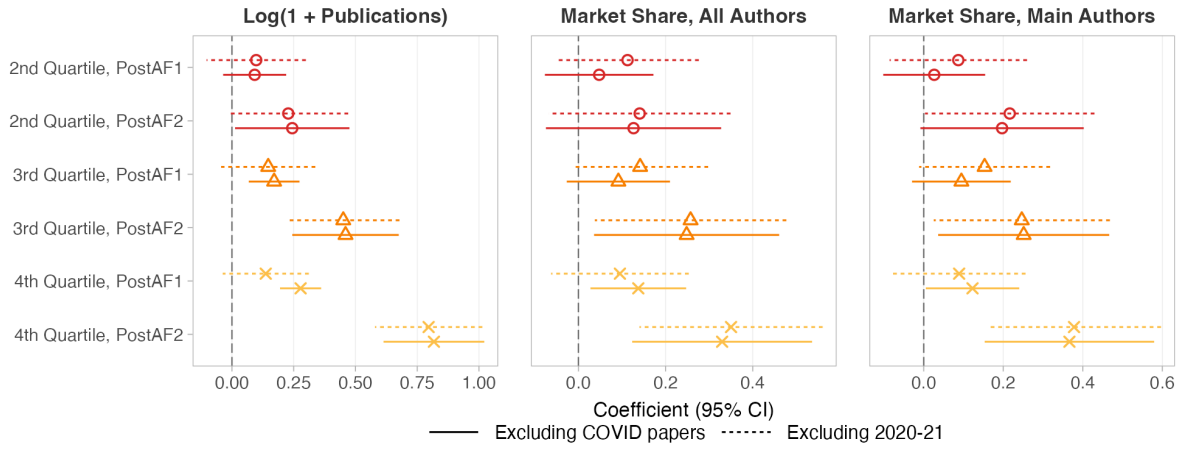
Figure A11: **Timing of the quartile gains across the two AlphaFold releases, excluding COVID-19 research and the pandemic years**

The figure separates the estimates in Figure A10 into the period following the release of AlphaFold 1 and the period following the release of the AlphaFold Protein Structure Database in July 2021. It reports the quartile × PostAF1 and quartile × PostAF2 estimates for two restricted samples: one excluding publications assigned to COVID-19 topics, and one excluding the pandemic years 2020–2021. Point estimates are shown with 95% confidence intervals, with standard errors clustered by university. The first quartile is the reference group. Panel (A) covers all protein publications, Panel (B) structural-biology publications, and Panel (C) non-structural protein publications.

(A) All protein publications



(B) Structural biology publications



(C) Non-structural protein publications

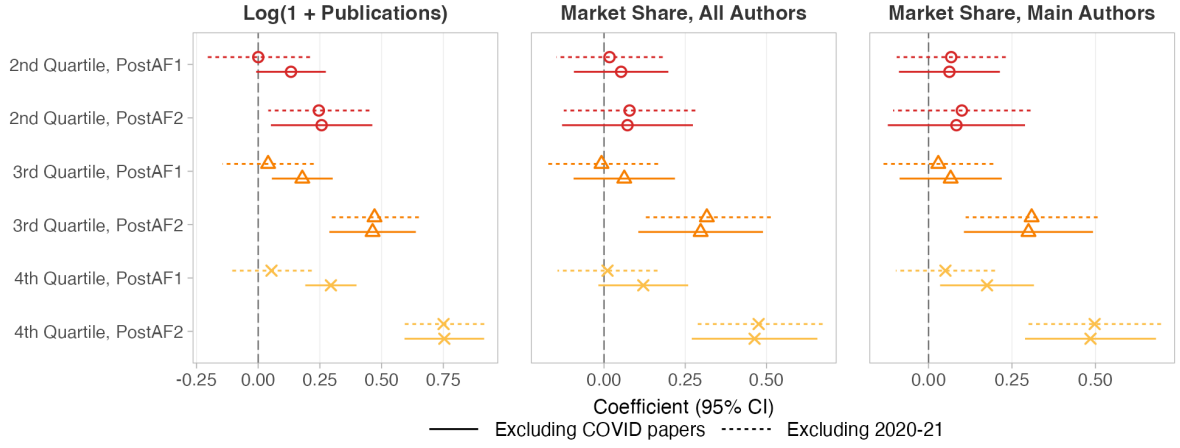
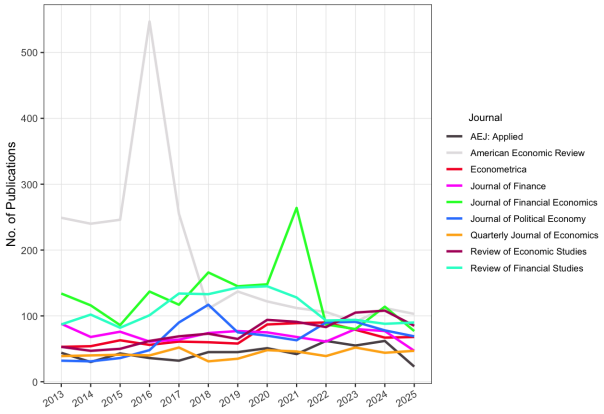


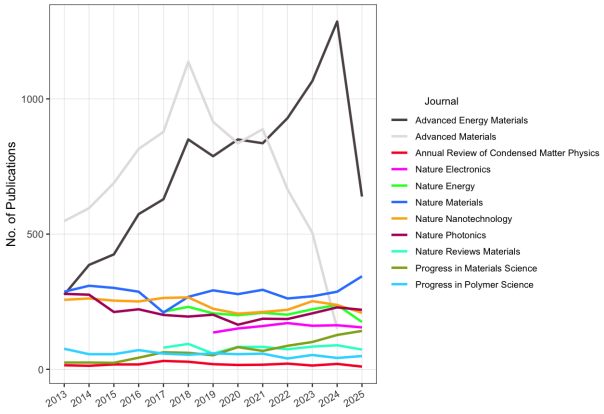
Figure A12: **Distribution of top-journal publications in other fields over time**

The figure reports the annual number of publications in each top journal in our sample, defined as journals with SJR ≥ 10, for economics and finance in Panel (A), materials science in Panel (B), and social sciences in Panel (C).

(A) Economics & finance



(B) Materials science



(C) Social sciences

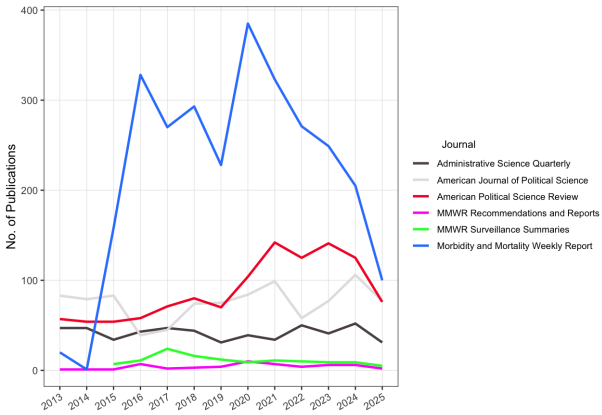
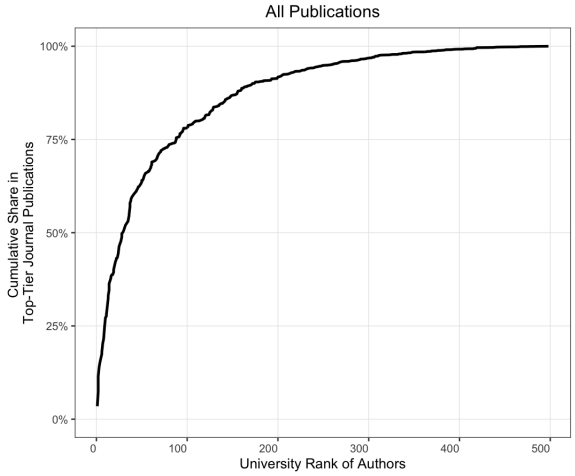


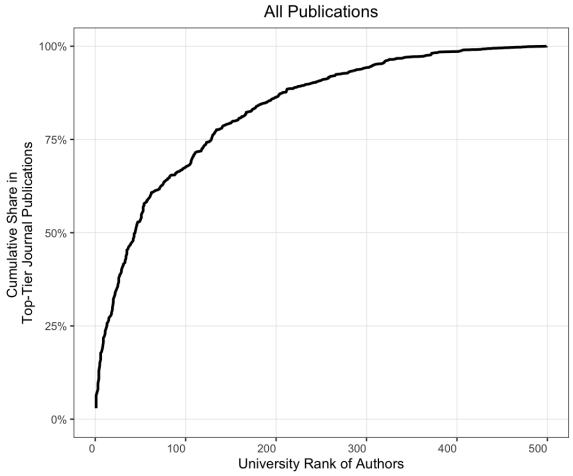
Figure A13: **Market shares of top 500 universities in top-journal publications in other fields**

The figure plots the cumulative distribution of top-journal publication shares across universities. University rankings are based on each university's Total Normalized Citation Score (TNCS) in a given year. For economics and finance in Panel (A) and social sciences in Panel (C), we use the Leiden Ranking category "social sciences and humanities." For materials science in Panel (B), we use "physical sciences and engineering." As in protein research, the top 500 universities capture nearly all top-journal publication counts. The distributions in all three fields show substantial concentration among top universities.

(A) Economics & finance



(B) Materials science



(C) Social sciences

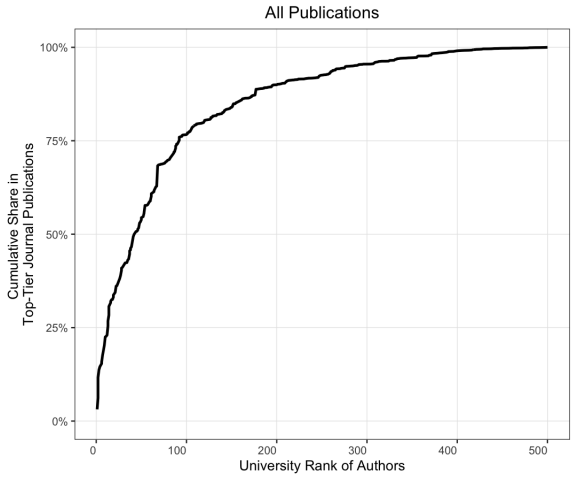


Figure A14: **Bottom-quartile changes in top-journal protein publications by protein stratum, period, and subfield**

Each point is a quartile-by-period-by-stratum interaction from the stacked specification in Table 6. A paper is structure-available when at least half of its linked proteins had an experimental structure, or a homology template of at least 30 percent sequence identity, before the release of the AlphaFold protein structure database on 22 July 2021, and has no prior protein information otherwise. A positive value means the quartile's gain relative to the top quartile is larger on structure-available papers than on no-prior-structure papers. Bars are 95% confidence intervals.

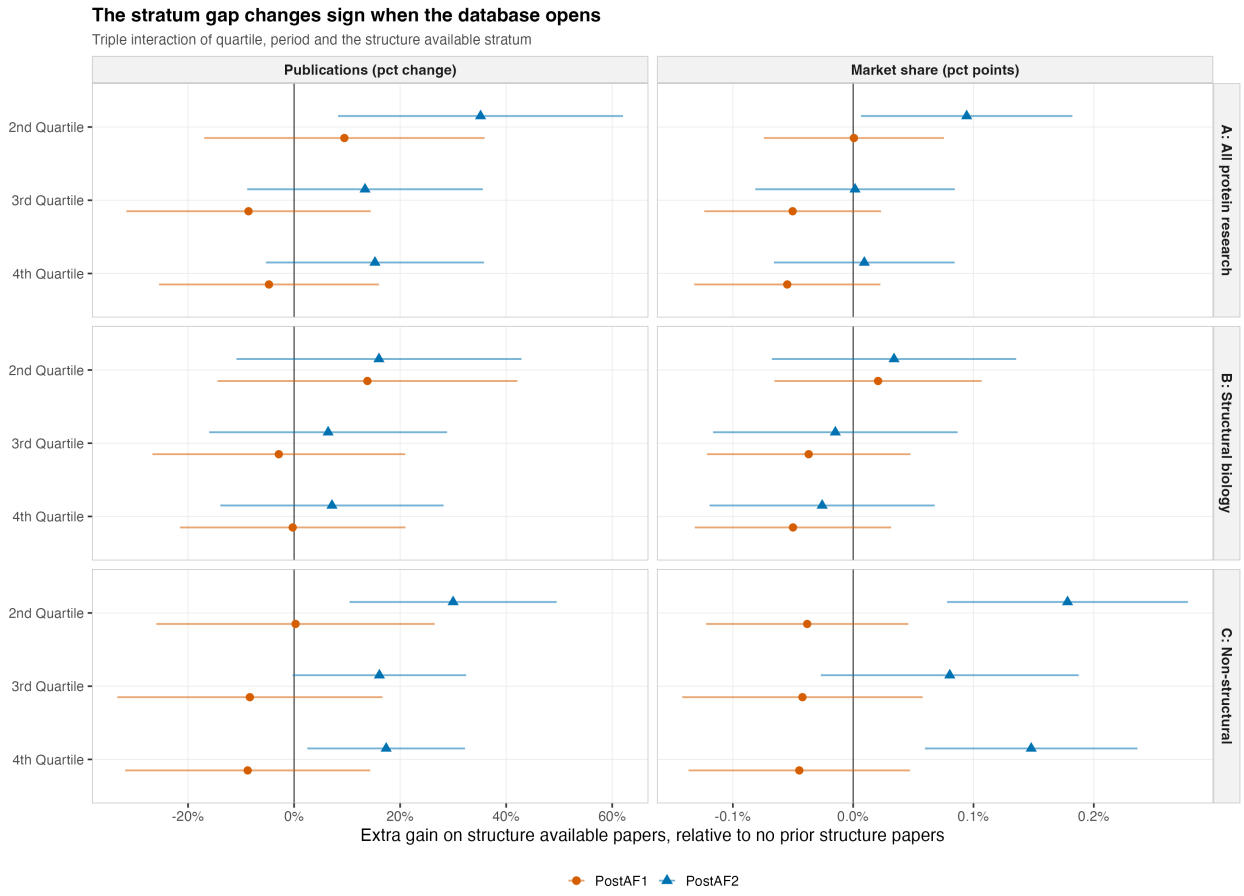


Figure A15: **Lower-quartile differentials in top-journal protein research under alternative definitions of a no-prior-structure paper**

Each point is the additional gain of bottom-quartile universities on structure-available papers, measured relative to the top quartile and to papers without prior protein information before AlphaFold. Figure A14 plots the same quantity under the main definition, in which a paper counts as having no prior structure when fewer than half of its linked proteins had an experimental structure or a usable homology template before the AlphaFold database release. The top row of each panel repeats the main definition, under which a paper has no prior information when more than half of its linked proteins lack it. The three rows below raise the required share, first to the top tercile of the no-prior-structure distribution, then to three quarters of a paper's linked proteins, and finally to all of them. The remaining row applies a different criterion, counting a paper as having no prior information when none of its linked proteins had an experimental structure of its own, whether or not a homology template was available.

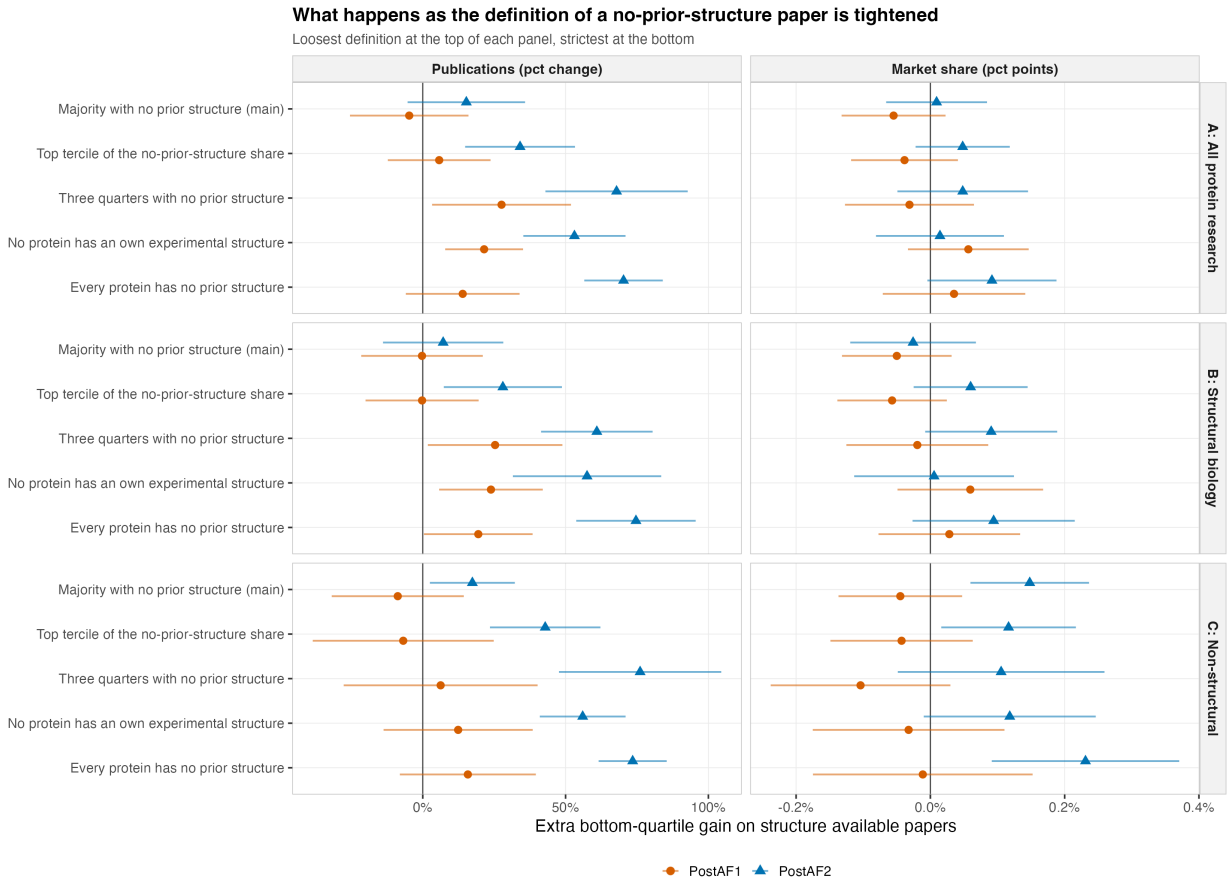
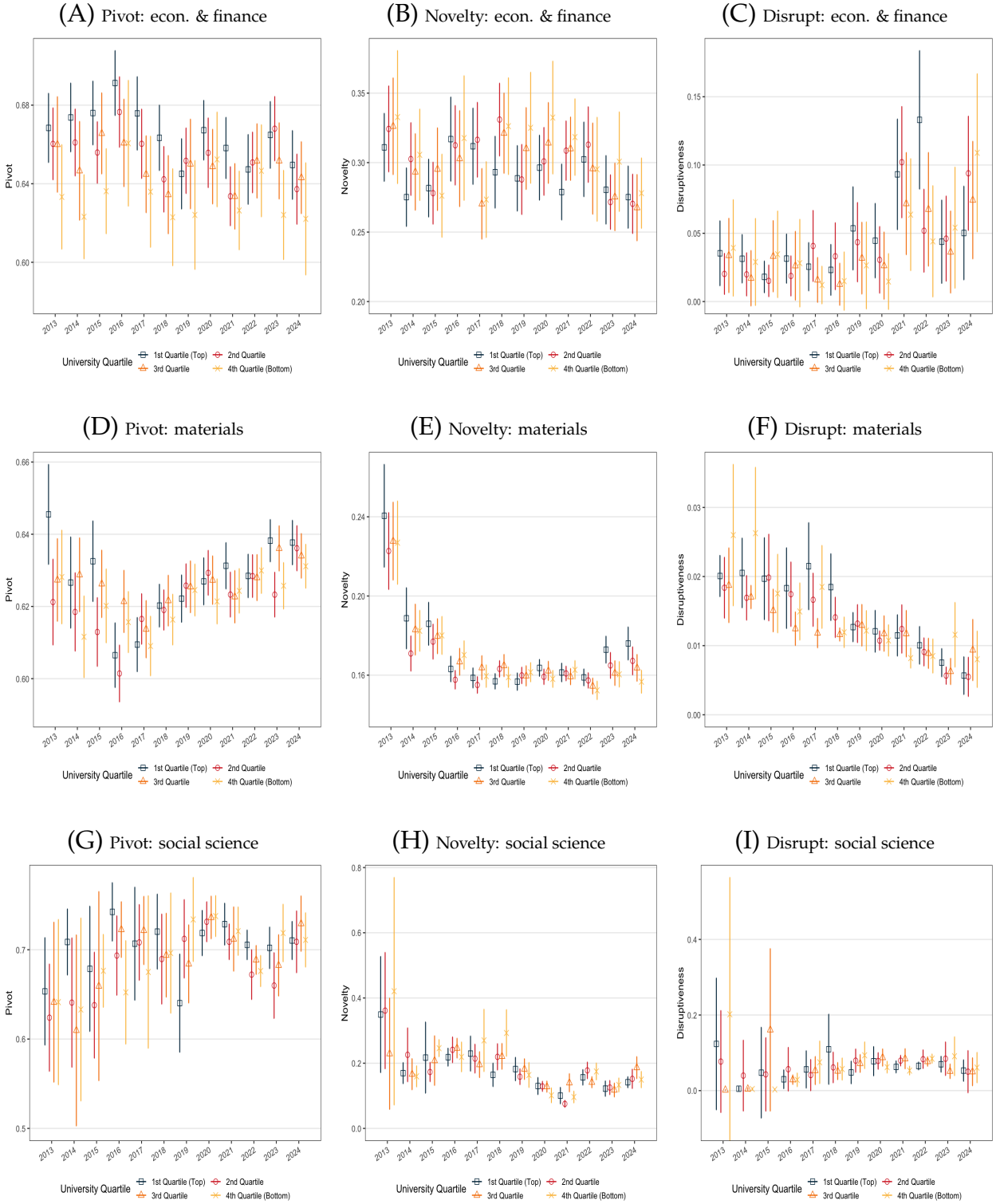


Figure A16: **Research direction measures in top-journal publications by university quartile in other fields**

The figure reports pivot scores, novelty, and disruptiveness measures based on three-year publication histories of authors in top journals, defined as journals with SJR ≥ 10. Universities are ordered by field-specific pre-AlphaFold TNCS and grouped into consecutive quartiles with approximately equal pre-AlphaFold top-journal publication shares. Columns report pivot scores, novelty, and disruptiveness from left to right. Rows correspond to economics and finance, materials science, and social sciences from top to bottom.



A.2 Supplementary Tables

Table A1: **Other AI tools developed for protein structure prediction**

Table lists some of the most prominent AI tools that were developed following AlphaFold 's success. Though most of them are based on the same core transformer models like AlphaFold , many of them focus on different tasks and/or use different training data.

AI Product	Release Date	Open-Source	Creator(s)	Key Feature
RoseTTAFold	July 2021	Yes	University of Washington	Predicts Protein-nucleic acid complexes
EvoDiff	September 2023	Yes	Microsoft	Applies diffusion models to get protein design from evolutionary information
ESMFold	November 2022	Yes	Meta AI	Uses amino acid sequence only for faster predictions. ESM atlas is a database with 617 million predicted protein structures
ProGen	January 2023	No	Salesforce AI Research	Synthesizes new protein sequences with desired properties from natural language inputs
BaseFold	February 2023	Yes	Basecamp Research	Uses proprietary protein samples in training data
RFDiffusion	July 2023	Yes	University of Washington	Applies diffusion models to focus on protein design

Table A2: **Top journals in protein research**

Table lists all journals with an SJR above ten which are used to constitute the sample of publication in the analysis throughout.

Journal Name	ISSN	Mean SJR (2013–18)	Mean Impact Factor (2013–18)	Publisher
Nature reviews. Molecular cell biology	1471-0072	28.16	39.74	Nature Portfolio
Cell	0092-8674	27.21	32.02	Cell Press
Nature genetics	1061-4036	23.01	28.02	Nature Portfolio
Annual rev. of biochemistry	1545-4509	22.12	28.44	Annual Reviews
Nature	1476-4687	18.46	41.24	Nature Portfolio
Nature methods	1548-7091	18.14	27.13	Nature Portfolio
Annual review of cell and dev. biology	1530-8995	14.97	14.94	Annual Reviews
Nature cell biology	1476-4679	14.63	18.37	Nature Portfolio
Cell stem cell	1934-5909	13.70	22.49	Elsevier BV
Molecular cell	1097-4164	13.47	14.24	Elsevier BV
Science	1095-9203	13.08	39.10	AAAS
Annual review of genetics	1545-2948	12.95	11.98	Annual Reviews
Cell metabolism	1550-4131	11.21	18.79	Cell Press
Genes & development	1549-5477	11.06	10.84	Cold Spring Harbor
Nature structural & molecular biology	1545-9993	10.82	12.72	Nature Portfolio
Trends in cell biology	0962-8924	10.41	14.39	Elsevier BV
Annual review of physiology	0066-4278	10.04	17.22	Annual Reviews

Table A3: **Descriptive statistics: top protein publications and university characteristics before AlphaFold**

Table presents summary statistics at the university-year level for the period 2013–18, representing the years preceding the December 2018 CASP13 result. The sample is restricted to protein-related publications in top journals with an SJR score of 10 or higher, authored by researchers affiliated with the top 500 universities prior to the CASP13 result. Each cell reports the mean of the respective variable, measured annually during the pre-AlphaFold period, with universities grouped into quartiles.

Variable (*Pre AlphaFold estimates*)	1st (reference group)	2nd	3rd	4th
Structural biology publications				
Number of Publications per Year per University	8.58	2.39	1.75	0.80
Market Share (% publications)	0.37	0.10	0.08	0.03
Market Share (% of authors on publication)	0.39	0.08	0.06	0.02
Non-structural protein publications				
Number of Publications per Year per University	7.88	2.58	1.59	0.88
Market Share (% publications)	0.36	0.12	0.07	0.04
Market Share (% of authors on publication)	0.37	0.10	0.05	0.03
University characteristics				
Faculty Size (Biology)	614	308	267	178
Share Structural Biologists (%)	5.98	4.90	5.19	3.93
Share Female (%)	31.04	32.96	31.46	34.23
Share White (%)	28.91	26.11	23.62	22.61
Share Asian (%)	51.09	50.15	55.80	54.69
Share Black (%)	12.68	12.92	8.96	9.43
Share affiliated in US (%)	45.01	22.60	16.93	21.87
Share affiliated in Europe (%)	38.04	54.28	48.36	44.56
Share affiliated in Asia (%)	20.68	19.77	33.26	26.14
Share affiliated with Govt (%)	5.76	9.58	6.59	7.18
Share affiliated with Research Lab (%)	14.26	16.73	12.06	13.31
Share affiliated with Industry (%)	3.22	2.70	2.70	1.99
Share affiliated with Medical Inst (%)	16.97	16.92	16.46	14.46
Share HPC Access (%)	50.90	32.26	31.00	9.07
HPC power (petaFlop/s)	351.11	205.55	231.27	54.18
Number of Universities	12	36	88	364

Table A4: **List of universities in the top quartile**

These 12 universities together make up approximately 25% of top journal publications before AlphaFold (year 2013-2018)

Name	Country
Harvard University	USA
Johns Hopkins University	USA
University of California, San Francisco	USA
University of Toronto	Canada
University of Pennsylvania	USA
University of Washington	USA
University College London	UK
Stanford University	USA
Duke University	USA
University of Michigan–Ann Arbor	USA
University of California, Los Angeles	USA
University of Oxford	UK

Table A5: **Structural biology top publications around AlphaFold: robustness checks**

PostAF1 is a binary variable set to one for papers published after the December 2018 CASP13 result. The sample is limited to structural biology papers published in top journals with an SJR score of 10 or higher and to authors affiliated with the top 500 universities prior to the CASP13 result. Universities are ranked based on their average TNCS in the period preceding CASP13. The highest ranked universities placed in the top quartile (*University Tier = 1st Quartile*) are used as the reference group. Standard errors are clustered by university, and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

(A) Publications

Dependent Variable:	Log(1 + No. of Publications)				
Model:	(1)	(2)	(3)	(4)	(5)
University Tier = 2nd Quartile × PostAF1	0.2444***	0.2443***	0.2351***	0.2418***	0.2309***
	(0.0172)	(0.0173)	(0.0145)	(0.0197)	(0.0166)
University Tier = 3rd Quartile × PostAF1	0.3555***	0.3557***	0.3764***	0.3425***	0.3605***
	(0.0211)	(0.0210)	(0.0262)	(0.0272)	(0.0322)
University Tier = 4th Quartile × PostAF1	0.5849***	0.5845***	0.5598***	0.5678***	0.5392***
	(0.0321)	(0.0320)	(0.0334)	(0.0379)	(0.0412)
Controls: Share Structural Biologists		✓			✓
Controls: HPC Power			✓		✓
Controls: Author Characteristics				✓	✓
Year FE	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓
Observations	5,500	5,500	5,500	5,500	5,500
Adjusted R^2	0.780	0.781	0.805	0.785	0.809

(B) Market share (% publications, all authors)

Dependent Variable:	Market Share (% Publications, All Authors)				
Model:	(1)	(2)	(3)	(4)	(5)
University Tier = 2nd Quartile × PostAF1	0.1387***	0.1394***	0.1406***	0.1322***	0.1342***
	(0.0019)	(0.0018)	(0.0014)	(0.0049)	(0.0037)
University Tier = 3rd Quartile × PostAF1	0.1433***	0.1445***	0.1738***	0.1404***	0.1644***
	(0.0028)	(0.0025)	(0.0219)	(0.0059)	(0.0216)
University Tier = 4th Quartile × PostAF1	0.2827***	0.2813***	0.2837***	0.2584***	0.2497***
	(0.0043)	(0.0050)	(0.0232)	(0.0105)	(0.0271)
Controls: Share Structural Biologists		✓			✓
Controls: HPC Power			✓		✓
Controls: Author Characteristics				✓	✓
Year FE	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓
Observations	5,500	5,500	5,500	5,500	5,500
Pseudo R^2	0.904	0.904	0.906	0.917	0.921

(C) Market share (% publications, main authors)

Dependent Variable:	Market Share (% Publications, Main Authors)				
Model:	(1)	(2)	(3)	(4)	(5)
University Tier = 2nd Quartile × PostAF1	0.1482***	0.1480***	0.1291***	0.1683***	0.1480***
	(0.0020)	(0.0024)	(0.0029)	(0.0036)	(0.0043)
University Tier = 3rd Quartile × PostAF1	0.1409***	0.1405***	0.1543***	0.1578***	0.1623***
	(0.0032)	(0.0036)	(0.0160)	(0.0063)	(0.0145)
University Tier = 4th Quartile × PostAF1	0.2700***	0.2664***	0.2688***	0.2560***	0.2420***
	(0.0044)	(0.0058)	(0.0209)	(0.0111)	(0.0242)
Controls: Share Structural Biologists		✓			✓
Controls: HPC Power			✓		✓
Controls: Author Characteristics				✓	✓
Year FE	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓
Observations	5,500	5,500	5,500	5,500	5,500
Pseudo R^2	0.899	0.901	0.903	0.915	0.919

Table A6: **Non-structural protein top publications around AlphaFold: robustness checks**

PostAF1 is a binary variable set to one for papers published after the release of AlphaFold 1 in December 2018. The sample is limited to non-structural protein papers published in top journals with an SJR score of 10 or higher and to authors affiliated with the top 500 universities prior to the release of AlphaFold 1. Universities are ranked based on their average TNCS in the period preceding the release of AlphaFold 1. The highest ranked universities placed in the top quartile (*University Tier = 1st Quartile*) are used as the reference group. Standard errors are clustered by university, and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

(A) Publications

Dependent Variable:	Log(1 + No. of Publications)				
Model:	(1)	(2)	(3)	(4)	(5)
University Tier = 2nd Quartile × PostAF1	0.2912***	0.2909***	0.2931***	0.2757***	0.2767***
	(0.0201)	(0.0201)	(0.0194)	(0.0281)	(0.0276)
University Tier = 3rd Quartile × PostAF1	0.4052***	0.4043***	0.4183***	0.3913***	0.4018***
	(0.0185)	(0.0185)	(0.0215)	(0.0240)	(0.0289)
University Tier = 4th Quartile × PostAF1	0.5963***	0.5957***	0.5572***	0.5766***	0.5361***
	(0.0279)	(0.0279)	(0.0259)	(0.0347)	(0.0340)
Controls: Share Structural Biologists		✓			✓
Controls: HPC Power			✓		✓
Controls: Author Characteristics				✓	✓
Year FE	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓
Observations	5,500	5,500	5,500	5,500	5,500
Adjusted R^2	0.760	0.761	0.787	0.765	0.791

(B) Market share (% publications, all authors)

Dependent Variable:	Market Share (% Publications, all authors)				
Model:	(1)	(2)	(3)	(4)	(5)
University Tier = 2nd Quartile × PostAF1	0.0876***	0.0869***	0.0943***	0.0840***	0.0904***
	(0.0028)	(0.0023)	(0.0061)	(0.0050)	(0.0083)
University Tier = 3rd Quartile × PostAF1	0.1482***	0.1488***	0.1834***	0.1392***	0.1715***
	(0.0030)	(0.0032)	(0.0195)	(0.0084)	(0.0280)
University Tier = 4th Quartile × PostAF1	0.2505***	0.2501***	0.2628***	0.2387***	0.2470***
	(0.0044)	(0.0044)	(0.0143)	(0.0074)	(0.0227)
Controls: Share Structural Biologists		✓			✓
Controls: HPC Power			✓		✓
Controls: Author Characteristics				✓	✓
Year FE	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓
Observations	5,500	5,500	5,500	5,500	5,500
Pseudo R^2	0.884	0.883	0.888	0.899	0.902

(C) Market share (% publications, main authors)

Dependent Variable:	Market Share (% Publications, main authors)				
Model:	(1)	(2)	(3)	(4)	(5)
University Tier = 2nd Quartile × PostAF1	0.1246***	0.1242***	0.1288***	0.1142***	0.1164***
	(0.0024)	(0.0017)	(0.0052)	(0.0070)	(0.0096)
University Tier = 3rd Quartile × PostAF1	0.1532***	0.1529***	0.1893***	0.1398***	0.1707**
	(0.0024)	(0.0019)	(0.0195)	(0.0131)	(0.0312)
University Tier = 4th Quartile × PostAF1	0.2840***	0.2832***	0.3096***	0.2767***	0.2962***
	(0.0041)	(0.0038)	(0.0185)	(0.0117)	(0.0300)
Controls: Share Structural Biologists		✓			✓
Controls: HPC Power			✓		✓
Controls: Author Characteristics				✓	✓
Year FE	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓
Observations	5,500	5,500	5,500	5,500	5,500
Pseudo R^2	0.881	0.880	0.885	0.897	0.901

Table A7: **Protein publications in top journals around AlphaFold: alternate estimators**

PostAF1 is a binary variable set to one for papers published after the December 2018 CASP13 result. The sample is limited to all protein research papers (both structural biology and non-structural protein research) published in top journals with an SJR score of 10 or higher and to authors affiliated with the top 500 universities prior to the CASP13 result. Universities are ranked based on their average TNCS in the period preceding CASP13. The highest ranked universities placed in the top quartile (*University Tier = 1st Quartile*) are used as the reference group. Columns (1)–(2) report Poisson pseudo-maximum-likelihood (PPML) (Santos and Tenreyro, 2006) estimates of annual publication and author-publication counts per university; columns (3)–(4) report fractional-response (Papke and Wooldridge, 1996) logit estimates of annual university-level publication market shares for all authors and for main authors, respectively; and column (5) reports an OLS estimate of the market share measured in percentage points. For the fractional-response columns, average marginal effects (AME) on the market share are reported in percentage points beneath the coefficients. Panel A covers all protein research papers, Panel B is restricted to structural biology, and Panel C to non-structural protein research. All specifications include university and year fixed effects. Standard errors are clustered by university, and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

(A) All protein publications

Estimator:	PPML (Santos and Tenreyro, 2006)		Frac. logit (Papke and Wooldridge, 1996)		OLS
Dependent Variables:	No. of Publications	No. of Author-Publications	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)	Market Share (% Publications, All Authors)
Model:	(1)	(2)	(3)	(4)	(5)
University Tier = 2nd Quartile × PostAF1	0.0491***	0.0520***	0.0644***	0.0983***	0.2545***
	(0.0021)	(0.0108)	(0.0035)	(0.0040)	(0.0179)
University Tier = 3rd Quartile × PostAF1	0.0988***	0.0572**	0.1299**	0.1368***	0.3177***
	(0.0213)	(0.0226)	(0.0242)	(0.0210)	(0.0224)
University Tier = 4th Quartile × PostAF1	0.1567***	0.0115	0.2085***	0.2273***	0.3166***
	(0.0240)	(0.0280)	(0.0238)	(0.0237)	(0.0359)
Year FE	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓
Observations	5,500	5,500	5,500	5,500	5,500
Adjusted/Pseudo R^2	0.95	0.92	0.93	0.93	0.88
AME: University Tier = 2nd Quartile (pp)			0.52 (0.59)	1.02 (0.87)	
AME: University Tier = 3rd Quartile (pp)			1.09 (0.73)	1.45 (0.95)	
AME: University Tier = 4th Quartile (pp)			1.82 (0.94)	2.52 (1.33)	

(B) Structural biology publications

Estimator:	PPML (Santos and Tenreyro, 2006)		Frac. logit (Papke and Wooldridge, 1996)		OLS
Dependent Variables:	No. of Publications	No. of Author-Publications	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)	Market Share (% Publications, All Authors)
Model:	(1)	(2)	(3)	(4)	(5)
University Tier = 2nd Quartile × PostAF1	0.1192***	0.1266***	0.1342***	0.1480***	0.3273***
	(0.0031)	(0.0089)	(0.0037)	(0.0043)	(0.0155)
University Tier = 3rd Quartile × PostAF1	0.1438***	0.0159	0.1644***	0.1623***	0.3774***
	(0.0197)	(0.0253)	(0.0216)	(0.0145)	(0.0232)
University Tier = 4th Quartile × PostAF1	0.2187***	0.0917***	0.2497***	0.2420***	0.3791***
	(0.0278)	(0.0330)	(0.0271)	(0.0242)	(0.0393)
Year FE	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓
Observations	5,500	5,500	5,500	5,500	5,500
Adjusted/Pseudo R^2	0.94	0.91	0.92	0.92	0.87
AME: University Tier = 2nd Quartile (pp)			1.12 (0.77)	1.69 (1.19)	
AME: University Tier = 3rd Quartile (pp)			1.39 (0.87)	1.87 (1.25)	
AME: University Tier = 4th Quartile (pp)			2.20 (1.15)	2.90 (1.65)	

(C) Non-structural protein publications

Estimator:	PPML (Santos and Tenreyro, 2006)		Frac. logit (Papke and Wooldridge, 1996)		OLS
Dependent Variables:	No. of Publications	No. of Author-Publications	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)	Market Share (% Publications, All Authors)
Model:	(1)	(2)	(3)	(4)	(5)
University Tier = 2nd Quartile × PostAF1	0.0766***	0.1022***	0.0904***	0.1164***	0.2666***
	(0.0073)	(0.0140)	(0.0083)	(0.0096)	(0.0216)
University Tier = 3rd Quartile × PostAF1	0.1378***	0.0165	0.1715***	0.1707**	0.3264***
	(0.0268)	(0.0349)	(0.0280)	(0.0312)	(0.0221)
University Tier = 4th Quartile × PostAF1	0.1903***	0.0565	0.2470***	0.2962***	0.3173***
	(0.0249)	(0.0368)	(0.0227)	(0.0300)	(0.0347)
Year FE	✓	✓	✓	✓	✓
University FE	✓	✓	✓	✓	✓
Observations	5,500	5,500	5,500	5,500	5,500
Adjusted/Pseudo R^2	0.93	0.90	0.90	0.90	0.85
AME: University Tier = 2nd Quartile			0.75 (0.75)	1.24 (1.04)	
AME: University Tier = 3rd Quartile			1.48 (0.95)	1.86 (1.17)	
AME: University Tier = 4th Quartile			2.22 (1.18)	3.45 (1.83)	

Table A8: **Protein publications in top journals around AlphaFold: excluding review journals**

The table repeats Table 1 without the papers published in the six review journals among the 17 top journals. The six journals are *Nature Reviews Molecular Cell Biology*, *Trends in Cell Biology*, *Annual Review of Biochemistry*, *Annual Review of Cell and Developmental Biology*, *Annual Review of Genetics* and *Annual Review of Physiology*. Most articles in the six journals are reviews commissioned by the journal editors. Publication counts and market shares are recomputed over the remaining papers. *PostAF1* is a binary variable set to one for papers published after the December 2018 CASP13 result, with 2013–2018 as the reference period. The sample includes structural biology and non-structural protein research papers by authors affiliated with the top 500 universities. Universities are ranked by their average TNCS in 2013–2018. The top quartile (*University Tier = 1st Quartile*) is the reference group. Column (1) is estimated by OLS. Columns (2) and (3) are estimated by fractional logit. All columns include the university-level controls of Table 1 with university and year fixed effects. Standard errors are clustered by university and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

(A) All protein publications

Dependent Variables:	Log(1 + No. of Publications)	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)
Model:	(1)	(2)	(3)
University Tier = 2nd Quartile × PostAF1	0.12*	0.06	0.09
	(0.07)	(0.07)	(0.07)
University Tier = 3rd Quartile × PostAF1	0.23***	0.13*	0.13*
	(0.06)	(0.07)	(0.07)
University Tier = 4th Quartile × PostAF1	0.37***	0.20***	0.22***
	(0.06)	(0.07)	(0.07)
Year FE	✓	✓	✓
University FE	✓	✓	✓
Observations	5,500	5,500	5,500
R^2/Pseudo R^2	0.83	0.93	0.93

(B) Structural biology publications

Dependent Variables:	Log(1 + No. of Publications)	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)
Model:	(1)	(2)	(3)
University Tier = 2nd Quartile × PostAF1	0.18**	0.13	0.13
	(0.09)	(0.08)	(0.08)
University Tier = 3rd Quartile × PostAF1	0.29***	0.16**	0.16*
	(0.08)	(0.08)	(0.08)
University Tier = 4th Quartile × PostAF1	0.45***	0.24***	0.23***
	(0.08)	(0.08)	(0.08)
Year FE	✓	✓	✓
University FE	✓	✓	✓
Observations	5,500	5,500	5,500
R^2/Pseudo R^2	0.81	0.92	0.92

(C) Non-structural protein publications

Dependent Variables:	Log(1 + No. of Publications)	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)
Model:	(1)	(2)	(3)
University Tier = 2nd Quartile × PostAF1	0.19**	0.08	0.11
	(0.08)	(0.08)	(0.08)
University Tier = 3rd Quartile × PostAF1	0.29***	0.16**	0.16**
	(0.07)	(0.08)	(0.07)
University Tier = 4th Quartile × PostAF1	0.43***	0.24***	0.29***
	(0.06)	(0.07)	(0.07)
Year FE	✓	✓	✓
University FE	✓	✓	✓
Observations	5,500	5,500	5,500
R^2/Pseudo R^2	0.79	0.90	0.89

Table A9: **Protein publications in top journals around AlphaFold: AF1 versus AF2, excluding review journals**

The table repeats Table 2 without the papers published in the six review journals listed in Table A8. Publication counts and market shares are recomputed over the remaining papers. *PostAF1/AF2* is a categorical variable with the pre-period as the reference group, where *PostAF1* is set to one for papers published after the December 2018 CASP13 result but before the July 2021 open release of AlphaFold 2's code and database, and *PostAF2* is set to one for papers published after that open release. The reference period is 2013–2018. The sample includes structural biology and non-structural protein research papers by authors affiliated with the top 500 universities. Universities are ranked by their average TNCS in 2013–2018. The top quartile (*University Tier = 1st Quartile*) is the reference group. Column (1) is estimated by OLS. Columns (2) and (3) are estimated by fractional logit. All columns include the university-level controls of Table 1 with university and year fixed effects. Standard errors are clustered by university and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

(A) All protein publications

Dependent Variables:	Log(1 + No. of Publications)	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)
Model:	(1)	(2)	(3)
University Tier = 2nd Quartile × PostAF1	0.07	0.04	0.06
	(0.07)	(0.07)	(0.07)
University Tier = 2nd Quartile × PostAF2	0.20**	0.09	0.13
	(0.09)	(0.09)	(0.09)
University Tier = 3rd Quartile × PostAF1	0.11*	0.03	0.04
	(0.06)	(0.06)	(0.06)
University Tier = 3rd Quartile × PostAF2	0.40***	0.29***	0.28***
	(0.08)	(0.09)	(0.09)
University Tier = 4th Quartile × PostAF1	0.15***	0.07	0.10
	(0.05)	(0.06)	(0.06)
University Tier = 4th Quartile × PostAF2	0.70***	0.41***	0.40***
	(0.08)	(0.09)	(0.09)
Year FE	✓	✓	✓
University FE	✓	✓	✓
Observations	5,500	5,500	5,500
R^2/Pseudo R^2	0.83069	0.93222	0.93055

(B) Structural biology publications

Dependent Variables:	Log(1 + No. of Publications)	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)
Model:	(1)	(2)	(3)
University Tier = 2nd Quartile × PostAF1	0.15*	0.12	0.09
	(0.08)	(0.08)	(0.08)
University Tier = 2nd Quartile × PostAF2	0.22*	0.13	0.21*
	(0.12)	(0.11)	(0.11)
University Tier = 3rd Quartile × PostAF1	0.18***	0.11*	0.11
	(0.07)	(0.07)	(0.07)
University Tier = 3rd Quartile × PostAF2	0.45***	0.25**	0.24**
	(0.12)	(0.11)	(0.11)
University Tier = 4th Quartile × PostAF1	0.23***	0.18***	0.14**
	(0.06)	(0.07)	(0.07)
University Tier = 4th Quartile × PostAF2	0.78***	0.34***	0.37***
	(0.11)	(0.11)	(0.11)
Year FE	✓	✓	✓
University FE	✓	✓	✓
Observations	5,500	5,500	5,500
R^2/Pseudo R^2	0.82	0.92	0.92

(C) Non-structural protein publications

Dependent Variables:	Log(1 + No. of Publications)	Market Share (% Publications, All Authors)	Market Share (% Publications, Main Authors)
Model:	(1)	(2)	(3)
University Tier = 2nd Quartile × PostAF1	0.16*	0.09	0.12
	(0.08)	(0.08)	(0.08)
University Tier = 2nd Quartile × PostAF2	0.25**	0.08	0.10
	(0.10)	(0.10)	(0.10)
University Tier = 3rd Quartile × PostAF1	0.18**	0.08	0.08
	(0.07)	(0.08)	(0.08)
University Tier = 3rd Quartile × PostAF2	0.46***	0.30***	0.29***
	(0.09)	(0.10)	(0.10)
University Tier = 4th Quartile × PostAF1	0.22***	0.10	0.16**
	(0.06)	(0.07)	(0.07)
University Tier = 4th Quartile × PostAF2	0.73***	0.46***	0.48***
	(0.08)	(0.10)	(0.10)
Year FE	✓	✓	✓
University FE	✓	✓	✓
Observations	5,500	5,500	5,500
R^2/Pseudo R^2	0.80	0.90	0.90

Table A10: **Construction of the protein-linkage sample for the three estimation panels**

This table traces each estimation panel from its top-journal papers to the proteins those papers study. *Linked to proteins* counts papers for which at least one UniProt accession was recovered. A linked paper is *Structure available* when at least half of its proteins had prior known information before the AlphaFold Protein Structure Database release on 22 July 2021, and *No prior structure* otherwise. *Distinct proteins* counts unique accessions pooled over the panel's linked papers. *Rebuild check (ρ)* is the Pearson correlation, across university-year cells, between the publication counts aggregated from the author-affiliation table used for the linkage and the counts in the estimation panel of Table 1.

Panel	Top-journal papers	Linked to proteins	Linked papers by prior structure		Distinct linked proteins	Rebuild check (ρ)
			Structure available	No prior structure		
Panel A: All protein research	10,581	2,859	1,388	1,471	48,314	0.9996
Panel B: Structural biology	5,409	1,625	813	812	43,666	0.9992
Panel C: Non-structural protein research	5,172	1,234	575	659	38,304	0.9998

Table A11: **Prior protein information available before the AlphaFold database release**

This table classifies the proteins linked to top-journal papers in Table A10 by the structural information available for them before the AlphaFold Protein Structure Database release on 22 July 2021. A protein is *Own structure* if an experimental structure of that protein was deposited in the Protein Data Bank before that date. It is *Template only* if no such structure exists but another protein with at least 30% sequence identity to it was solved before that date. It is otherwise *No prior structure* if neither holds. The three classes are mutually exclusive such that every linked protein falls into one of them. *Mean structural coverage* is the share of residues spanned by the best pre-release structure or template. *With an AlphaFold model* counts proteins for which the database provides a predicted structure. *Mean pLDDT* averages, over the proteins in the row that have an AlphaFold model, the model's mean predicted local distance difference test score across residues. The score runs from 0 to 100 and measures how reliable the predicted position of a residue is, with values above 70 conventionally read as confident.

Structural status	Proteins	Share of proteins (pct)	Mean structural coverage (pct)	Median length (aa)	With an AlphaFold model	Mean pLDDT
Own structure	8,926	18.5	100.0	428.0	8,634	79.8
Template only	7,768	16.1	60.9	494.0	7,183	76.6
No prior structure	31,620	65.4	2.0	428.0	27,945	80.0
Total	48,314	100.0	—	—	43,762	—

Table A12: **Protein-linkage coverage by university quartile and by period**

This table reports how often the protein linkage of Table A9 succeeds, separately by the pre-AlphaFold publication-share quartile of the university. The unit is the paper–university pair, so a paper co-authored across several universities enters once for each. *Linked to protein* counts pairs whose paper yielded at least one UniProt accession, and *Coverage* is that count as a share of all pairs in the quartile. The final three columns give the same share within each period, where *Pre-AlphaFold* covers 2013 to 2018, *PostAF1* runs from the December 2018 CASP13 result to the July 2021 open release of AlphaFold 2's code and database, and *PostAF2* covers the period after that open release. Coverage varies by 2.3 percentage points across quartiles, which bounds the extent to which linkage success tracks university rank.

Quartile	Paper-university pairs	Linked to Protein	Coverage (%)	Pre-AlphaFold (%)	PostAF1 (%)	PostAF2 (%)
1	5509	1513	27.5	28.3	25.8	25.4
2	6459	1709	26.5	27.9	23.9	23.2
3	7256	1949	26.9	28.1	26.1	22.4
4	8941	2255	25.2	26.5	22.4	23.3

Table A13: **Heterogeneity by prior structural-biology authorship**

No structural biologist pre-AF is an indicator that the university had no author whose departmental affiliation is in structural biology, computational biology, or biophysics on any of its top-journal protein papers published before AlphaFold 1. Each column adds a triple interaction between quartile, period, and that indicator to the corresponding stratum-specific specification. A positive triple means that the lower-quartile differential was larger where no such author had previously been observed.

Dependent Variable:	Log(1 + No. of publications)					
	Structure available			No prior structure		
	All protein research	Structural biology	Non-structural	All protein research	Structural biology	Non-structural
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
2nd Quartile × PostAF1 × No structural biologist pre-AF	0.0907	0.1	0.11	-0.06	0.04	0.05
	(0.16)	(0.21)	(0.15)	(0.21)	(0.22)	(0.14)
3rd Quartile × PostAF1 × No structural biologist pre-AF	0.04	0.04	0.05	-0.07	-0.04	0.00
	(0.09)	(0.08)	(0.08)	(0.09)	(0.09)	(0.06)
4th Quartile × PostAF1 × No structural biologist pre-AF	0.06**	0.05**	0.03	0.04	0.04	0.05
	(0.02)	(0.02)	(0.04)	(0.03)	(0.03)	(0.05)
2nd Quartile × PostAF2 × No structural biologist pre-AF	0.5***	0.52***	0.33***	0.11	0.18	0.06
	(0.1)	(0.13)	(0.11)	(0.11)	(0.15)	(0.11)
3rd Quartile × PostAF2 × No structural biologist pre-AF	0.30***	0.29***	0.1	0.17**	0.18**	0.2*
	(0.08)	(0.07)	(0.07)	(0.07)	(0.08)	(0.06)
4th Quartile × PostAF2 × No structural biologist pre-AF	0.13***	0.12***	0.12***	0.1***	0.10***	0.10***
	(0.02)	(0.02)	(0.04)	(0.02)	(0.02)	(0.04)
Fixed-effects						
University	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	5,500	5,500	5,500	5,500	5,500	5,500
Adjusted R^2	0.60	0.59	0.51	0.6	0.58	0.53

Table A14: **Changes in structural-biologist authorship by university quartile**

The dependent variable is an indicator taking the value one if a top-journal publication has at least one structural biologist among its authors. A negative estimate means that universities from a given quartile are less likely to include a structural biologist on their top-journal publications than first quartile; relative to the pre-AF gap with the top quartile.

Dependent Variable:	Paper has a structural biologist		
	All protein research	Structural biology	Non-structural
Model:	(1)	(2)	(3)
Variables			
2nd Quartile × PostAF1	0.02	-0.03	0.03
	(0.04)	(0.05)	(0.04)
3rd Quartile × PostAF1	-0.02	-0.05	0.03
	(0.04)	(0.05)	(0.04)
4th Quartile × PostAF1	-0.02	-0.02	0.06
	(0.05)	(0.06)	(0.05)
2nd Quartile × PostAF2	-0.05	-0.02	-0.00
	(0.05)	(0.05)	(0.04)
3rd Quartile × PostAF2	-0.08	-0.09	-0.05
	(0.05)	(0.06)	(0.04)
4th Quartile × PostAF2	-0.10**	-0.09*	-0.07*
	(0.05)	(0.05)	(0.04)
Fixed-effects			
University	Yes	Yes	Yes
Year	Yes	Yes	Yes
Fit statistics			
Observations	7,373	6,103	4,537
Adjusted R^2	0.11	0.12	0.12

B Sample Selection and Methods

B.1 Publications

Data on publications are obtained from OpenAlex.[18] Each publication is assigned a unique identifier, and its metadata include the title of the article, the abstract, citation counts, the list of authors with their institutional affiliations and departmental affiliation, and the range of topics covered. We use OpenAlex rather than Scopus or Web of Science because it is the only source that combines (i) free, fully snapshotted access ensuring reproducibility, (ii) coverage of preprints, which is material in the post-AlphaFold period, (iii) author and institution disambiguation, and (iv) topic classifications required for our domain definitions. Direct comparisons of OpenAlex with the proprietary databases have since been published: Alperin et al. (2024) find that the OpenAlex record set contains that of Scopus in full, while Culbert et al. (2025) report reference coverage for 2015–22 publications comparable to that of Scopus and Web of Science.

B.2 Journals

We select journals in 'biochemistry, genetics, molecular biology' (2,170 journals) and 'Multidisciplinary' (175 journals), and label those with an average SJR greater than 10 between 2013 and 2018 as 'top journals' (there are 17 top journals in total, as listed in Table A2). Figure A4 shows that the number of publications in the 17 top journals is relatively stable in our sample period. We split the articles published in these journals into three groups labeled as follows: i) 'Structural biology', ii) 'Non-structural protein', and iii) 'Non-protein.' A paper is classified as structural biology if it has an author who is a structural biologist or if the paper's list of topics includes one of the 238 topics we identify as structural biology. If a paper does not satisfy this condition but has one of the 1,066 topics of protein research, then it falls in the second group. All other papers are assigned to the third group.

We measure journal quality using the SCImago Journal Rank (SJR), a prestige metric that weights citations according to the importance of the citing journals. Computed from Scopus data, SJR incorporates both citation volume and the prestige of citing sources to identify the most influential journals. As the SJR of each journal varies from year to year,

[18] OpenAlex is maintained by the nonprofit OurResearch, which launched it in January 2022 as a fully open successor to the Microsoft Academic Graph after Microsoft stopped updating that database at the end of 2021. The catalog aggregates bibliographic metadata from Crossref, ORCID, PubMed, and open repositories such as arXiv and Zenodo, which gives it coverage of preprints alongside published articles. OpenAlex provides API access and full database snapshots for large-scale analysis.

we rank the journals using the average SJR during the six-year period preceding the release of AlphaFold (i.e., 2013–2018).

We choose SJR over alternatives such as the Journal Impact Factor (JIF) for four reasons. First, SJR is a prestige-weighted measure (PageRank-style on the Scopus citation graph), so it captures journal standing rather than raw citation volume and is less sensitive to mechanical inflation from review articles or self-citation. Second, SJR is field-normalized, which permits us to pool biochemistry/molecular-biology and multidisciplinary outlets on a single comparable scale. Third, SJR is computed from Scopus, with publicly documented methodology, and is freely accessible, ensuring reproducibility (JIF is paywalled and proprietary). Fourth, averaging over the six-year pre-AlphaFold window (2013–2018) removes year-to-year noise and avoids any post-treatment contamination of the journal set. The resulting top-17 set is dominated by Nature, Science, Cell, and other outlets that are uncontroversially 'top' under any standard alternative metric (JIF, h5-index, expert consensus), so the choice of SJR is unlikely to drive our results.

B.3 University Rankings

The center for Science and Technology Studies (CWTS) at Leiden University produces the *Total Normalized Citation Score* (TNCS) of about one thousand universities around the world. We compute the average TNCS score of each university between 2013 and 2018 (the pre-AlphaFold period), in the category "biomedical and health sciences".[19]

We rely on the CWTS Leiden Ranking for institutional stratification for several reasons. First, it is produced by an independent academic center with a publicly documented methodology and is the standard reference in the scientometrics literature. Second, the TNCS is normalized at the subfield level, so universities are not penalized for working in subfields with structurally lower citation rates. Third, the TNCS is size-dependent (MNCS × #pubs), which matches what we want to capture: an institution's *capacity* to produce top-journal output, not its per-paper average. Fourth, restricting to "biomedical and health sciences" aligns the ranking with the protein-research scope of our analysis. Fifth, averaging over the six-year pre-AlphaFold window yields a fixed, time-invariant ranking, which rules out any mechanical sorting of universities in or out of quartiles

[19] CWTS computes the TNCS annually for each university as follows: First, each publication is assigned to a biomedical subfield. Then, the citation count for each publication is compared to the average in the subfield in that year, resulting in a normalized citation score, where 1 represents the overall average. The *Mean Normalized Citation Score* (MNCS) is then calculated by averaging these normalized scores across all the publications in the university. Finally, the TNCS is obtained by multiplying the MNCS by the total number of publications, providing a size-dependent indicator of the university's overall citations. This method thus accounts for differences in citation practices between subfields and across years.

caused by AlphaFold itself. Using the TNCS ranking first, rather than journal market share, ensures that our analysis based on top-journal share also aligns with more traditional university rankings. As a robustness check, we also rank universities by their pre-AlphaFold top-journal market share; results are similar.

We match authors of each publication in our sample to universities based on their affiliation. In our data, some authors are not affiliated with universities, but instead with research centers, private companies, or hospitals. These researchers are dropped from the sample.

B.4 Measuring Research Output

We define the annual publication count for university u in year t as the number of protein research articles published in our selected top journals with at least one author affiliated with university u.

Let

$$\mathcal{P}_t = \{i : \text{article } i \text{ is published in a top journal in year } t\},$$

and for each article i, let

$$A_i = \{a_{i,1}, a_{i,2}, \ldots, a_{i,n_i}\}$$

denote its ordered set of authors. Denote by $u(a)$ the university assigned to author a (i.e., the highest ranked among any multiple affiliations).

Then, the publication counts are defined as follows:

1. All authors count:

$$P_{u,t}^{all} = \sum_{i \in \mathcal{P}_t} \mathbf{1}\Big\{\exists\, k \in \{1, \ldots, n_i\} : u(a_{i,k}) = u\Big\},$$

2. Main authors count:

$$P_{u,t}^{main} = \sum_{i \in \mathcal{P}_t} \mathbf{1}\Big\{u(a_{i,1}) = u \quad \text{or} \quad u(a_{i,n_i}) = u\Big\}.$$

These definitions ensure that each article contributes at most one count per university, regardless of the number of affiliated authors.

B.5 Additional Data

We construct several author- and university-level covariates to use as controls in our regressions.

OpenAlex does not track demographic characteristics or detailed expertise fields of authors. We therefore infer authors' ethnicities using the *rethnicity* package in R, which predicts ethnicity from first names via probabilistic models trained on demographic data. Gender is determined using the *genderizeR* package in R, based on first names. Finally, we classify authors as structural biologists if their primary institutional affiliation is in a department specializing in structural biology, computational biology, biophysics, or closely related fields (e.g., department names containing keywords such as "structural biology," "computational biology," "biophysics," or "protein folding").

To examine heterogeneity related to pre-existing computational resources (a key institutional barrier to conducting protein research that AlphaFold may substitute for), we incorporate data on university-level access to high-performance computing (HPC). HPC plays a pivotal role in modern protein research, enabling complex simulations, large-scale data analyses, and computationally intensive methods such as those traditionally employed in structural biology. Access to HPC resources is crucial for improving researchers' productivity and enabling the exploration of a broader range of research topics (McKeon and Boudou, 2024). We use data from Top500.org as a proxy for the computational power available at academic institutions. Since 1993, Top500.org has systematically tracked HPC facilities at universities worldwide on a biannual basis, providing detailed metrics on total processing power (quantified by the number of processor cores) and computational speed (measured in petaflops per second). This dataset allows us to assess the link between a university's HPC capabilities and its research output.

B.6 Empirical Design

We rank universities by their cumulative share of protein publications in our list of top journals during the pre-AlphaFold period and divide them into quartiles, such that each quartile accounts for approximately 25% of all such publications during this period. The top quartile comprises the top 12 universities. The second quartile includes the next 36 universities, the third quartile the next 88, and the bottom quartile comprises the bottom 364.

In our main regression specification outlined in equation 4, we include university-level controls for faculty size, author demographics, structural biology expertise, institutional type, and geographic location that vary over time (summarized in Table A3) to account

for changes in author composition over time. The dependent variable y_{ut} is one of three variables:

1. The log number of publications in top journals from university u at time t: $y_{ut} = log(1 + P^{all}_{u,t})$
2. The market share of university u at time t: $y_{ut} = MktS^{all}_{ut}$
3. The market share of university u at time t taking into account only main authors is: $y_{ut} = MktS^{main}_{u,t}$

B.7 Measures of Research Direction and Impact

We employ three metrics to quantify changes in research direction: *pivot score*, *novelty*, and *disruptiveness*. These measures capture distinct aspects of topical focus, research novelty, and impact, allowing us to evaluate how AlphaFold influenced the direction and originality of protein research across university quartiles.

Pivot Score

The *pivot score* measures the extent to which a researcher's new work diverges from their previous research portfolio. We compute this score using cosine similarity between the distribution of journals cited in a new paper and those cited within the researcher's prior work.

$$\Phi^j_i = 1 - \frac{R^j_i \cdot R_i}{||R^j_i||||R_i||} \tag{5}$$

where R^j_i is a vector representing citation frequencies to journals in focal paper j by researcher i, and R_i aggregates citation frequencies to journals across all prior papers by the same researcher i. The score ranges from 0 (no departure from prior work) to 1 (complete shift in research direction).

Novelty

Novelty quantifies a paper's originality by assessing the rarity of concepts in its references, drawing on OpenAlex's concept classifications. OpenAlex tags each paper with a list of concepts, assigning each a relevance score (ranging from 0 to 1) that indicates the concept's

centrality to the paper. Each concept also includes a works count reflecting its overall prevalence in the literature, which we use to measure rarity in our novelty metric.[20]

$$N = \frac{1}{|C|} \sum_{c \in C} \frac{1}{w_c} \tag{6}$$

where C is the set of unique concepts from the paper's references with relevance score > 0.5, and w_c is the works count of concept c. The 0.5 threshold ensures only concepts meaningfully connected to the references contribute to the score. The inverse weighting $\frac{1}{w_c}$ assigns greater importance to rarer concepts, so papers referencing niche topics with low works counts (e.g., 50 papers) receive higher novelty scores than those relying on widely studied topics (e.g., 10,000 papers). For example, a paper on neural networks might have high relevance for "machine learning" (relevance = 0.9) but low relevance for "agriculture" (relevance = 0.2), with only the former contributing to the novelty calculation.

Disruptiveness

We measure a paper's *disruptiveness* using the "Disruption Index", which quantifies the extent to which a paper disrupts its field by analyzing citation patterns. Higher values indicate that subsequent works cite the focal paper while largely ignoring its references (signaling the creation of new research directions), whereas low values reflect consolidation of existing lines of inquiry.

$$\text{DI} = \frac{i_{\text{est}} - j_{\text{est}}}{i_{\text{est}} + j_{\text{est}} + k_{\text{est}}} \tag{7}$$

where i_{est} represents papers citing only the focal paper (indicating standalone impact), j_{est} represents papers citing only its references (reflecting influence of prior work), and k_{est} represents papers citing both (showing overlapping influence). These estimates are calculated from: (i) A, the focal paper's citation count; (ii) B, the number of papers citing at least one reference; and (iii) p_k, the proportion of sampled citing papers that also cite a reference. The estimates follow: $i_{\text{est}} = (1 - p_k) \cdot A$, $k_{\text{est}} = p_k \cdot A$, and $j_{\text{est}} = B - k_{\text{est}}$.

The Disruption Index ranges from -1 to 1. Positive values indicate disruptive papers that attract citations largely independent of their references (thus overshadowing prior work), while negative values denote consolidating papers whose citations tend to include both the focal paper and its predecessors. Zero values represent neutral impact.

[20]The relevance score is computed by OpenAlex using machine learning techniques applied to metadata such as abstracts, titles, and citation patterns. Scores above 0.5 indicate concepts that are significant descriptors of the paper's content rather than peripheral topics.

Scaling of measures for comparability

To ensure temporal comparability and account for evolving trends in protein research, such as shifts in concept prevalence, citation patterns, and publication volumes, we normalize novelty and disruptiveness measures quarterly across all universities as follows:

$$\text{Scaled Measure} = \frac{\text{Measure} - \min(\text{Measure})}{\max(\text{Measure}) - \min(\text{Measure})}$$

Here, min(Measure) and max(Measure) denote the minimum and maximum values of the given metric observed across all universities in a given quarter. This min-max normalization rescales the raw measures to the $[0, 1]$ interval within each quarter. The pivot score, being derived from cosine similarity, is inherently normalized and requires no further scaling.

B.8 Sample Construction for Placebo Analysis

To replicate our analysis in other fields and confirm whether the observed effects are specific to AlphaFold or part of broader trends in academic publishing or concurrent AI advancements, we select four additional high-skill research domains: microbiology (excluding protein research), economics and finance, materials science, and social sciences. These fields exhibit similar hierarchies in research output but lacked a transformative open-access AI tool equivalent to AlphaFold during the study period. The data collection procedures mirror those used for protein research, with field-specific adjustments as needed.

Data sources

As with our protein research analysis, we draw on three primary data sources:

1. **Journal rankings**: Journal quality is measured using the SCImago Journal Rank (SJR), which accounts for both citation volume and the prestige of citing journals. We collect yearly SJR data from 2013 to 2024 for journals in each field. Since 2025 SJR rankings were not available at the time of analysis, we use the 2024 rankings for publications in 2025.

2. **Publications**: We obtain publication metadata including article titles, publication dates, author information, institutional affiliations, citation data, and topic classifications from OpenAlex. Publications are extracted separately from OpenAlex for the top journals in each field, ranked by their annual SJR as determined in Step 1.

3. **University rankings**: We construct university rankings using the Total Normalized Citation Score (*TNCS*). The rankings cover approximately 1,000 universities worldwide and are reported annually across multiple academic fields.

Journals selection

For each field, we identify top journals using SCImago's subject category classifications and rank them by their SJR scores for each year, selecting those with the highest values.

- **Microbiology excluding protein research**: Publications classified as microbiology after excluding protein research, using the same top-journal and university-quartile construction as the main analysis.
- **Materials science**: Journals classified under "Materials Science" subject areas in SCImago, with an SJR threshold of 10 or higher, consistent with our protein research sample.
- **Economics**: Journals classified under the "Economics, Econometrics and Finance" subject areas in SCImago. We apply an SJR threshold of 10 or higher to identify top journals.
- **Social sciences**: Journals classified under the "Social Sciences" category in SCImago. To account for lower typical citation rates in the social sciences, which result in generally lower SJR scores compared to fields like protein or materials science, we select journals with an SJR of 8 or higher to capture a comparable set of high-impact outlets.

We compute journal rankings annually based on their SJR scores, with journals ranked within each year using dense ranking (ties receive the same rank). To maintain field specificity, we exclude publications from general interest or review-oriented journals (e.g., *Journal of Economic Perspectives*, *Annual Review of Economics* and *Journal of Economic Literature* in economics). We also exclude journals primarily focused on adjacent disciplines, even if they appear in the field's SCImago category. For example, in the social sciences category, we exclude the *Review of Economics and Statistics*, which is a leading economics journal. The resulting selection of top journals for these fields, along with their annual publication volumes over time, is presented in Figure A12.

University rankings

These are constructed using field-specific TNCS scores from the CWTS Leiden Rankings:

- For materials science, we use the "Physical sciences and engineering" category.
- For economics and social sciences, we use the "Social sciences and humanities" category.

This ranking employs full-counting methodology, which assigns a full weight of one to each publication regardless of the number of co-authoring institutions. The distribution of top-journal publications based on our ranking scheme can be found in Figure A13. The cumulative distribution of top-journal publications across university ranks in these fields, based on our pre-AlphaFold ranking scheme, is illustrated in Figure A13. The figure reveals persistent stratification in publication shares across the top 500 universities in these fields, underscoring that lower-ranked universities do not hold significant publication shares in top journals. This concentration aligns with the Matthew effect observed in protein research prior to AlphaFold's release.

As in protein research, we fix university quartiles using pre-AlphaFold top-journal publication shares and examine how the shares of these fixed groups evolve in the post-AlphaFold period[21].

Author-institution matching

For each publication in the sample, we extract author-level information including institutional affiliations from OpenAlex. Authors with multiple affiliations are assigned to the highest-ranked institution among their affiliations based on the fixed, field-specific pre-AlphaFold TNCS ranking. This ensures each author-publication observation is associated with a single university quartile.

Sample coverage

The final samples span publications from 2013 to 2025 for authors affiliated with the top 500 universities in each field. Publications are restricted to those appearing in journals with an SJR at or above the field-specific threshold throughout the sample period. The

[21] One might view the public release of ChatGPT in November 2022 as a potential discrete shock for fields such as economics or social sciences. If ChatGPT had triggered significant shifts in university publication shares in top journals, particularly gains by lower-quartile universities, these changes would be visible in our placebo analysis because universities remain assigned to the quartile determined before AlphaFold . Post-release publication shares are therefore not mechanically reclassified using later rankings.

distribution of publications across university quartiles is computed annually, enabling the construction of market share time series comparable to those in our main analysis on protein research.

Research quality metrics

To investigate whether the shifts in research practices observed in protein research also occurred in non-protein fields, we compute the same three metrics *pivot score*, *novelty*, and *disruptiveness* for publications in each field. All these metrics are computed using identical methodologies to those employed for protein research (see Section B.7 for detailed definitions).

C Identifying the Proteins Studied by Each Publication

The main results compare post-AlphaFold changes in top-journal protein publication across university quartiles. This section supplements them by splitting the sample according to what was already known pre-AlphaFold about the proteins each paper studies. A protein has prior structural information if, before the AlphaFold database release on 22 July 2021, it had its own experimental structure or at least one usable homology template.[22] A paper is deemed to have prior protein structure available when at least half of its linked proteins meet this condition, and no prior protein information otherwise. We ask whether the post-AlphaFold changes across quartiles differ between these two kinds of papers.

Both definitions presume a known set of proteins for each paper, which publications may or may not report. A paper may study a single protein, several, or none, and the proteins it studies are named in prose rather than recorded in any field of its bibliographic record. Recovering that set is the substantive task of this section, which describes how we link publications to proteins, construct the paper-level measure of prior protein information, and merge it into the estimation sample.

Formally, let $\mathcal{D}$ denote the set of publications in the sample. Proteins are identified by their accession number in UniProtKB, the reference database of protein sequences and annotations. We use only the human-reviewed section of UniProtKB, known as Swiss-Prot, for our purposes. Let $\mathcal{U}$ denote the set of these reviewed accessions. The first object we construct is a correspondence:

$$\mathcal{P} : \mathcal{D} \longrightarrow 2^{\mathcal{U}}, \qquad p \longmapsto \mathcal{P}(p) \subseteq \mathcal{U}, \tag{8}$$

assigning to each paper p the (possibly empty) set of proteins it studies. The main difficulty here is that publications do not report accession numbers in any systematic machine-readable form. Proteins can be referred to by gene symbols, common names, synonyms, and abbreviations, and the same string frequently denotes different proteins in different organisms. We therefore identify proteins through three complementary channels—curator-linked literature, machine-normalized entity annotations, and an ensemble of named-entity recognition models applied to abstracts—and take their union. The channels differ in what they require of a paper and in how directly they establish a link, with each one linking papers to proteins in unique ways. Coverage matters directly for the analysis. A paper that cannot be linked to any protein drops out of every protein-level specification.

[22] Proteins with similar sequences fold into similar shapes. A homology template is an experimentally solved structure of a related protein that can stand in for the target, and the percentage sequence identity between the two represents how far it can be trusted. We treat a template as usable at 30 percent identity or above, the conventional threshold below which a template no longer supports molecular replacement.

Unlinked papers therefore shrink the estimation sample rather than merely adding noise. We therefore treat coverage as an object to be documented rather than assumed, and report the corresponding diagnostics.

Identifier resolution in publications

We begin from the set of OpenAlex works published in the 17 top journals identified in Section 2 and listed in Table A2. For each work we use its external identifiers (PubMed identifier and DOI) and its abstract. Almost all publications in our sample carry a PubMed identifier, so the two identifier-based channels described below apply to nearly the entire sample; the third channel does not require one.

An important point to note is that OpenAlex does not distribute abstracts as running text. It stores an inverted index that maps each token to the positions at which the token occurs. Writing $\mathcal{I}_p = \{(w, T_w^p)\}_{w \in V_p}$, where V_p is the vocabulary of the abstract of paper p and $T_w^p \subseteq \{1, \ldots, n_p\}$ is the set of positions at which token w appears, the abstract is recovered as the sequence $a_p = \big(a_p(1), \ldots, a_p(n_p)\big)$ defined by

$$a_p(t) = w \iff t \in T_w^p, \qquad \text{with} \qquad n_p = \sum_{w \in V_p} \big|T_w^p\big|. \tag{9}$$

Before performing entity recognition we normalize the reconstructed text by converting to lower-case, collapsing repeated whitespaces, and stripping non-alphanumeric characters. We denote the normalized abstract by $\tilde{a}_p$.

The protein universe

The "universe" of proteins $\mathcal{U}$ is the manually reviewed section of UniProtKB, known as Swiss-Prot, which contains just over 570,000 entries curated manually by expert biocurators (The UniProt Consortium, 2025). Each of its entries describes a single protein and is labeled by a unique accession number. The terms $entry$, $accession$ and $protein$ therefore refer to the same object throughout this section. We exclude TrEMBL, the computationally annotated section, whose entries have not been reviewed. Restricting attention to human-reviewed entries is consequential for the linkage. It keeps the set of candidate proteins fixed and biologically meaningful, and prevents a single ambiguous name from matching thousands of unreviewed sequence records.

Rather than querying the UniProt API once per paper or once per protein mention, we download the entire reviewed index once through the bulk stream endpoint `https://rest.uniprot.org/uniprotkb/stream?query=reviewed:true&format=tsv`, with the *fields* parameter

Table C1: Fields retrieved from the reviewed UniProtKB index

Field	Content	Role in the analysis
Accession	Primary UniProt identifier	Unit of the protein-level data
Gene name	Primary gene symbol	Matching; reporting
Protein name	Recommended full name	Matching; reporting
Organism	Source species	Disambiguation across species
Sequence length	Number of amino acids	Protein characteristics
Molecular mass	Mass in daltons	Protein characteristics
First public date	Date the entry entered UniProtKB	Entry age
PDB cross-references	Accessions of linked PDB entries	Key to PDB entries
AlphaFold DB link	Linked predicted structure	Availability of a prediction
NCBI Gene identifier	Cross-reference to NCBI Gene	Channel 2 linkage key
Cited PubMed IDs	Curated literature references	Channel 1 linkage key

Notes. The *fields* parameter of the stream request is set to *accession*, *gene_primary*, *protein_name*, *organism_name*, *length*, *mass*, *date_created*, *xref_pdb*, and *xref_alphafolddb*; *xref_geneid* and *lit_pubmed_id* are retrieved in two further streams. Cross-references and cited PubMed identifiers are multi-valued and are expanded into separate rows in the cross-reference table described in Section C. PDB cross-references carry no dates: deposit and release dates are obtained from the Protein Data Bank, and candidate homology templates with their sequence identities from SWISS-MODEL and UniProt.

set to the columns listed in Table C1. Two further streams are run against the same endpoint with *fields=accession,xref_geneid* and *fields=accession,lit_pubmed_id*. The first returns each accession's cross-references to NCBI Gene, the gene catalog maintained by the National Center for Biotechnology Information. Those cross-references are the linkage key for Channel 2 below. The second returns the PubMed identifiers cited by the curated entry, which are the linkage key for Channel 1. Both fields are multi-valued, which is why each is retrieved on its own and expanded to one row per pair. The three streams were downloaded in May 2026, and every step described below runs against that stored snapshot.

Linkage channels

Channel 1: curator-linked literature. Each reviewed entry carries a list of publications a curator read and used as evidence when annotating the protein. Inverting this list gives, for each paper, the proteins whose entries cite it. Candidate papers are screened on three criteria: that the protein is a central object of the study rather than a passing mention, that the findings reported are experimental rather than computationally inferred, and that they are not already recorded in the entry from an earlier paper. Reviews are excluded, since they report no new experimental evidence.

Let L_i denote the set of PubMed identifiers cited by the curated entry of protein i, and let q_p denote the PubMed identifier of paper p. The curator-linked channel assigns to p every protein whose entry cites it

$$\mathcal{P}^{\text{cur}}(p) = \left\{ i \in \mathcal{U} \ : \ q_p \in L_i \right\}, \tag{10}$$

This is the most precise of the three channels. A link exists only because a curator has judged that the paper reports experimental evidence about that protein. Its recall is bounded by two features of curation. The first is selectivity: an entry cites the papers that established its annotations, not every paper that studies the protein. The second is timing. A paper enters L_i only once a curator has processed it, and the resulting lag falls hardest on the most recent publications, consisting mainly of those in the post-AlphaFold window.

Channel 2: normalized gene annotations. A curator-linked citation is the most direct evidence that a paper studies a protein, but it exists only where a biocurator has read the paper. Curation may also lag the publication by a long period. The AlphaFold database was released in July 2021, meaning the papers on which the analysis depends are precisely those least likely to have been curated. Coverage from Channel 1 alone would therefore drop significantly in the post-AlphaFold period, with the resulting measurement error correlated with the timing that identifies the effect rather than averaging out across periods.

The second channel is used to overcome this issue. It draws on PubTator3, a resource maintained by NCBI that locates gene and protein names in text and maps them to NCBI Gene identifiers, applied to roughly 36 million PubMed abstracts and 6 million open-access full texts in PubMed Central and refreshed weekly (Wei et al., 2024). Two features here suit our purpose very well. First, annotation is applied to a huge corpus of publications rather than a small curated subset, a weekly refresh rate which places recent papers on the same footing as older ones. Second, PubTator3 reports each mention as an NCBI Gene identifier rather than as the string it found in the text. Every reviewed entry records its own NCBI Gene cross-references. Mapping an annotated gene into $\mathcal{U}$ is therefore an exact lookup, not a match on names and synonyms, and the homonymy and cross-species ambiguity that affect string matching quality do not arise. For papers with an open-access full text in PubMed Central, the procedure draws on the body of the paper rather than the abstract alone.

Let G_p denote the set of gene identifiers PubTator3 annotates in paper p, and let $A(g) \subseteq \mathcal{U}$ denote the reviewed entries cross-referenced to gene g, read off the accession-to-gene cross-references downloaded with the index. The annotation channel assigns to p every protein cross-referenced to a gene as:

$$\mathcal{P}^{\text{tm}}(p) = \bigcup_{g \in G_p} A(g). \tag{11}$$

This matching comes at the cost of over-inclusion. A gene mentioned in passing enters G_p on an unweighted basis alongside a focal gene, making $\mathcal{P}^{\text{tm}}(p)$ a superset of the proteins

that paper p actually studies. The paper-level share of linked proteins with prior structural information, is calculated as a sample average over $\mathcal{P}^{\text{tm}}(p)$. Including unstudied proteins in this set pulls the paper's average toward the unconditional sample mean. Because a paper is assigned to the structure-available stratum when this share reaches 0.5, over-inclusion introduces classification noise near the cutoff, attenuating the observed differential and rendering our estimates of the stratum gap conservative.

Channel 3: entity recognition over abstracts. The first two channels yield a non-empty $\mathcal{P}^{\text{cur}}(p)$ only for papers a curator has cited, and a non-empty $\mathcal{P}^{\text{tm}}(p)$ only for papers PubTator3 has annotated. The third channel described here places no such requirement on the paper. It covers papers without a PubMed identifier, as well as papers that PubMed indexes but that no database has linked to a protein. Instead, the entity recognition approach identifies candidate proteins from the abstract by applying an ensemble of K biomedical named-entity recognition models. These models are distributed by the *scispaCy* repository, and are applied to the normalized abstract $\tilde{a}_p$ (Neumann et al., 2019). The models used are listed in Table C2. Each is trained on a different annotated corpus and labels a different set of entity types. For instance, the model trained on the BioNLP 2013 Cancer Genetics corpus labels protein families and protein domains in addition to gene products, whereas the model trained on JNLPBA labels proteins and genes alone. A mention labeled by one model is therefore not necessarily labeled by another, and the union defined in Equation (12) may sometimes retain more mentions than any single model returns.

Let $\mathcal{E}_k(\cdot)$ denote the set of labeled spans returned by model k and $\mathcal{L}$ the set of protein-relevant labels. The candidate mention set is defined as:

$$M_p = \bigcup_{k=1}^{K} \left\{ s \ : \ (s, \ell) \in \mathcal{E}_k(\tilde{a}_p), \ \ell \in \mathcal{L} \right\}. \tag{12}$$

Each mention s is matched against the recommended names, gene symbols, and synonyms recorded in the reviewed index, and the entry with the highest match score is retained as the protein the mention denotes:

$$\rho(s) = \arg\max_{i \in \mathcal{U}} \ \omega(s, i), \qquad \mathcal{P}^{\text{ner}}(p) = \left\{ \rho(s) \ : \ s \in M_p \text{ with a match} \right\}, \tag{13}$$

where $\omega(s, i)$ is the UniProtKB relevance score of entry i for query string s and only the best-ranked entry is retained. Two restrictions in this step matter a great deal in practice. Limiting the search to reviewed entries and keeping a single best match prevents one

Table C2: Named-entity recognition models and retained labels

Model	Annotation corpus	Retained entity labels
`en_ner_bionlp13cg_md`	BioNLP 2013 Cancer Genetics	GENE_OR_GENE_PRODUCT, AMINO_ACID, PROTEIN_FAMILY_OR_GROUP, PROTEIN_DOMAIN_OR_REGION
`en_ner_jnlpba_md`	JNLPBA molecular biology abstracts	PROTEIN, GENE
`en_ner_craft_md`	CRAFT full-text biomedical articles	GGP (gene or gene product)
`en_ner_bc5cdr_md`	BioCreative V chemical–disease	none retained
`en_core_sci_scibert`	Scientific text (SciBERT encoder)	general entity spans

Notes. The ensemble is applied to the normalized abstract and the retained spans are unioned across models. The chemical–disease model contributes no labels to $\mathcal{L}$: its spans identify chemicals and diseases, which are excluded from the candidate set rather than resolved against UniProtKB. Retained spans are subsequently resolved by lexical search restricted to reviewed entries, keeping the best-ranked match per mention.

mention from generating links to the counterpart of the same protein in every species that has one, which would inflate $|\mathcal{P}(p)|$ without adding information. Additionally, restricting the retained labels to $\mathcal{L}$ serves the same purpose. The models also detect cell types, chemicals, and organism names, which are discarded rather than resolved against the reviewed index. The entity-recognition channel is used because any link it establishes rests on a name appearing in an abstract rather than on a curator's judgment or a normalized annotation. It is the only channel that does not require the paper to have been indexed and annotated elsewhere, and can therefore link a paper to one or more proteins that the other two methods may have left unlinked.

Paper p is linked to protein i if at least one of the three channels assigns i to p:

$$\mathcal{P}(p) = \mathcal{P}^{\text{cur}}(p) \;\cup\; \mathcal{P}^{\text{tm}}(p) \;\cup\; \mathcal{P}^{\text{ner}}(p), \tag{14}$$

A paper linked to the same protein by more than one channel contributes only one unique (paper, protein) pair. For any subset of channels C, define the coverage rate:

$$c(C) = \frac{1}{|\mathcal{D}|} \sum_{p \in \mathcal{D}} \mathbf{1}\left\{ \bigcup_{c \in C} \mathcal{P}^{c}(p) \neq \emptyset \right\}, \tag{15}$$

the share of publications linked to at least one protein. Because (14) is a union, coverage is monotone in the set of channels. The incremental contribution of the k-th channel,

$$\Delta_k = c\big(\{1, \ldots, k\}\big) - c\big(\{1, \ldots, k-1\}\big) \;\geq\; 0, \tag{16}$$

is weakly positive by construction. Taking the channels in the order above, each Δ_k is strictly positive in our data: every channel links papers that the preceding ones leave

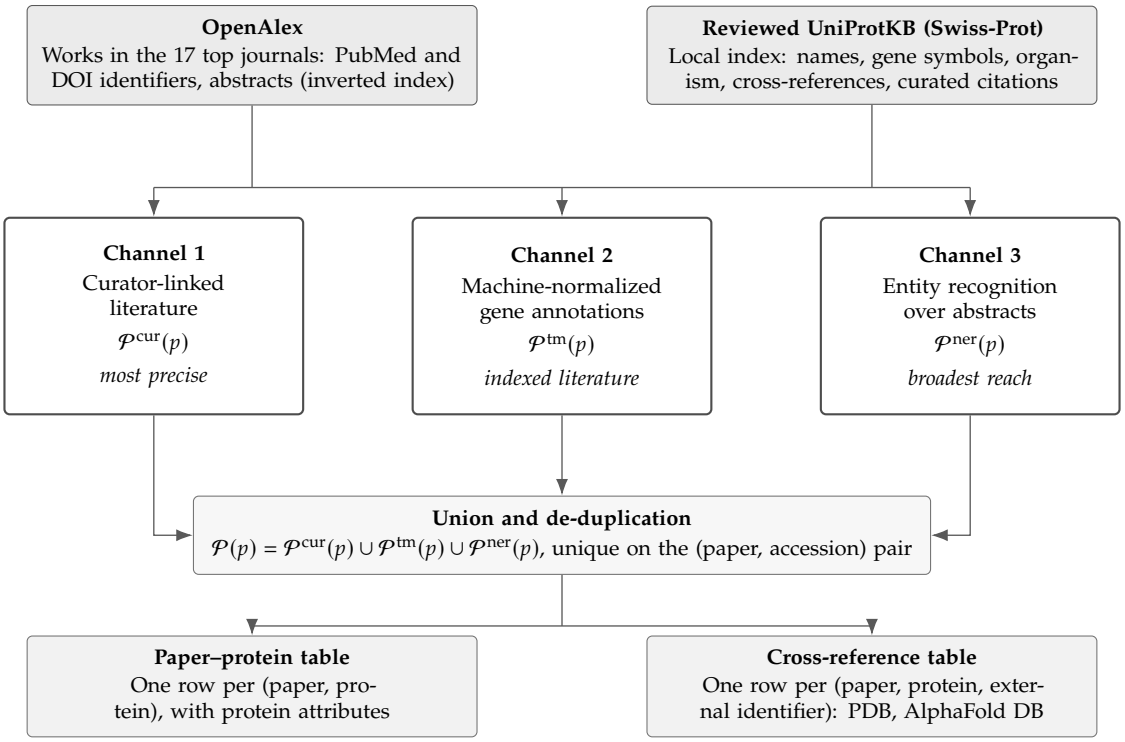


Figure C1: Construction of the publication–protein linkage

Notes. Both data sources feed all three linkage channels. Channels are ordered from the most precise (curator-linked citations, in which a biocurator has judged that the paper provides evidence about the protein) to the most permissive (named-entity recognition over abstracts, resolved lexically against reviewed UniProtKB entries). A paper is linked to a protein if *any* channel establishes the link, so coverage is weakly increasing in the set of channels used.

unlinked. We therefore run all three rather than restricting the linkage to curator-established links.

Output tables and join keys

Figure C1 summarizes the paper–protein linkage procedure using the three channels described above. The procedure produces two tables. The first records one row per paper-protein pair, together with the protein's attributes: accession, gene symbol, recommended name, date of first public release, sequence length, and molecular mass. The second records one row per paper-protein-external identifier tuple, covering cross-references to the Protein Data Bank (PDB) and to the AlphaFold Protein Structure Database. The second table is the input to the treatment: the PDB cross-references the table contains supply the deposit and release dates from which the pre-AlphaFold structural classification in (17) to (19) is built. The AlphaFold links the table records are, however, not used in that classification, and serve only to attach the predicted structure and its confidence score to each protein.

The sources we combine record the same publication or protein in different identifier

formats, and a merge would treat two formats of one identifier as two distinct records.[23] We therefore reconfigure every identifier into a consistent canonical form before merging. Publication identifiers are reduced to the OpenAlex work identifier by upper-casing and extracting the leading `W`-prefixed numeric token. Accessions are upper-cased and stripped of the isoform suffix. This step matters because UniProt reports sequence attributes per isoform, and without it a single protein would enter a paper several times and inflate $|\mathcal{P}(p)|$. PDB identifiers are truncated to their four-character entry code. Rewriting the identifiers once, rather than at each merge, keeps them identical at the protein, paper, and institution levels.

Pre-AlphaFold structural difficulty

We are interested in the structural information available for a protein *before* AlphaFold. Let τ denote the release date of the AlphaFold database, 22 July 2021. Each PDB entry is dated by the earliest of its deposit date, its initial release date, and the release date recorded with its linked publication, and only entries dated before τ are counted. Protein i is considered to be *solved* if it has its own experimental structure released before τ. Otherwise, its structural information comes from homology: SWISS-MODEL and UniProt record candidate template structures for each protein, along with the percentage of matching amino acids (sequence identity) between the target protein and the template. Writing $H_i(\tau)$ for the set of templates of protein i whose structures were released before τ, and id_{ij} for the proportion of identical amino acids between protein i and template j, the best pre-AlphaFold structural coverage is defined as

$$s_i = \max\left\{ 100 \cdot \mathbf{1}\{i \text{ solved before } \tau\},\ \max_{j \in H_i(\tau)} \mathrm{id}_{ij} \right\} \in [0, 100], \tag{17}$$

with $s_i = 0$ for a protein with neither its own structure nor any dated template. The corresponding continuous difficulty measure is

$$d_i = 1 - \frac{s_i}{100} \in [0, 1], \tag{18}$$

equal to zero for a fully solved protein and to one for a protein about whose structure nothing was known. We also assign each protein to a discrete difficulty class, using the thresholds conventional in structural biology—50 percent identity for a template close enough to be used with little adjustment, and 30 percent for the weakest template that still

[23] For example, a publication may appear as `W2741809807` in one source and as `https://openalex.org/W2741809807` in another. A protein may appear as `P12345` in one and as the isoform `P12345-2` in another.

supports molecular replacement:

$$\kappa_i = \begin{cases} \text{Solved} & \text{own structure released before } \tau, \\ \text{Close homologue} & \text{not solved, } \max_j \mathrm{id}_{ij} \geq 50, \\ \text{Distant homologue} & \text{not solved, } 30 \leq \max_j \mathrm{id}_{ij} < 50, \\ \text{No template} & \max_j \mathrm{id}_{ij} < 30. \end{cases} \tag{19}$$

Of the four classes, *no template* is the one where AlphaFold's marginal contribution is the largest: a protein in that class had neither its own experimental structure nor a usable homology template before τ. Two further attributes are attached to each accession and play no part in the classification. The first is the AlphaFold confidence score ($pLDDT$), which the AlphaFold database reports for each predicted structure. The second is *sequence length*, the number of amino acids in the protein's chain.[24]

Aggregating protein attributes to paper-level measures

The difficulty measure d_i and the class κ_i are defined for individual proteins. Using this information at the paper level, the level at which protein content enters the analysis, requires aggregating the protein-level table to papers. The number of proteins to aggregate over varies across papers: the linkage returns one for some, several dozen for others, none for a substantial share. Papers in the last group have no protein attribute to carry forward. They drop from every specification that uses these measures, a loss documented in Table A10. For the remainder, the paper-level measure is a weighted mean of protein-level attributes. The weights can be assigned in two ways.

The first option is to use unweighted aggregation, taking the paper–protein pair as the unit of observation. Each pair is counted once, so that a paper enters the aggregate as many times as it has linked proteins. A paper linked to twenty proteins therefore enters the aggregate twenty times over, against once for a paper linked to a single protein. The second option is fractional weighting, which takes the paper as the unit of aggregation. Each of a paper's protein links receives a weight $1/|\mathcal{P}(p)|$. Every paper then carries the same total weight, whatever the number of proteins linked to it.

We use fractional weighting for two reasons. The first concerns the unit in which the outcome is measured. The dependent variable is an institution's share of top-journal

[24] A protein is a chain of amino acids, and its length is the count of this chain, and is the most direct measure of protein size available. Longer chains were historically harder to solve experimentally, which is why length enters the analysis as a control. UniProt records a length for each isoform of a protein, so we take the median across the isoforms of an accession.

protein publications, in which each paper counts once whatever its protein content. The institution-year measure defined below inherits that convention: it averages the paper-level measure over an institution's linked publications, giving each publication the same weight. Under unweighted aggregation that average would instead be a mean across proteins. Papers reporting many accessions would consequently dominate the measure of protein content, while contributing no more than any other paper to the outcome it is meant to explain. The second reason concerns the count $|\mathcal{P}(p)|$ itself. It measures the reach of the three protein identification channels as much as the breadth of the paper. A paper a curator has already processed carries links that an uncurated paper of identical content lacks. The annotation channel adds a protein whenever a gene appears in the text, even if it is mentioned in passing in the paper. Neither feature reflects what the paper truly studies, although both may end up inflating $|\mathcal{P}(p)|$. Under unweighted aggregation a paper enters in proportion to its protein count, which would leave institution-year exposure dominated by the papers the linkage happened to cover most thoroughly. The measure would then describe the reach of the procedure rather than the structural content of the research.

Our estimates come from comparing papers with different values of σ_p^κ and $\bar{d}_p$, which is precisely where error of this kind falls. Our decision to use fractional weighting removes $|\mathcal{P}(p)|$ from that comparison. Each link accordingly receives weight defined as:

$$\omega_{pi} = \frac{1}{|\mathcal{P}(p)|}, \qquad \text{so that} \qquad \sum_{i \in \mathcal{P}(p)} \omega_{pi} = 1. \tag{20}$$

For any protein-level attribute x_i, the corresponding paper-level measure is the weighted mean across the proteins the paper studies,

$$\bar{x}_p = \sum_{i \in \mathcal{P}(p)} \omega_{pi}\, x_i. \tag{21}$$

Figure C2 illustrates the construction. Applying (21) to the class indicators of (19) gives the class shares:

$$\sigma_p^\kappa = \sum_{i \in \mathcal{P}(p)} \omega_{pi}\, \mathbf{1}\{\kappa_i = \kappa\}, \qquad \sum_\kappa \sigma_p^\kappa = 1, \tag{22}$$

which describe a paper's protein content as a distribution across the four structural classes rather than as a single label. Applying (21) to the continuous measure d_i gives mean difficulty $\bar{d}_p$. We complement these measures with statistics computed directly from a paper's protein set: i) the best coverage across its proteins given as $s_p^{\max} = \max_{i \in \mathcal{P}(p)} s_i$, ii) two indicators recording whether any of those proteins is solved and whether any has no

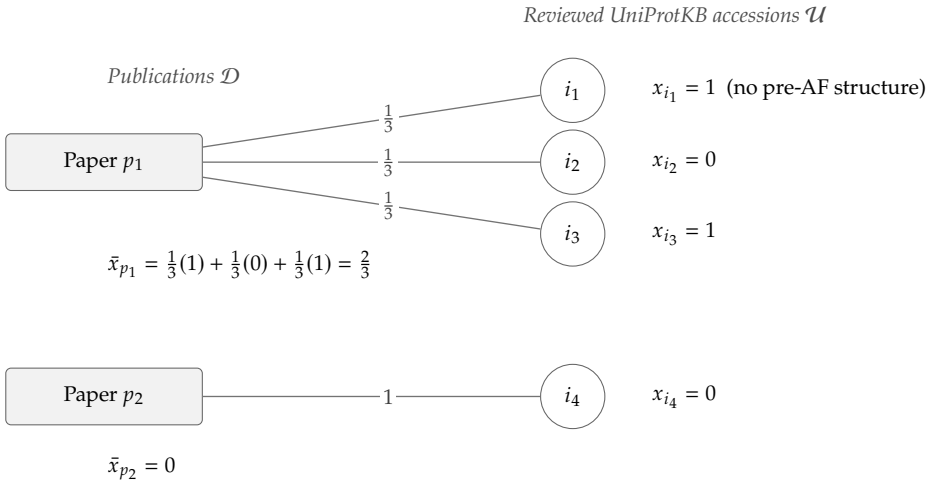


Figure C2: Paper–protein linkage, fractional weights, and paper-level aggregation

Notes. Each publication is linked to the set of proteins identified by the union of the three channels. Edge weights are $\omega_{pi} = 1/|\mathcal{P}(p)|$, so every paper contributes a total weight of one regardless of how many proteins it studies. Paper-level measures are weighted means of protein-level attributes, as in equation (21). In the illustration, x_i indicates that protein i had no experimental structure deposited in the Protein Data Bank before 22 July 2021.

template, iii) the number of distinct accessions gives the size of the set, and iv) the sequence lengths of the linked proteins summarized by their mean, median, maximum, and sum. Some specifications call for a binary treatment. For these we use an indicator that at least half of a paper's proteins have no template, $\mathbf{1}\{\sigma_p^{\text{no template}} \geq 0.5\}$, or the dominant class $\kappa_p^* = \arg\max_\kappa \sigma_p^\kappa$. Our preferred measure is the continuous no-template share $\sigma_p^{\text{no template}}$, which retains the within-paper variation that the majority indicator and the dominant class both discard.

Merging with the estimation sample

The estimation sample is a panel of institutions, in which the outcome is an institution's share of top-journal protein publications and the treatment group is the pre-AlphaFold publication-share quartile of the institution. Protein exposure is measured at the paper level. Merging the two therefore requires a step through the authorship records, which give for each publication its authors and their institutional affiliations. Each institution carries the quartile assigned in Section 2. Figure C3 shows the three joins required for this process: proteins to publications on the accession, publications to institutions through author affiliations, and institutions to the estimation panel on institution and year.

Suppose $\mathcal{A}$ denotes the set of triples (p, u, t) for which an author of publication p in year t is affiliated with institution u. Let

$$\mathcal{D}_{ut} = \{\, p \in \mathcal{D} \;:\; (p, u, t) \in \mathcal{A} \,\}, \qquad \mathcal{D}_{ut}^{L} = \{\, p \in \mathcal{D}_{ut} \;:\; \mathcal{P}(p) \neq \emptyset \,\} \tag{23}$$

be, respectively, the publications of institution u in year t and the subset of them for which protein content is observed. For each institution-year we take h_{ut}, the average of the paper-level measure across the institution's linked top journal protein publications in that year:

$$h_{ut} = \begin{cases} \dfrac{1}{\left|\mathcal{D}^L_{ut}\right|} \displaystyle\sum_{p \in \mathcal{D}^L_{ut}} h_p & \text{if } \mathcal{D}^L_{ut} \neq \emptyset, \\ 0 & \text{otherwise,} \end{cases} \tag{24}$$

where h_p stands for any of the paper-level measures of Section C the specification uses: the class shares σ_p^κ, mean difficulty $\bar{d}_p$, or one of the two scalar summaries. Our preferred choice is the no-template share $\sigma_p^{\text{no template}}$.

Two features of Equation (24) deserve attention. First, institution-years without a single linked publication are retained in the panel with $h_{ut} = 0$ rather than dropped, so that the estimation sample is the same one used for the headline specification in equation (4) and the coefficients remain comparable to it. The alternative, which is to treat institution-years with no linked publication as a separate group rather than assigning them $h_{ut} = 0$, is examined in Section C, where the headline quartile gradient is re-estimated on each set of institution-years in turn. Second, a publication with authors at several institutions contributes to the measure of every institution on its author list. This mirrors the main specification, in which a paper counts once for every institution with at least one author on it. The merge is an inner join on the publication key. Papers in the linkage file that fall outside the top-journal sample are therefore discarded at this step.

The protein-derived controls are constructed in the same way. The log number of proteins per paper and the log maximum protein length are averaged over an institution's linked publications. This ensures that the controls and h_{ut} are constructed at the institution-year level and from the same pool of papers. The merged panel supplies the institution-year exposure measures used in the reported stratum-specific specifications.

Implementation

The protein linkage procedure described above in this section is implemented in Python. Three design choices are worth recording because they bear on reproducibility. First, all reference data are obtained through one-time bulk downloads and stored locally. Matching therefore involves no network calls, and the result is deterministic conditional on the stored snapshot. Second, the remaining external queries are batched, at fifty identifiers per request to OpenAlex and one hundred per request to PubTator3. Each is issued through

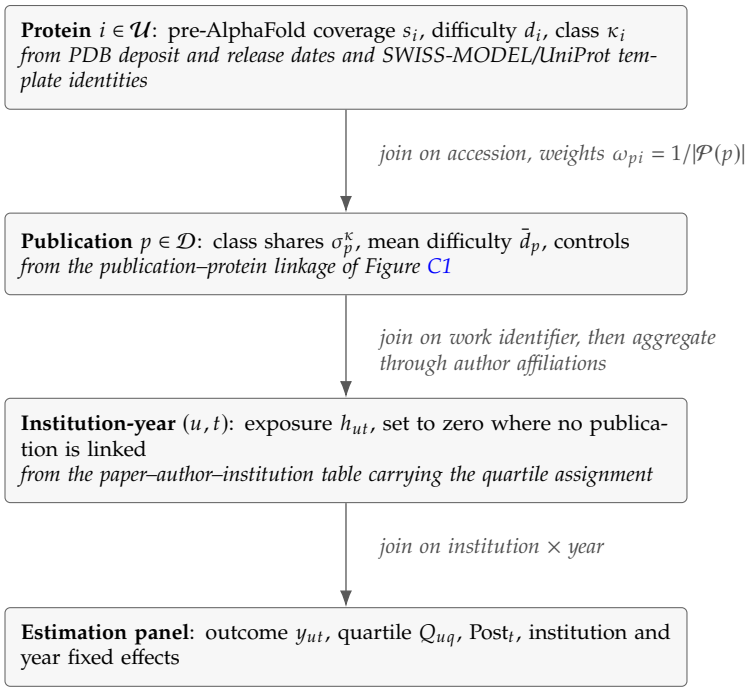


Figure C3: From protein structural status to the estimation panel

Notes. Each step is a join on a normalized key: protein accessions stripped of isoform suffixes, OpenAlex work identifiers, and institution by year. Publications are aggregated to institutions through author affiliations, so a co-authored paper contributes to the exposure of every institution on its author list. Institution-years in which no publication can be linked to a protein are kept in the panel with exposure set to zero, so that the estimation sample coincides with the one used for the headline specification.

a session that retries with exponential backoff on rate-limit and server errors, so that transient failures do not silently truncate the sample. Third, every stage writes to disk: the cached reference indices, the resolved identifier map, the retrieved annotations, and a log of completed publications. The procedure can be interrupted and resumed, and re-running it returns the same result.

Coverage, validation, and limitations

Note that the proteins a paper studies are recovered from text and external databases rather than read off its bibliographic record. Recovery succeeds for 27 percent of the papers in our sample, which is high for a task of this kind. Publications report no accession numbers, so every link must be reconstructed from gene symbols, common names, synonyms, and abbreviations, many of which denote different proteins in different organisms. No single channel comes close to this rate on its own: curator-verified links exist only for papers a biocurator has already processed, and annotation requires the paper to have been indexed. Coverage reaches this level only because the three channels are used and combined eventually.

A paper that cannot be linked to any protein has no value of σ_p^κ and leaves the estimation

sample, raising the question of what the surviving papers represent. Two alternative scenarios must be considered. If the linkage rate is the same across the quartiles of a paper's institutions, then the linked sample is a random subsample of the protein research in our main sample, and the cost is precision alone. Alternatively, if the linking rate differs by quartile, the resulting matched sample over-represents some institutions relative to others. The diagnostics below test which case holds by estimating whether a paper's quartile predicts its being linked, and by comparing linked and unlinked papers on observables the quartiles do not capture.

Whether coverage varies with quartile is a question about the data, and we test it directly by estimating

$$\mathbf{1}\{\mathcal{P}(p) \neq \emptyset\} = \delta_t + \sum_{q=2}^{4} \pi_q \, Q_{pq} + u_p, \tag{25}$$

with year fixed effects, where Q_{pq} assigns paper p to the quartile of its highest-ranked institution. If π_q is small and statistically indistinguishable from zero, knowing a paper's quartile tells us nothing about whether it was linked, which is the first of the two cases above. A quartile assignment places a paper in one of four groups by the relative rank of its institutions, and papers within a group may still differ in team size, journal, and subfield, all of which bear on how many proteins a paper studies and how likely a curator is likely to have processed it. We therefore compare linked and unlinked papers on journal rank, team size, number of institutions, structural-biology presence, access to high-performance computing, and AlphaFold use, summarizing each comparison by the normalized difference:

$$\mathrm{ND}(x) = \frac{\bar{x}^L - \bar{x}^U}{\sqrt{(\hat{\sigma}_L^2 + \hat{\sigma}_U^2)/2}}, \tag{26}$$

where L and U index linked and unlinked papers, and values below 0.25 in absolute value are conventionally taken to indicate balance.

Table A12 reports *coverage* – the share of paper–university pairs linked to at least one protein – by university quartile and by period. Coverage is generally consistent across university quartiles, ranging from 27.5% in the first quartile to 25.2% in the fourth, a narrow gap of just 2.3 percentage points. Furthermore, pairs from third-quartile universities are linked a bit more frequently than those from the second quartile, ruling out a strictly linear relationship between university rank and coverage. Across all quartiles, coverage declines slightly from the pre-AlphaFold period to PostAF1 and PostAF2. The curation

and indexing lags described earlier explain this decline, as the most recent publications are added to the curated record last. Because this drop is common to all four quartiles, it is absorbed by the year fixed effects in our protein-level specifications. Compared to the first quartile, no other quartile's change in coverage between the pre-AlphaFold and post-AlphaFold periods exceeds 2.8 percentage points. For the fourth quartile—the group driving the estimated gains—this differential is only 1.6 percentage points post-AlphaFold 1 and 0.3 percentage points post–AlphaFold 2.

Our paper-protein linking procedure therefore retains papers at nearly the same rate in every quartile within each period. Papers carrying a value of σ_p^κ are representative of the top-journal protein research in our main sample along the two dimensions the specification compares: university quartile and period. Since protein coverage is stable across quartiles and periods, a quartile-by-period coefficient in the protein-level specifications tracks how much a quartile's output of papers in a given protein stratum moves relative to the first quartile, rather than reflecting any selection effects in the linkage process.

D Literature & Background

D.1 Concentration in Scientific Research

Scientific research is highly concentrated, with the top 1% of scientists obtaining 20% of the citations (Nielsen and Andersen, 2021) and 50% of funding (Bol et al., 2018; Madsen and Aagaard, 2020). The pros and cons of this concentration are actively debated (Aagaard et al., 2020). Advocates for a more equitable distribution argue that concentrating funds stifles scientists' creativity. For example, superstar scientists may act as gatekeepers and hinder the emergence of new paradigms (Azoulay et al., 2019). Similarly, in grant reviews, it is found that homophily among senior scientists can penalize the more novel research proposals (Li, 2017; Nicholson and Ioannidis, 2012; Packalen and Bhattacharya, 2018). In contrast, critics point to economies of scale and efficient resource use by more productive scientists.

Research on the topic has focused on explaining this concentration. A prevailing explanation is that scientific concentration arises partly because of the winner-takes-all reward system in science: renowned scientists receive a disproportionate share of funding and credit for their research. This enhances their productivity and reputation, thereby further solidifying their dominant position—a phenomenon known as the "Matthew effect" (Merton, 1968). This theory has been empirically validated across diverse contexts, including scientific productivity, funding disparities, and career outcomes (Azoulay et al., 2014; Bol et al., 2018; Ghirelli et al., 2023).

Related to our work are papers examining the effects of external shocks on competition in science. The hiring of star researchers in lower-ranked institutions boosts the productivity of future recruits without affecting existing researchers (Agrawal et al., 2017). The deaths of star scientists negatively impact their collaborators but benefit others in the same field (Azoulay et al., 2019; 2010). The influx of Soviet mathematicians decreased career opportunities for prominent American mathematicians following the USSR's collapse (Borjas and Doran, 2012). The introduction of Bitnet improved collaboration and publication outcomes for mid-tier researchers located near top scientists (Agrawal and Goldfarb, 2008). The introduction of CRISPR, a novel gene-editing tool, initially improved the prospects of institutions already at the cutting edge of genetic research (Thompson and Zyontz, 2017). The free availability of Microsoft's Kinect democratized motion-sensing technology, increasing the competitiveness of outsider researchers and the diversity of ideas (Furman and Teodoridis, 2020). Reduced costs for accessing genetically modified mice led to new researchers entering the field and to more diverse research being produced (Murray et al., 2016). More recently, AI tools across the natural sciences have been shown to

expand individual scientists' impact (through increased publications, citations, and earlier leadership roles) but contract the collective focus of science by shrinking the range of explored topics and shifting toward data-rich established areas (Hao et al., 2026).

D.2 Structural Biology

Structural biology is a branch of protein research, alongside genomics, drug discovery, and related areas. Its primary objective is to identify the three-dimensional shapes of proteins. Knowledge of protein structure is key to understanding how proteins interact with other molecules (Berg et al., 2002) and is fundamental for drug development (Anderson, 2003). Before 2018, most protein structures were determined by X-ray crystallography (Shi, 2014). The cost of this technique varies between $250,000 for soluble human proteins, with a 25% success rate, and $1,000,000 for membrane proteins, with a 10% success rate (Stevens, 2003).

Proteins are crystallized by preparing a concentrated protein solution to induce crystal formation. This process, essential for obtaining high-quality crystals suitable for X-ray crystallography, is complex and highly variable. It typically requires extensive trial-and-error screening of numerous conditions, including temperature, pH, precipitant types and concentrations, and chemical additives such as salts or polymers (Chayen and Saridakis, 2008). Once suitable crystals are obtained, they are exposed to high-intensity X-ray beams at a synchrotron facility, producing a diffraction pattern. This diffraction data is then processed to generate an electron density map, which reveals the positions of atoms within the protein. Scientists build and iteratively refine a proposed 3D atomic model of the protein structure to best fit the electron density data. This process is iterative and can take months, requiring multiple adjustments to accurately capture structural details of the protein (Garman, 2014).

An alternative to X-ray crystallography is cryo-electron microscopy (cryo-EM). Cryo-EM freezes proteins at cryogenic temperatures, maintaining their natural structures. An electron beam captures 2D images that are computationally assembled into a 3D model. Cryo-EM has gained popularity for determining structures of large molecular complexes, flexible proteins, and membrane-bound proteins that are challenging to crystallize (Cheng et al., 2015; Nogales, 2016).

D.3 The Advent of AlphaFold

AlphaFold was developed by DeepMind, a London-based AI laboratory and Google subsidiary. The first model, AlphaFold 1, was developed in 2018 and used a neural network

trained on experimentally determined 3D protein structures from the Protein Data Bank (PDB). The DeepMind team was not the first to attempt protein-structure prediction using deep learning, but it was the first to succeed at this level. The model used a larger and more computationally efficient neural network than prior models, enabling unprecedented accuracy in protein-folding predictions (AlQuraishi, 2019). AlphaFold processed amino-acid relationships and inferred structural data faster and more accurately than traditional methods (Senior et al., 2020); it consequently won the 13th Critical Assessment of Protein Structure Prediction (CASP13) in December 2018.

The results of CASP13 validated the efficacy of deep learning approaches in protein structure prediction, marking a breakthrough after years of limited progress with traditional methods. The success of AlphaFold 1 sparked the development of more advanced deep learning models to predict protein structures more accurately. The associated research paper was subsequently published in the journal Nature in January 2020 (Senior et al., 2020).

However, AlphaFold 1's precision remained insufficient for reliable predictions of complex proteins. In 2020, DeepMind introduced AlphaFold 2, which achieved near-experimental accuracy across a broader range of protein targets. It combined evolutionary data, structural templates, and a revised deep-learning infrastructure featuring an attention mechanism. Compared with its predecessor, AlphaFold 2 better captured spatial dependencies between amino acids and achieved precision comparable to experimental methods (Jumper et al., 2021). In November 2020, it won CASP14 by a record margin.[25]

On 15 July 2021, the paper introducing AlphaFold 2 was published in Nature (Jumper et al., 2021). The publication was accompanied by the release of open-source code and a searchable database containing predicted structures for nearly 1 million proteins. This database has since expanded to over 214 million structures, covering virtually all catalogued proteins, including the entire human proteome. In contrast, prior to AlphaFold, experimentally determined structures were available for only about 17% of residues in the human proteome (Tunyasuvunakool et al., 2021). The paper has garnered over 55,300 citations (as of September 2026), and its principal authors were awarded the 2024 Nobel Prize in Chemistry.

Since the release of AlphaFold 2, several deep-learning tools have emerged to address structural-protein challenges; Table A1 provides examples. On 8 May 2024, DeepMind

[25] AlphaFold 2 scored above 90 for around two-thirds of proteins. This score measures the similarity between a predicted structure, such as one produced by AlphaFold, and an experimentally determined structure; 100 denotes a complete match. AlphaFold 2 achieved a record overall score of 92.4, indicating high precision in protein-structure prediction. See https://www.guinnessworldrecords.com/world-records/642132-highest-score-at-the-casp-competition.

announced an enhanced version, AlphaFold 3. This version can predict highly complex protein structures and protein–protein interactions. The development of AlphaFold and other models is summarized in Figure A5.

D.4 Role of AlphaFold in Structural Biology

AlphaFold can serve as both a complement and a substitute for experimental methods in structural biology. AlphaFold predictions can either bypass experimental work entirely or automate part of the experimental process.

As a substitute, AlphaFold enables researchers to obtain predicted structures without costly experiments (Tunyasuvunakool et al., 2021). The release of AlphaFold 2's open-access database dramatically expanded the availability of predicted protein structures, enabling researchers to access them readily without running the model themselves. However, exclusive reliance on AlphaFold without experimental validation risks inaccuracies. AlphaFold 2, in particular, can struggle with novel or large protein complexes, yielding erroneous predictions even in regions with high confidence scores (Terwilliger et al., 2024).

As a complement to experimental methods, AlphaFold accelerates and reduces the cost of structure discovery. Its predictions provide more comprehensive starting hypotheses, enabling researchers to construct models that better align with experimental data. They also facilitate faster, cheaper iteration over large sets of potential structures, allowing more refinement and yielding finer interpretations of the data (Ruff and Pappu, 2021).

Beyond substituting for experiments, AlphaFold 2 also extends experimental work to large protein assemblies, as Appendix D.5 shows for the human nuclear pore complex.

D.5 Protein Structure Determination Before and After AlphaFold

The comparison of the PostAF1 and PostAF2 periods in Section 4.1 highlights two distinct effects of the AlphaFold releases, namely improvements in predictive accuracy and broader access to highly accurate predictions. Figure D1 illustrates these improvements by tracing one protein, the enzyme BT1044 of the gut bacterium *Bacteroides thetaiotaomicron* (UniProt ID: Q8A8X3), which was evaluated in CASP (the field's biennial benchmark competition in which research groups blindly predict structures that have been experimentally solved but not yet publicly released), across the three regimes defined in Table 2. Before AlphaFold 1, determining the enzyme's three-dimensional structure required the use of experimental methods, with the substantial resource and time costs described in Section D.2. Under AlphaFold 1, the released codebase could generate predictions only for proteins that were in the evaluation set of the CASP13 competition in December 2018 (Google DeepMind,

2021). Applying AlphaFold 1 methods to other proteins therefore required researchers to reconstruct the pipeline from its published technical description, as discussed in Section 4.1. Subsequently, with the launch of AlphaFold 2 and the open-access AlphaFold Protein Structure Database in July 2021, highly accurate structural predictions for this enzyme and nearly all other cataloged proteins became freely available for immediate download (Varadi et al., 2022).

In addition to expanding access, both AlphaFold releases significantly increased predictive accuracy, with the second release yielding by far the larger gain (see Figure D2, panel A). CASP evaluates novel prediction methods submitted by researchers worldwide on a set of target proteins, scoring each domain of a target separately by the *GDT score*, which ranges from 0 to 100 and measures how closely the predicted positions of individual amino acids (*residues*) match their positions in the experimentally determined structure.[26] On the 43 of the 45 hardest CASP13 domains that AlphaFold 1 predicted, its median GDT score was 58.9, compared with 52.4 for the best other group, which AlphaFold 1 outscored on 32 of the 43 domains. On the 38 hardest CASP14 domains, AlphaFold 2 reached a median GDT score of 87.6, compared with 61.3 for the best other group, which AlphaFold 2 outscored on all 38 domains.

Panel (B) of Figure D2 illustrates how AlphaFold 2 expands the scope of structures that researchers can resolve by combining predictions with experimental methods. The human nuclear pore complex, one of the largest protein assemblies in human cells, regulates the transport of molecules between the nucleus and the surrounding cytoplasm. Its 2015 structure, determined using experimental methods alone, covered mainly the two outer rings and contained 154,800 amino-acid residues per pore (PDB ID: 5A9Q) (von Appen et al., 2015). In 2022, Mosalaganti et al. (2022) fitted AlphaFold 2 predictions for 25 constituent proteins into *tomography maps* obtained from human cells.[27] The resulting structure (PDB ID: 7R5J) contained 624,240 residues per pore, four times as many as the 2015 structure, and covered more than 90% of the pore's scaffold, about twice the coverage of earlier models (Mosalaganti et al., 2022). The example illustrates how AlphaFold 2 can extend the reach of experimental methods, enabling researchers to resolve more complete structures of complex protein assemblies.

[26] A *residue* is an amino acid that has joined a protein chain by forming peptide bonds with neighboring amino acids.

[27] Tomography maps are three-dimensional reconstructions of cellular structures obtained from multiple two-dimensional images collected at different viewing angles.

Figure D1: How a protein structure was obtained in each period

Each row corresponds to one period of Table 2. The three structures show the same protein at the same scale: the secreted endoglycosidase BT1044 of the gut bacterium *Bacteroides thetaiotaomicron* (CASP13 target T1005, UniProt Q8A8X3). The top row shows the experimental structure of the protein evaluated prior to CASP13 (PDB 6Q64). The middle row shows the blind prediction submitted by Google DeepMind (using AlphaFold 1) for the same protein. The bottom row shows the AlphaFold DB entry for this protein, colored by the AlphaFold 2 residue-level confidence score (pLDDT). This entry includes residues 1–38, which were missing from the original experimental structure and are predicted by AlphaFold 2 with low confidence. Protein structures are rendered with Mol* (**sehnal2021molstar**).

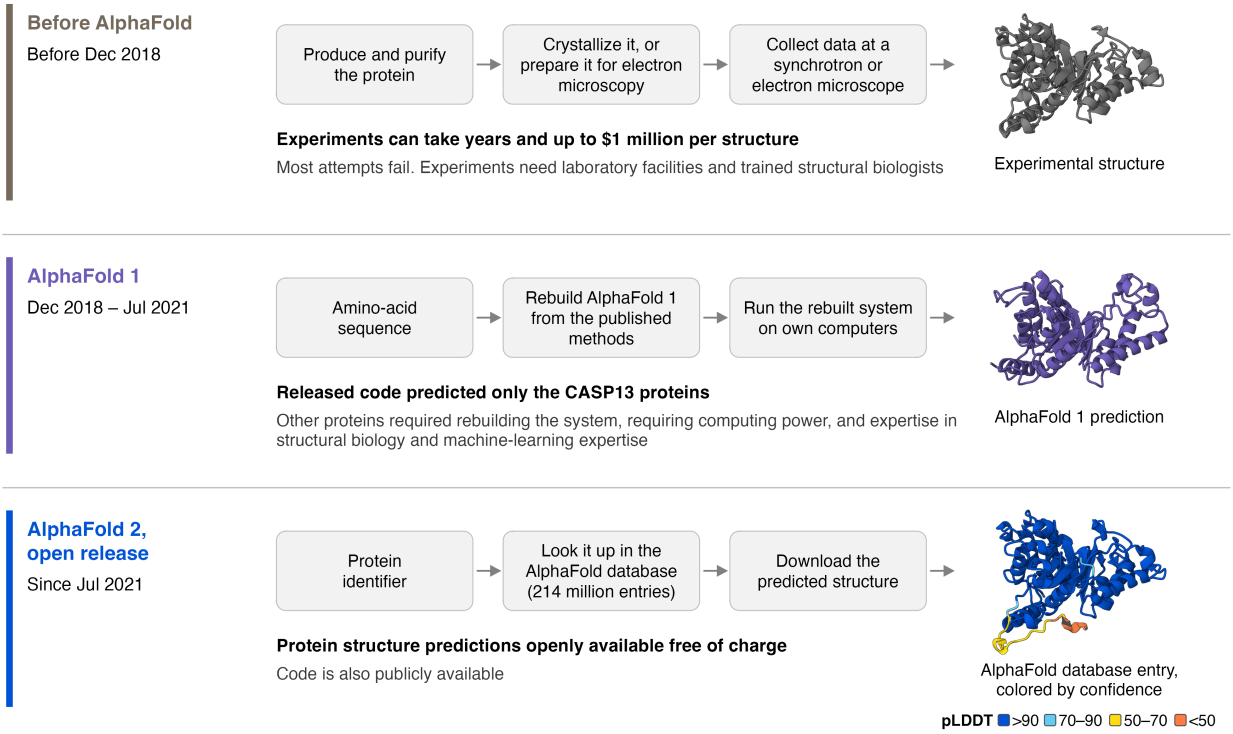
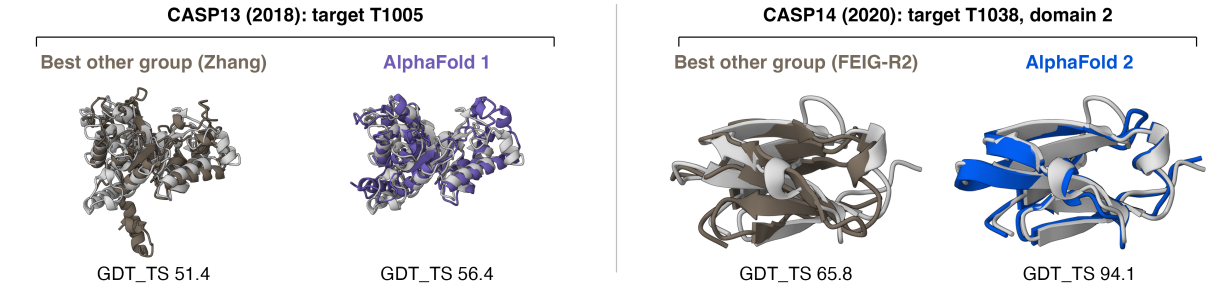


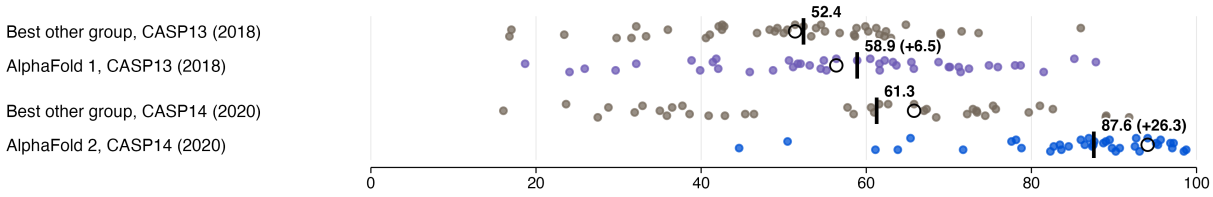
Figure D2: Accuracy of predicted structures and a frontier application

Panel (A) compares the prediction accuracy of AlphaFold 1 in CASP13 (2018) and AlphaFold 2 in CASP14 (2020) against the top-performing runner-up group. The analysis focuses on the most challenging targets in each competition, comprising 45 *domains* (parts of a protein that fold independently) in CASP13 and 38 in CASP14, where related experimental structures provided little to no guidance. AlphaFold 1 submitted predictions for 43 of the 45 CASP13 domains, whereas AlphaFold 2 predicted all 38 CASP14 domains. Accuracy is measured using the GDT_TS score of each group's top-ranked model, based on official results (Protein Structure Prediction Center, 2018; 2020). Each dot represents one domain and each bar indicates the median score achieved by the submitting group. Values in parentheses show AlphaFold's lead over the runner-up group. Across all individual targets, AlphaFold 1 outperformed each of the top five competing groups on at least 32 domains, while AlphaFold 2 surpassed everyone else on at least 37 domains. The 3D structure insets highlight representative domains near the median performance scores, shown as circles on the chart. Panel (B) compares the partially resolved 2015 structure of the human nuclear pore complex (PDB 5A9Q), one of the largest protein assembly in human cells, with the updated 2022 structure (PDB 7R5J) at the same scale. Gray shading in the 2022 model highlights regions that were missing in 2015 but were subsequently completed using AlphaFold 2 predictions.

(A) Accuracy of predicted structures on the hardest targets of the CASP competitions

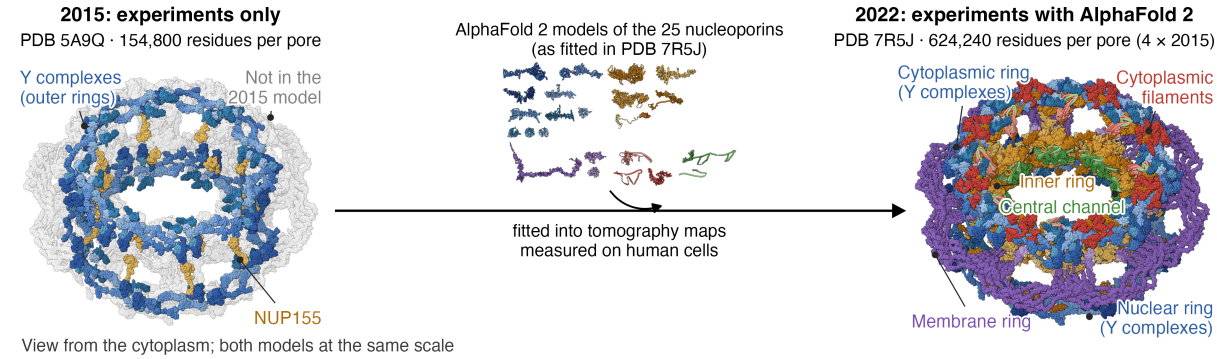


Gray: experimental structure. Color: the group's first blind prediction for the target, superposed on the experimental structure.



Each dot is one target domain. Bar: median over domains, with the gain over the best other group on the same domains in parentheses. Circle: the domain drawn above.

(B) A frontier application: the human nuclear pore complex before and after AlphaFold 2



Nucleoporins (the proteins that make up the nuclear pore complex), grouped by subcomplex

Outer rings (Y complexes)				Inner ring		Central channel	Cytoplasmic filaments	Membrane ring	Not in the 2015 model
NUP160	NUP96	NUP37	ELYS	NUP205	NUP93	NUP54	NUP358	NUP210	Scaffold
NUP133	NUP85	SEC13		NUP188	NUP35	NUP58	NUP214	NDC1	
NUP107	NUP43	SEH1		NUP155	NUP98	NUP62	NUP88	ALADIN	

D.6 Descriptive Statistics

Table A3 presents descriptive statistics on university characteristics by quartile for the pre-AlphaFold period, highlighting significant disparities in factors such as structural biology expertise, HPC access, author demographics, faculty size, institutional types, and geographic affiliations. Top-quartile universities dominate top-journal publications in both structural biology and non-structural protein research. On average, they produce 8.58 structural biology papers per university annually, compared to 2.39, 1.75, and 0.80 for the second, third, and bottom quartiles, respectively. Similar disparities are evident in non-structural protein research. Consequently, top-quartile universities hold substantially larger market shares, in terms of both publication volume and authorship contributions. For structural biology, the average top-quartile university accounts for 0.37% of all top publications, versus 0.10%, 0.08%, and 0.03% for the second, third, and bottom quartiles per year, respectively. Patterns are comparable for non-structural protein publications and authorship shares.

Table A3 indicates that top-quartile universities have more resources at their disposal. Their average faculty size is nearly double that of the second quartile and more than triple that of the bottom quartile. Access to HPC resources also varies. Top-quartile universities have a higher rate of HPC access, 50.9% versus 9% among bottom-quartile universities, and greater computing power conditional on access, 351 versus 54 petaflops, respectively.

Geographically, U.S.-affiliated authors predominate in the top quartile (45%), while European affiliations are most common in the second (54%), third (48%), and bottom (44%) quartiles. Asian affiliations increase notably in the bottom two quartiles (33% and 26%, respectively) relative to the top (21%). Additionally, top-quartile universities have more researchers affiliated with industry but fewer with government labs. There are minor differences across quartiles in other dimensions. For example, author demographics are relatively consistent: approximately one-third of authors are women across all quartiles, with around half classified as Asian and one-quarter as white.

E Additional Results

E.1 AlphaFold Adoption

We approximate the use of AlphaFold or other equivalent AI systems at the publication level. We classify a publication as using AlphaFold if i) it explicitly mentions the term "AlphaFold " or an analogous AI tool in its abstract, or ii) it cites the seminal papers that introduced AlphaFold 1 and 2 (Jumper et al., 2021; Senior et al., 2020). Note that our method for identifying AlphaFold usage provides neither a lower nor an upper bound on actual adoption rates. A paper might mention or cite AlphaFold without employing it (false positive), while another might use it without any explicit mention or citation (false negative). That said, we believe that our measure underestimates AlphaFold adoption more than it overestimates it, as we search only abstracts for mentions and expect many papers to reference AlphaFold predominantly in the main text.

Figure A6 illustrates the proportion of top-journal publications mentioning or citing AlphaFold in both structural biology and non-structural protein research, as well as across subfields of structural biology. Adoption remains modest and stable from 2018 through 2020, before surging sharply following the July 2021 release of the AlphaFold 2 database. Structural biology exhibits rapid uptake thereafter, reaching approximately 30% by 2023 (Figure A6(A)). Non-structural protein research shows similar trends, albeit with lower overall magnitudes. Restricting the sample to papers in protein structure prediction and macromolecular crystallography (subfields where AlphaFold directly substitutes for or complements traditional computational and experimental methods) yields even higher adoption, approaching 60% by the end of 2023 (Figure A6(B)).

Adoption rates also differ across university quartiles. Figure A7(A) displays the proportion of top-journal publications mentioning or citing AlphaFold for each quartile. In structural biology, the third and bottom quartiles account for the largest shares of AlphaFold -using articles by 2023, 36% and 31%, respectively, outpacing the top and second quartiles at 21% and 12%. The pattern is more nuanced in non-structural protein research. Figure A7(B) shows that all quartiles had similar proportions of AlphaFold -using papers from 2020 to 2022, while the bottom quartile had a higher adoption rate in 2019, after the release of AlphaFold 1. By the final sample year, 2023, third- and bottom-quartile shares rose to 30% and 34%, respectively, while top- and second-quartile shares declined to 16% and 20%.

Finally, Figure A3 shows a marked increase in the number of AI-based tools submitted to CASP competitions, particularly following AlphaFold 2's success in CASP14 in 2020. This trend is consistent with the increasing use of AI methods for protein-structure prediction

within the structural-biology community.

E.2 Dynamic Estimates

We complement the baseline fixed-effects regressions from equation (4) with disaggregated yearly estimates using identical controls, where the post-AlphaFold dummy is split into annual indicators. Figures A8 and A9 plot the corresponding difference-in-differences coefficients, revealing distinct temporal pattern in top-journal protein publication counts and market shares across university quartiles.

Post-AlphaFold, publication volumes in the lower quartiles increase relative to the top quartile, with a sharp spike in 2020, a temporary dip in 2021, and steady growth thereafter. These effects are stronger and more statistically precise in structural biology than in broader protein research.

Market share estimates display analogous temporal dynamics but with more variation and wider confidence intervals. The 2020 spike coincides with the COVID-19 pandemic, which may confound interpretation. Nonetheless, the coefficients from 2021 onward are consistent with the gradual adoption of AlphaFold documented in Section E.1.

E.3 Publication Outcomes and Research Patterns Outside Protein Science

To rule out the possibility that our main results stem from general trends in scientific publishing or the rise of general-purpose AI tools, we replicate our analysis using publication data from four additional fields: microbiology (excluding protein research), economics and finance, materials science, and social sciences. These fields exhibit similar pre-AlphaFold hierarchies, with a small number of elite universities dominating top-journal publications, as illustrated by the cumulative distribution functions in Figure A13. This figure highlights the persistent concentration of top-journal publications among the highest-ranked institutions, akin to the patterns observed in protein research before the release of AlphaFold .

As detailed in Section 3.1, Figure 3 further confirms the specificity of AlphaFold's impact on scientific publishing: unlike in protein research, we observe no comparable shifts in top-journal market shares across university quartiles in these other fields during the sample period.

Additionally, we compute the same qualitative indicators of research practices—pivot scores, novelty, and disruptiveness—for these fields as in Section 5. Figure A16 shows no

notable changes over the full period. In contrast to protein research, where lower-quartile institutions display greater novelty, pivot, and citation impact, these placebo fields exhibit stable patterns. This contrast is consistent with the observed protein-research patterns reflecting AlphaFold rather than broader academic or technological trends.

Data Availability

Publication and citation data were sourced from OpenAlex using its premium API (full snapshots, accessed for the 2013–2023 window). Journal rankings were obtained from Scimago Journal Rank (SJR), with the per-journal score averaged over 2013–2018. University rankings were derived from the CWTS Leiden Ranking system, using the Total Normalized Citation Score (TNCS) in the "biomedical and health sciences" category, averaged over 2013–2018.

For the publication–protein linkage described in Section C, we retrieved the OpenAlex record of every work published in the 17 top journals, including its PubMed identifier, DOI, and abstract. Requests were issued against the OpenAlex works endpoint in batches of fifty identifiers using the `openalex:` filter and were submitted through the polite pool, so that the full sample was resolved in a few hundred requests rather than one request per paper.

Protein records were taken from the reviewed section of UniProtKB, known as Swiss-Prot. Rather than querying the UniProt API once per paper or once per protein mention, we downloaded the entire reviewed index once through the bulk stream endpoint, and stored it locally. Two further streams retrieved the mapping from accessions to NCBI Gene identifiers and the list of PubMed identifiers cited by each curated entry; the curator-linked channel is equivalent to the UniProt query `lit_pubmed:$q_p$ AND reviewed:true`.

Machine-normalized gene and protein annotations were retrieved from PubTator3, in batches of one hundred PubMed identifiers, through the BioC export endpoint. The named-entity recognition models applied to abstracts are those distributed with scispaCy.

Experimental structures and their deposit and release dates were obtained from the Protein Data Bank, candidate homology templates and their sequence identities from SWISS-MODEL and UniProt, and predicted structures and per-residue confidence scores from the AlphaFold Protein Structure Database (Varadi et al., 2022). All reference data were obtained through one-time bulk downloads and stored locally, so that matching involved no network calls and the result is deterministic conditional on the snapshot. All remaining external queries were batched and issued through a session that retries with exponential backoff on rate-limit and server errors, so that transient failures do not silently truncate the sample.

Appendix References

Aagaard, Kaare, **Kladakis**, Alexander, and **Nielsen**, Mathias, "Concentration or dispersal of research funding?", *Quantitative Science Studies*, 2020, *1* (1), 117–149.

Agrawal, Ajay and **Goldfarb**, Avi, "Restructuring Research: Communication Costs and the Democratization of University Innovation", *American Economic Review*, 2008, *98* (4), 1578–1590.

Agrawal, Ajay, **McHale**, John, and **Oettl**, Alexander, "How stars matter: Recruiting and peer effects in evolutionary biology", *Research Policy*, 2017, *46* (4), 853–867.

Alperin, Juan Pablo, **Portenoy**, Jason, **Demes**, Kyle, **Larivière**, Vincent, and **Haustein**, Stefanie, *An analysis of the suitability of OpenAlex for bibliometric analyses*, arXiv Working Paper 2404.17663, arXiv, 2024.

AlQuraishi, Mohammed, "AlphaFold at CASP13", *Bioinformatics*, 2019, *35* (22), 4862–4865.

Anderson, Amy, "Structure-based functional design of drugs: from target to lead compound", *Methods in Molecular Biology*, 2003, *230*, 375–384.

Azoulay, Pierre, **Fons-Rosen**, Christian, and **Graff Zivin**, Joshua, "Does Science Advance One Funeral at a Time?", *American Economic Review*, 2019, *109* (8), 2889–2920.

Azoulay, Pierre, **Graff Zivin**, Joshua, and **Wang**, Jialan, "Superstar Extinction", *Quarterly Journal of Economics*, 2010, *125* (2), 549–589.

Azoulay, Pierre, **Stuart**, Toby, and **Wang**, Yanbo, "Matthew: Effect or Fable?", *Management Science*, 2014, *60* (1), 92–109.

Berg, Jeremy, **Tymoczko**, John, and **Stryer**, Lubert, *Biochemistry*, WH Freeman, 2002.

Bol, Thijs, **Vaan**, Mathijs de, and **Rijt**, Arnout van de, "The Matthew effect in science funding", *Proceedings of the National Academy of Sciences*, 2018, *115* (19), 4887–4890.

Borjas, George and **Doran**, Kirk, "The Collapse of the Soviet Union and the Productivity of American Mathematicians", *Quarterly Journal of Economics*, 2012, *127* (3), 1143–1203.

Chayen, Naomi and **Saridakis**, Emmanuel, "Protein crystallization strategies for structural genomics", *Nature Methods*, 2008, *5* (2), 147–153.

Cheng, Yifan, **Grigorieff**, Nikolaus, **Penczek**, Pawel, and **Walz**, Thomas, "A primer to single-particle cryo-electron microscopy", *Cell*, 2015, *161* (3), 438–449.

Culbert, Jack, **Hobert**, Anne, **Jahn**, Najko, **Haupka**, Nick, **Schmidt**, Marion, **Donner**, Paul, and **Mayr**, Philipp, "Reference coverage analysis of OpenAlex compared to Web of Science and Scopus", *Scientometrics*, 2025, *130* (4), 2475–2492.

Furman, Jeffrey and **Teodoridis**, Florenta, "Automation, Research Technology, and Researchers' Trajectories: Evidence from Computer Science and Electrical Engineering", *Organization Science*, 2020, *31* (2), 330–354.

Garman, Elspeth, "Developments in X-ray Crystallographic Structure Determination of Biological Macromolecules", *Science*, 2014, *343* (6175), 1102–1108.

Ghirelli, Corinna, **Havari**, Enkelejda, **Meroni**, Elena, and **Verzillo**, Stefano, *The Long-Term Causal Effects of Winning an ERC Grant*, ssrn: 4437664, (visited on 09/22/2026), 2023.

Google DeepMind, *AlphaFold 2: Source Code and Documentation*, https://github.com/google-deepmind/alphafold, 2021.

Hao, Qianyue, **Xu**, Fengli, **Li**, Yong, and **Evans**, James, "Artificial Intelligence Tools Expand Scientists' Impact but Contract Science's Focus", *Nature*, 2026, *649* (8099), 1237–1243.

Jumper, John, **Evans**, Richard, **Pritzel**, Alexander, **Green**, Tim, **Figurnov**, Michael, **Ronneberger**, Olaf, **Tunyasuvunakool**, Kathryn, **Bates**, Russ, **Žídek**, Augustin, **Potapenko**, Anna, **Bridgland**, Alex, **Meyer**, Clemens, **Kohl**, Simon, **Ballard**, Andrew, **Cowie**, Andrew, **Romera-Paredes**, Bernardino, **Nikolov**, Stanislav, **Jain**, Rishub, **Adler**, Jonas, **Back**, Trevor, **Petersen**, Stig, **Reiman**, David, **Clancy**, Ellen, **Zielinski**, Michal, **Steinegger**, Martin, **Pacholska**, Michalina, **Berghammer**, Tamas, **Bodenstein**, Sebastian, **Silver**, David, **Vinyals**, Oriol, **Senior**, Andrew, **Kavukcuoglu**, Koray, **Kohli**, Pushmeet, and **Hassabis**, Demis, "Highly accurate protein structure prediction with AlphaFold", *Nature*, 2021, *596* (7873), 583–589.

Li, Danielle, "Expertise versus Bias in Evaluation: Evidence from the NIH", *American Economic Journal: Applied Economics*, 2017, *9* (2), 60–92.

Madsen, Emil and **Aagaard**, Kaare, "Concentration of Danish research funding on individual researchers and research topics: Patterns and potential drivers", *Quantitative Science Studies*, 2020, *1* (3), 1159–1181.

McKeon, John and **Boudou**, Justine, *Innovation Under Resource Constraints: Supercomputing in Scientific Research*, SSRN Working Paper, 2024.

Merton, Robert, "The Matthew Effect in Science", *Science*, 1968, *159* (3810), 56–63.

Mosalaganti, Shyamal, **Obarska-Kosinska**, Agnieszka, **Siggel**, Marc, **Taniguchi**, Reiya, **Turoňová**, Beata, **Zimmerli**, Christian, **Buczak**, Katarzyna, **Schmidt**, Florian, **Margiotta**, Erica, **Mackmull**, Marie-Therese, **Hagen**, Wim, **Hummer**, Gerhard, **Kosinski**, Jan, and **Beck**, Martin, "AI-based structure prediction empowers integrative structural analysis of human nuclear pores", *Science*, 2022, *376* (6598), eabm9506.

Murray, Fiona, **Aghion**, Philippe, **Dewatripont**, Mathias, **Kolev**, Julian, and **Stern**, Scott, "Of Mice and Academics: Examining the Effect of Openness on Innovation", *American Economic Journal: Economic Policy*, 2016, *8* (1), 212–252.

Neumann, Mark, **King**, Daniel, **Beltagy**, Iz, and **Ammar**, Waleed, "ScispaCy: Fast and Robust Models for Biomedical Natural Language Processing", *Proceedings of the 18th BioNLP Workshop and Shared Task*, Florence, Italy: Association for Computational Linguistics, 319–327, 2019.

Nicholson, Joshua and **Ioannidis**, John, "Conform and be funded", *Nature*, 2012, *492* (7427), 34–36.

Nielsen, Mathias and **Andersen**, Jens, "Global citation inequality is on the rise", *Proceedings of the National Academy of Sciences*, 2021, *118* (7), e2012208118.

Nogales, Eva, "The Development of Cryo-EM into a Mainstream Structural Biology Technique", *Nature Methods*, 2016, *13* (1), 24–27.

Packalen, Mikko and **Bhattacharya**, Jay, *Does the NIH Fund Edge Science?*, Working Paper, National Bureau of Economic Research, 2018.

Papke, Leslie and **Wooldridge**, Jeffrey, "Econometric Methods for Fractional Response Variables with an Application to 401(k) Plan Participation Rates", *Journal of Applied Econometrics*, 1996, *11* (6), 619–632.

Protein Structure Prediction Center, *CASP13: Results Tables and Target Structures*, https://predictioncenter.org/download_area/CASP13/, 2018.

— *CASP14: Results Tables and Target Structures*, https://predictioncenter.org/download_area/CASP14/, 2020.

Ruff, Kiersten and **Pappu**, Rohit, "AlphaFold and implications for intrinsically disordered proteins", *Journal of Molecular Biology*, 2021, *433* (20), 167208.

Santos, Silva and **Tenreyro**, Silvana, "The Log of Gravity", *The Review of Economics and Statistics*, 2006, *88* (4), 641–658.

Senior, Andrew, **Evans**, Richard, **Jumper**, John, **Kirkpatrick**, James, **Sifre**, Laurent, **Green**, Tim, **Qin**, Chongli, **Žídek**, Augustin, **Nelson**, Alexander, **Bridgland**, Alex, et al., "Improved protein structure prediction using potentials from deep learning", *Nature*, 2020, *577* (7792), 706–710.

Shi, Yigong, "A milestone in protein crystallography: The 100th anniversary of X-ray diffraction", *Science China Life Sciences*, 2014, *57* (9), 856–857.

Stevens, Raymond, "The Cost and Value of Three-Dimensional Protein Structure", *Drug Discovery World*, 2003, *4* (3), 35–48.

Terwilliger, Thomas, **Liebschner**, Dorothee, **Croll**, Tristan, **Williams**, Christopher, **McCoy**, Airlie, **Poon**, Billy, **Afonine**, Pavel, **Oeffner**, Robert, **Richardson**, Jane, **Read**, Randy, and **Adams**, Paul, "AlphaFold predictions are valuable hypotheses and accelerate but do not replace experimental structure determination", *Nature Methods*, 2024, *21* (1), 110–116.

The UniProt Consortium, "UniProt: the Universal Protein Knowledgebase in 2025", *Nucleic Acids Research*, 2025, *53* (D1), D609–D617.

Thompson, Neil and **Zyontz**, Samantha, *Decomposing the "Tacit Knowledge Problem:" Codification of Knowledge and Access in CRISPR Gene-Editing*, SSRN Working Paper, 2017.

Tunyasuvunakool, Kathryn, **Adler**, Jonas, **Wu**, Zachary, **Green**, Tim, **Zielinski**, Michal, **Žídek**, Augustin, **Bridgland**, Alex, **Cowie**, Andrew, **Meyer**, Clemens, **Laydon**, Agata, et al., "Highly accurate protein structure prediction for the human proteome", *Nature*, 2021, *596* (7873), 590–596.

Varadi, Mihaly, **Anyango**, Stephen, **Deshpande**, Mandar, **Nair**, Sreenath, **Natassia**, Cindy, **Yordanova**, Galabina, **Yuan**, David, **Stroe**, Oana, **Wood**, Gemma, **Laydon**, Agata, **Žídek**, Augustin, **Green**, Tim, **Tunyasuvunakool**, Kathryn, **Petersen**, Stig, **Jumper**, John, **Clancy**, Ellen, **Green**, Richard, **Vora**, Ankur, **Lutfi**, Mira, **Figurnov**, Michael, **Cowie**, Andrew, **Hobbs**, Nicole, **Kohli**, Pushmeet, **Kleywegt**, Gerard, **Birney**, Ewan, **Hassabis**, Demis, and **Velankar**, Sameer, "AlphaFold Protein Structure Database: massively expanding the structural coverage of protein-sequence space with high-accuracy models", *Nucleic Acids Research*, 2022, *50* (D1), D439–D444.

von Appen, Alexander, **Kosinski**, Jan, **Sparks**, Lenore, **Ori**, Alessandro, **DiGuilio**, Amanda, **Vollmer**, Benjamin, **Mackmull**, Marie-Therese, **Banterle**, Niccolo, **Parca**, Luca, **Kastritis**, Panagiotis, **Buczak**, Katarzyna, **Mosalaganti**, Shyamal, **Hagen**, Wim, **Andres-Pons**, Amparo, **Lemke**, Edward, **Bork**, Peer, **Antonin**, Wolfram, **Glavy**, Joseph, **Bui**, Khanh, and **Beck**, Martin, "In situ structural analysis of the human nuclear pore complex", *Nature*, 2015, *526* (7571), 140–143.

Wei, Chih-Hsuan, **Allot**, Alexis, **Lai**, Po-Ting, **Leaman**, Robert, **Tian**, Shubo, **Luo**, Ling, **Jin**, Qiao, **Wang**, Zhizheng, **Chen**, Qingyu, and **Lu**, Zhiyong, "PubTator 3.0: an AI-powered literature resource for unlocking biomedical knowledge", *Nucleic Acids Research*, 2024, *52* (W1), W540–W546.